UNIT 1 INTRODUCTION TO SETS

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1.0 INTRODUCTION

Sometimes, we deal with some types of collections e.g.

i) Collection of books in a university library..

ii) Collection of natural numbers which are factors of, say, 20 or any other natural number.

A set is also a collection of objects, but it is a well-defined collection (we will learn more about this in Sec 1.2). Consider the collection of states in India. We know that there are 28 states in India, and this figure is exactly 28 (neither one more nor one less, as in January 2026). Also, if any number of people (with general knowledge of states in India) are asked to write the names of the states, the final list of everyone will contain the same 28 names (the order in which they write the names of the states does not matter). Such a well-defined collection is known as a set.

By **well defined** collection, we mean that given any object we must be able to know as to whether it belongs to the collection or does not belong to the collection.

In this unit, you will be introduced to the notations and terminology used for sets. The unit defines set, its various types, discusses hierarchy of sets, Venn diagrams, various operations on sets and finally, the unit is closed by giving an idea of some important and commonly used laws. Concepts related to sets are very elementary and are used directly or indirectly in many of the courses in this programme. So you need to understand the concepts discussed in this unit for further learning in this programme.

1.1 OBJECTIVE

After completing this unit, you should be able to:

- define a set,
- write a set in different forms,
- explain the types of sets,
- draw Venn diagrams,
- apply different operations on sets,
- get the idea of super sets and subsets,
- get an idea of some important laws related to sets, like commutative, associative, De-Morgan's laws, etc., and
- apply set theory concepts for problem-solving.

1.2 SETS

Conventionally, the concept of *number* and geometric concepts like *'straight line', triangle' and 'square'*. have been the most fundamental, and the very first of the concepts discussed in introductory mathematics. Since almost the middle of the previous century, the concept of ***'set'*** is considered as the most fundamental concept of mathematics and all other concepts, including even those of counting numbers and geometrical shapes, are discussed in terms of ***set***.

1.2.1 Concept of Set

Set is a basic concept like 'point', & 'straight line' in geometry and like 'number' in arithmetic, and hence, should be left undefined. However, in view of the fact, that 'set' is now-a-days considered as one of the most fundamental concepts in mathematics; to make a beginning in *new* mathematics, we just give a sort of working definition, and define some elementary operations on sets to be frequently used in the subsequent discussion.

Roughly, a set may be considered a collection of things or entities, such as names of animals, names of books, names of numbers, names of geometrical shapes, or any mixture of these. Any entity in a set is called a member, or an element, of that set. Further, in a set, **no member is written more than once**. Also, the entities must be ***distinctly identifiable***. For example, as the drops of water in a glass cannot be distinctly identified, hence, drops of water in a glass ***do not*** form a set.

More formally, we take a ***set as a well-defined collection of objects.*** *What is meant by 'well-defined' needs to be understood clearly.* ***A collection is said to be well-defined, if:***

(i) entities included in it are distinctly identifiable, and,

(ii) if we pick up any distinctly identifiable entity in the world, then we should be able to tell whether the picked-up entity is in (i.e. belongs to) the collection or not.

For example, let us consider the collection of ***all*** students of IGNOU who enrolled in the 2026 July session for the B.A. program. This is a well-defined collection, hence a set. But if we say the collection of ***some of*** those students of

IGNOU who enrolled in the 2026 July session for the B.A. program, then ***it is NOT a set.*** Also, if we say 'collection of all drops of water in this glass', then ***it is NOT a set,*** because, as mentioned earlier, a 'drop of water' is not distinctly identifiable. Thus, *not* ***any*** **collection is a set, only a well-defined collection is a set.**

A set is generally denoted by capital letters such as A, B, C, X, Y, Z, etc. and the objects which belong to the set are known as elements or members of the set and are generally denoted by small letters a, b, c, x, y, z, etc.

If 'a' is an element of a set A then we write $a \in A$ (read it as 'a' belongs to A). If 'a' is not an element of A then we write $a \notin A$ (read as 'a' does not belong to A).

Following example illustrates the term "well defined collection" or "set".

"a well-defined collection of distinct objects is called a set".

A set remains the same if some or all of its elements are repeated or rearranged.

For example, if a set contains the elements 0, 1, – 1 and another set contains the elements – 1, – 1, 1, 0, 0, 0, 1, 0, 1, then these two sets are nothing but represent the same set having three elements 0, 1, – 1.

Example 1: Consider the following collections and state reasons whether they form set or not.

(i) Collection of good cricketers in India.

(ii) Collection of honest students in a particular university in India.

(iii) Collection of natural numbers which are less than 5.

(iv) Collection of rich persons in India.

(v) Collection of letters of the word "ASSIGNMENT"

Solution:

(i) This collection does not form a set, because a given player may be good according to some person, but the same player may not be good according to some other person.

(ii) This collection does not form a set, because a student may be honest according to some person, but the same student may not be honest according to some other person.

(iii) Yes, this collection forms a set, and its elements are 1, 2, 3, and 4.

(iv) Richness is not a well-defined property, because, according to one person, a person may be rich while he/she may not be rich in view of some other person. So this collection does not form a set.

(v) It is a set, and its elements are A, S, I, G, N, M, E, T.

Notations

Like most of the mathematical concepts, sets have special notation. For illustration, the ***notation {0, 1, 2, 3, 4, 5, 6, 7, 8, 9***} is used to denote the set of all decimal digits, viz., 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

A set is generally represented by two methods: the **roster method and set-builder method.**

The notation for a set has at the leftmost position a left brace, denoted by *'{'*. Followed by one of the elements, say 0, then a comma and a blank-space, next another element say 1, followed by a comma and blank-space, and so on, till the last element is written, which is followed by the right-brace, denoted by *'}'*. Generally, a set is given a name, e.g., in this case the name of the set may be ***Num-Digits***, then we write Num-Digits= *{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}*. Any distinct name –the name which has not been used earlier for naming any other set –may be assigned. However, in general, the name assigned is mnemonic

(i.e., it should help remember its contents or members).

Member/element of a set is what is included, except commas, within braces. For example, 0 is a member, or element, of Num-Digits, 5 is a member of Num-Digits etc. Instead, we may also say 0 ***belongs to*** Num-Digits, 5 *belongs to* Num-Digits etc. However, we say 55 *does not belong to* Num-Digits.

Notation for member is the symbol $\in$**,** *called epsilon.* It is used to denote membership of a set. For the above example of Num-Digits, we will write 0 $\in$ Num-Digits, 9 $\in$ Num-Digits etc. Similarly, if we have a set called Vowels, with Vowels= {a,u,e,o,i}, then e $\in$Vowels, u$\in$Vowels etc. *Whenever x is a member of the set X, or equivalently, when x* $\in$X, *then we also say* ***'x belongs to X'. The notation for 'is not a member of' or for 'does not belong to 'is*** $\notin$**.** For example, for the definition of *Num-Digits given above, 55* $\notin$ Num-digits.

Extension of Set Notation: The problem with the set notation mentioned above arises when the set has either a very large or an infinite number of elements. Because then it is quite difficult, if not impossible, to specify all the elements required to use the notation. In such cases, if there is some pattern among the elements, then only some initial members are mentioned, followed by ***triple dots(...).*** For example, the set of numbers, each of which is less than 10,001 and is also square of some integer, may be denoted by {0, 1, 4, 9, …, 100, 121, …, 400, …., 10000}, or even more briefly by {0, 1, 4, 9, …, 10000} etc. It may be named as ***Sq_int_less_10001***.

1.2.2 Methods of Representing a Set

A set is generally represented by two methods as given below.

A. Roster Method

Dictionary meaning of 'Roster' is 'a list showing persons who perform their duties in turn'**.** As its meaning suggests, in this method each and every element is listed and put, separating by commas, in curly brackets. This method is also known as **Tabular Form** or **Listing Method.**

For example,

(i) If A is the set of vowels of English alphabets, then A = {a, e, i, o, u}

(ii) If N is the set of natural numbers, then N = {1, 2, 3, 4, 5, …}

(iii) If W is the set of whole numbers, then W = {0, 1, 2, 3, 4, 5, …}

(iv) If Z is the set of integers, then Z = {… , – 3, – 2, – 1, 0, 1, 2, 3,…}

(v) If E is the set of even natural numbers, then E = {2, 4, 6, 8, 10, 12, …}

(vi) If O is the set of odd natural numbers, then O = {1, 3, 5, …}

(vii) If P is the set of prime numbers, then
P = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, …}

Remark 1:

Throughout the course, we use N, W and Z for the sets of natural numbers, whole numbers, and integers, respectively.

B. Set-Builder Method

In this method, we consider one or more properties that are exclusive to the elements of a set, so that no other elements can be members of the set. This method is also known as the **Property Method** or **Rule Method.**

For example,

(i) Let A = {x : x is a vowel of English alphabet}, then elements of A are a, e, i, o, u and having exclusive property of being a vowel no other alphabet can be considered as an element of set A.

(ii) Let A = {x : x is a natural number and x is a multiple of 3}, then elements of A are 3, 6, 9, 12, … which have the exclusive property of being multiple of 3 and no other element can be consider as an element of A.

(iii) Let A = {x : x is a factor of 10 and x > 0}, then elements of A are 1, 2, 5, 10 and having exclusive property of being a factor of 10 and no other element can be consider as an element of A.

(iv) If $\mathbb{Q}$ is the set of rational numbers then $\mathbb{Q} = \left\{x : x = \frac{p}{q}, p, q \in Z, q \neq 0\right\}$

Rational number: A number which can be expressed in the form $\frac{p}{q}$ such that $p, q \in Z, q \neq 0$ is known as rational number. For example, $\frac{2}{3}, \frac{-4}{13}, etc.$

Remark 2:

(i) In terminology of set the symbol ":" used in each part above is read as "such that".

(ii) Throughout the course, the sets of rational numbers, irrational numbers, and real numbers will be denoted by $\mathbb{Q}$, I, and $\mathbb{R}$, respectively.

Irrational number: A number which cannot be expressed in the form $\frac{p}{q}$ where p, q are integers and q ≠ 0, is called an irrational number. For example, $\sqrt{2}, \sqrt{5}, \sqrt[3]{10}$, etc.

Note: Advantage of the second method, i.e. Set-Builder method lies in the fact that sometimes (or in some situations) we cannot list the elements of the set or even if we can list them, it may not be practical or feasible to do so.

For example, consider the set {x: x is a person who was born in 2010 in India}. Obviously, you will be more comfortable with the property method in this example.

Consider another set {x : x is a real number and 1 < x < 4}. This set cannot be described by the listing method, because the number of elements(Cardinality) in this set is uncountable.

Thus, the above two examples show that in some situations, either it is too difficult to describe the set by the listing method or it is impossible to describe it.

No doubt, there are some examples in which the listing method has its advantages .

For example, consider the set {28, Bihar, India}.
It is a set having three elements 28, Bihar and India.

Let us now consider some examples to clarify the ideas of the two methods discussed above.

Example 2: Write the following sets by the roster method:

(i) A = {x : x is a letter of the word "FUNCTION"}

(ii) B = {x : 2x + 5 < 17, x∈N}

(iii) C = {x : $x^2 - x - 12 = 0$, x∈N}

(iv) D = {x : $x^2 - 4x - 21 = 0$, $x^2 - 49 = 0$, x∈N}

Real number: A number which is either rational or irrational is called real number.

Solution:

(i) A = {F, U, N, C, T, I, O} [Repeated elements are written once only]

(ii) B = {x : 2x < 17 – 5 , x∈N} = {x : 2x < 12 , x∈N}

= {x : x < 6 , x∈N} = {1, 2, 3, 4, 5}

(iii) C = {x : x^2 – x – 12 = 0, x∈N}= {x : x^2 – 4x + 3x – 12 = 0, x∈N}

= {x : x(x – 4) + 3(x – 4) = 0, x∈N}= {x : (x – 4) (x + 3) = 0, x∈N}

= {x : x = 4, – 3, x∈N} = {4} $[\because -3 \notin N]$

(iv) D = {x : x^2 – 4x – 21 = 0, x^2 – 49 = 0, x∈N}

= {x : x^2 – 7x + 3x – 21 = 0, x^2 – 7^2 = 0, x∈N}

= {x : x (x – 7) + 3(x – 7) =0, (x – 7) (x + 7) = 0, x∈N}

= {x : (x – 7)(x + 3) = 0, x = 7, – 7, x∈N}

= {x : x = 7, – 3, x = 7, – 7, x∈N}

= {7}

∵ here are two properties one says x = 7, -3 and second says x = 7, -7, but in case we have more than one property, we take common element(s) between them, and in our case, it is 7.

Example 3: Express the following sets in the set-builder form:

(i) A = {3, 6, 9, 12, 15, 18, …}

(ii) B = {1, 2, 3, 5, 6, 10, 15, 30}

(iii) C = {5, 25, 125, 625, …}

(iv) D = {1, $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{27}$, $\frac{1}{81}$, $\frac{1}{243}$}

(v) E = {1, 4, 9, 16, 25, 36, 49, 64, 81, 100}

Solution:

(i) Here we see that elements in this set are multiple of 3 so in set-builder form it can be written as A = {x: x is a multiple of 3, x∈N}.

Similarly, by observing the pattern obeyed by the elements of other parts we can write them as given below.

(ii) B = {x: x is a factor of 30, x∈N}

(iii) C = {x: x =5^n, n∈N}

(iv) D = {x: x = $\frac{1}{3^n}, 0 \leq n \leq 5, n \in$W}

(v) E = {x : x = n^2, $1 \leq n \leq 10$, n∈N}

1.2.3 Relationships between Sets

For any two sets X and Y, the set relationships that we discuss are:

(i) *equality (=) of X and Y*

(ii) X is a *subset/ superset (⊆)* of Y

(iii) X is a proper *subset/superset(⊂)*of Y, and

(iv) is *not a subset* of **(⊄).**

1.2.3.1 Notations, Definitions and Examples

i) Two sets X and Y are said/defined as ***equal*** if every element of X is an element of Y, *and* every element of Y is an element of X.

The *notation X=Y* is used for 'equality of sets X and Y'

Example 4: Let X = {1, 2, 3, 4, 5}, and Y = {x: x is a natural number such that $1 \leq x \leq 5$}, and W= {3, 5, 2, 4, 1}.Then X = Y = W.

ii) X is said to be a ***subset*** of Y if every element of X is an element of Y. Instead of saying ***'X is a subset of Y'*** we may equivalently say ***'Y is a superset of X', or 'X is contained in Y' or 'Y contains X'.***

The notation $X \subseteq Y$ is used for 'X is a subset of Y'.

Example 5:

(i) Let X = {a, e, i}, and Y = set of all vowels in English language. Then $X \subseteq Y$.

(ii) Every set is its own subset, i.e. $X \subseteq X$ (can be easily verified by applying the definition).

iii)X is said to be a ***proper subset*** of Y if (i) every element of X is an element of Y, *and also* (ii) there is at least one element of Y which does not belong to X. Then, it can also be said that Y is a proper superset of X.

Definition: A set A is said to be a **proper subset** of a set B if A is a subset of B and A and B are not equal. We represent this by $A \subset B$.

For example, if A = {4,5,6} and B = {4,5,7,8,6}, then $A \subset B$; and if A = {Java, C, C++, Cobol} and B= {Java, C++}, then $A \supset B$.

Note: A set can have many subsets and many supersets.

For example, A = {1,2,3,4,5}, B= {2,3,4,5,6,7}, and C= {2,3}, then for C, A and B can be used as supersets.

Similarly, if X= {Ram, Rani, Sita Gita}, Y= {Rani}, and Z= {Sita}, then Y and Z both are subsets of X.

The notation $X \subset Y$ is used for 'X is a proper subset of Y' and also for 'Y is a proper superset of X'.

Example 6: Let X = set of even decimal digits = {0, 2, 4, 6, 8} and let Y = set of all decimal digits = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}. Then **$X \subset Y$, as every even decimal digit is also a digit. But $1 \in Y$, and $1 \notin X$.**

The notation **$X \not\subset Y$** denotes the fact that X is ***not a subset*** of Y. For showing $X \not\subset Y$, we need to show *only* that there is atleast *one* element of X which does not belong to Y. For example, Let X = {1, 5, 10, 15}, and Y = {1, 12, 14, 15}. To show **$X \not\subset Y$,** we need *only* to show that $5 \in X$, and $5 \notin Y$. We *need* not go further to say that $10 \in X$, and $10 \notin Y$. *Also, $X \not\subset X$, as condition (ii) for being a proper subset, is not satisfied.*

1.2.3.2 Establishing Set Relationships

Here, through a simple example, we demonstrate how to establish each of these set relationships. Later, we will consider some more complex examples of establishing these relationships, particularly regarding the equality (=).

Let us consider sets

A = {x: $x^2+3x+2= 0$}, the set of solutions of the equation $x^2+3x+2=0$.

B = { -1,-2}, C= { -2}, E = {0, -1,-2}

i) Easiest case is that of ***'is not a subset of':*** In our case,

E ⊄ C, as 0 ∉ E, but 0 ∉ C. Similarly, ***E ⊄ A, as*** 0 ∈ *E, but 0 is not a root of* $x^2+3x + 2 = 0$.

ii) Next, in order to show B ⊆ E, we need to consider, one by one, each element of B, and show that the considered element belongs to E also. In complex cases, we do not consider each element one by one, but take an arbitrary element.

iii) Further, in order to show that B is a ***proper subset*** of E (i. e.B ⊂ E), in addition to what we have done in (ii) above, we need to show the existence of an element of E, which does not belong to B. In our case 0 ∈ E, but 0 ∉ B.

iv) To show equality (=) of two sets, we need to repeat case (ii) two times, reversing the roles of the sets under consideration.

In our case, A=B, because -1, which belongs to B, is a root of $x^2+3x+2=0$, hence, belongs to A. Similarly, -2, which belongs to B, is a root of $x^2+3x+2 = 0$, hence, belongs to A. Conversely, as $x^2+3x + 2 = 0$ is a quadratic equation, it has exactly two roots, viz. -1, -2.

1.2.4 Cardinality of a Set

The number of elements in a finite set say A is called its Cardinality or cardinal number of a finite set and is denoted by n(A).

For example,

(i) If A = {x, y, z}, then n(A) = 3, i.e. cardinality of A is 3.

(ii) If B = {1,4, 2, 3, 9, 15}, then n(B) = 6, i.e. cardinality of B is 6.

1.2.5 Subsets

Before we relook at the subset, let us define the empty set.

The **empty set** (also known as the null set) is a unique concept in mathematics and set theory. It represents a set that contains **no elements**. It is denoted by ∅ or { } : A pair of curly braces with nothing between them.

Definition: A set A is said to be a **subset** of a set B if each element of A is also an element of B. In this case, B is called a **superset** of A. If A is a subset of B, we represent this by **A ⊆ B.**

As a statement in logic, we represent this situation as,

$A \subseteq B \Longleftrightarrow [x \in A \Rightarrow x \in B]$

'B contains 'A' or 'B' is a superset of A' is represented by B ⊇ A.

If A is not a subset of B, we write **A ⊈ B**.

For example,

i) if A ={4,5,6} and B = {4,5,7,8,6}, then A ⊆ B. But if C = {3,4} then C ⊈ B.

ii) If set A is set of all the cricketers of Indian Cricket Team and set B is all the bowlers of India cricket Team then B ⊆ A.

Remarks 2:

i) There is **no complete standardization of the notation of subset.** Some people use ⊂ Y, as every even decimal digit is also a digit. But 1 '⊂' to denote a subset, which may or may not be a proper subset. Then, they use the symbol ⊊ for proper subset. We could have used these symbols. But we would continue with symbols '⊆' and '⊂'denoting respectively 'subset' and 'proper subset'.

Next, we state, without proofs, some important results about the relation of subset

- A ⊆ A for every set A. Every set is a subset of itself.
- If A ⊆ B and B ⊆ C then A ⊆ C. We say that ⊆ is a *transitive relation* between sets.
- If $A \subseteq B$ and $B \subseteq A$ then $A = B$.

1.2.6 Power Set of a Set

Definitions: The power set of a set A is the set of all the subsets of A, and is denoted by P(A). Sometimes power set of a given set A is denoted by 2^A, because that is the Cardinality of the power set A.

Mathematically, P(A) = {x:x ⊆ A}.

Note that:

- If $|A|$ = n, then P(A) = 2^n. For example, the set {1, 2, 3} has 3 elements, and, the power set of A, viz. p(A) = {∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}} has 2^3 = 8 elements.

- ∅ ∈ P(A) and A ∈ P(A) for all sets A.

For example, if A={1}, then P(A)= {∅ , {1}} and if A={1,2}, then P(A)={ ∅ , {1}, {2}, {1,2}}
Similarly, if A = {1,2,3}, then P(A) = {∅, {1}, {2}, {3}, {1,2}, {1,3}, {2,3}, {1,2,3}}.

Definition: Any set which is a superset of all the set under consideration is known as the **universal set**. This is usually denoted by Ω, S or U.
For example, if A={1,2,3}, B = {3,4,6,9} and C = {0,1}, then we can take
U = {0,1,2,3,4,5,6,7,8,9} or U=N, or U=Z as the universal set.

Note that the universal set can be chosen arbitrarily for a given problem. But once chosen, it is fixed for the discussion of that problem.

Theorem 1: If A is a set with n elements, then $|P(A)| = 2^n$.
Proof: We shall prove this by mathematical induction.
For this, we first check if it is true for n=1. Then assuming that it is true for n=m, we prove it for the case n = m + 1. It will, the, follow that the result will be true $\forall$ n $\in$ N.
Step I: If A = 1, then $P(A) = 2 = 2^1$.
Step II: Assume that the theorem holds for all sets A of cardinality k, i.e. if |A| = k, then A has 2^k subsets.
Step III: Now consider any set $A = \{x_1, x_2, x_3, \ldots x_k, x_{k+1}\}$, with k+1 elements. Consider its subset $B = \{x_1, x_2, x_3, \ldots x_k\}$. now B has 2^k subsets, each being a subset of A. Now, take any such subset $\{x_{i1}, x_{i2}, \ldots x_{ir}\}$of B. Then $\{x_{i1}, x_{i2}, \ldots, x_{k+1}\}$ is a subset of A that is not a subset of B. So, for each of the 2^k subsets of B, we attach x_{k+1} to it to get 2^k more subsets of A.
You can see that this covers all the subsets of A.
So the number of subsets of $A = 2^k + 2^k = 2.2^k = 2^{k+1}$.
Hence the theorem.

Let us first consider some examples:

(i) If A = { }, then $\emptyset$ is only subset of A.
That is, there is only 1 ($=2^0$) subset of A.

(ii) If A = {a}, then possible subsets of A are $\emptyset$, {a}.
That is, there are only 2 ($=2^1$) subsets of A.

(iii) If A = {a, b}, then possible subsets of A are $\emptyset$, {a}, {b}, {a, b}.
That is, there are only 4 ($=2^2$) subsets of A.

(iv) If A = {a, b, c}, then possible subsets of A are $\emptyset$, {a}, {b}, {c}, {a, b}, {a, c}, {b, c}, {a, b, c}.
That is, there are only 8 ($=2^3$) subsets of A.

For example, in the above discussed cases (i) to (iv), the power set of A is given by

(i) P (A) = {$\emptyset$ }

(ii) P (A) = {$\emptyset$, {a}}

(iii) P (A) = {$\emptyset$, {a}, {b}, {a, b}}

(iv) P (A) = {$\emptyset$, {a}, {b}, {c}, {a, b}, {a, c}, {b, c}, {a, b, c}}

Check Your Progress -1

1) Give reasons whether the following collections are sets or not.

(i) Collection of intelligent students in a particular school.

(ii) Collection of good hockey players in India.

(iii) Collection of good actors in India.

(iv) Collection of vowels in the word "INDIA".

……

……

2) Describe the following sets by roster method:

(i) A = {x: x = 2n + 3, n ∈W}

(ii) (ii) B = {x : x = 7^n, 0 $\leq n \leq 3, n \in N$}

(iii) C = {x : x∈N and x∈W}

(iv) D = {x : x∈W and x∈Q}

……

……

3) Express the following sets in set-builder form:

(i) {5, 10, 15, 20, …} (ii) $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$ (iii) {2, 4, 6, 8, 10, …}

……

……

4) For the pair of sets X and Y, state whether X = Y or not, along with explanation for your response.

i) X = {2, 4, 6, 8, 10}, Y = {*x*: *x* is positive even integer and *x*≤ 10}

ii) X = {*x*: *x* is a multiple of 10}, Y = {10, 15, 20, 25, 30 …}

iii) X = {x: x is a letter in the word FOLLOW}, Y = {*y*: *y* is a letter in the word WOLF}

……

……

5) For X = {1, {2, 3}, 3, 4}, which ones of the following is/are true statement(s)?

(i) 2∈X, (ii) $\emptyset \in X$, (iii) $\emptyset \not\subset X$, (iv) $\emptyset \not\subset X$, (v) {2, 3} ∈ X , (vi) {2,3} ⊂ X, (vii) {{2,3}⊂ X

……

……

6) Give two proper subsets and two supersets of the set of vowels of the English alphabet.

……

……

7) Find the power set of the set A ={a,e,i,o,u}.

……

……

8) For which set A, is P(A) = 1 ?

……

……

9) If $A \subseteq B$, is $P(A) \subseteq P(B)$? Why?

……

……

10) $P(A) = P(B) \Longrightarrow A = B$. True or false? Why?

……

……..

1.3 TYPES OF SETS

We have seen that set is a well-defined collection of distinct objects. Also repetition of elements in a set is not allowed. So once a set is defined, automatically number of elements contained by it has also become fixed. In this section, we shall discuss the different names given to a set on the bases of the number of elements contained by the set. Equivalent and equal sets are also defined in this section.

Null Set or Empty Set

Consider a collection of those sons having their ages more than their respective fathers. Of course we will find no such son in this world. This type of collection is nothing but simply known as null set or **empty set** or **void set** in the terminology of sets.

It may appear strange, at least when heard for the first time, that a set may have no member at all, or the number of elements in a set may be zero. According to our definition of a set, the following is a set:
$S = \{x: x \text{ is an even integer, and } x^2 = 9\}$.
The collection S is well-defined, and hence, is a set. But there is no member which satisfies the conditions of its membership. Therefore, the number of members of S is zero. There are many combinations of conditions, which may make a set to have no member, Each of such sets is called a Null set, a Void set or an empty set. ***A null set is denoted by the standard notation*** $\emptyset$ or by {}.

Remarks 3*:* $\emptyset$ is a subset of every set X, i. e. $\emptyset \subseteq X$, for every set X.

The fact is easily established through contradiction, a common proof technique in Mathematics: Assume what is to be proved as wrong, and the reach a fact which conflicts with some earlier established fact. So, let $\emptyset \subseteq X$ be false. Then there must an $x \in \emptyset$, *and* $x \notin X$. But $x \notin \emptyset$ is not possible by definition of $\emptyset$. Hence, the assumption that '$\emptyset \subseteq X$ be false' is wrong. Hence the proof.

Let us now formally define null set.

A set is said to be null (or empty or void) if it has no element in it. Null set is denoted by $\emptyset$ or { }.

For example, $A = \{x : x \text{ is a natural number}, 1 < x < 2\}$ is a null set as there is no natural number between 1 and 2.

Proper subsets of a set X: As $\emptyset$ and a set X itself are subsets of the set X. These two subsets are not considered as proper subsets. All other subsets are considered as proper subsets.

Singleton Set

Consider the collection of mothers of a baby. Obviously, a baby has only one mother. This type of collection having a single element is known as singleton set in the terminology of sets. Thus, a singleton set is defined as follow.

A set is said to be singleton set if it contains only one element.

For example, A = {x : x is an even prime number} is a singleton set as there is only one even prime number, i.e. 2 .

Finite Set

A set is said to be finite set if either it is an empty set or it has a finite number of elements.

For example,

(i) A = {2, 5, 7, 15} is a finite set because it contains 4 elements, i.e. finite number of elements.

(ii) B = {1, 2, 3, 4, 5, …} is not a finite set as the number of elements in it are infinitely many.

(iii) C = {x : x +1 = 0, x∈N} = {x : x = – 1, x∈N}= { } is an empty set. So, it is a finite set.

Infinite Set

A set is said to be infinite if it is not finite.

For example, A = {1, 4, 9, 16, 25, 36, …} is an infinite set.

Equivalent Sets

Two finite sets A and B (say) are said to be equivalent if number of elements in both the sets are equal in numbers, i.e. n(A) = n(B) and we denote it by A ~ B (read as A is equivalent to B).

For example, if A = {a, b, c, d} and B= {2, 3, 5, 7}, then
A ~ B [∵ n(A) = n(B) = 4]

Equal Sets

Two sets A and B are said to be equal if every element of A is in B and every element of B is in A and is written as A = B.

If two sets A and B are not equal then we write A ≠ B.

Definition: Two sets A and B are equal if every element of A belongs to B and every element of B belongs to A. We represent this by A=B.

For example,

i) if A = {a, b, c, d} and B = {c, b, d, a}, then A = B as order of elements does not matter.

ii) if A= {1,2,3}, B={2,3,1}, then A ⊂ B and B ⊂ A, so that A = B.

iii) if A= {1,3,5,7,9,}, B={2,3,5} then A ≠ B.

Example 7: Give reasons whether the following statements are true or false:

(i) If A = {x : x is a vowel of English alphabet} and
B = {x : x is a natural number, 7 < x < 13}, then A = B

(ii) If A = {x : $x^2 = 9, x \in z$} and B = {3, – 3}, then A = B

(iii) If A = {x, y, z, w} and B = {d, e, 7, 9}, then A ~ B

(iv) If A = {x, x, y, z} and B = {x, y, z, w}, then A ~ B

Solution:

(i) Here, A = {a, e, i, o, u} and B = {8, 9, 10, 11, 12}.
Clearly, $A \neq B$ (as there elements are not same). Hence, the given statement is false.

(ii) Here, A = {3, – 3}

$$\left[\begin{array}{l} \because \text{ we know that if } x^2 = a \text{ then } x = \pm\sqrt{a} \\ \therefore x^2 = 9 \Rightarrow x = \pm\sqrt{9} = \pm 3 \end{array}\right]$$

and

B = {3, – 3}, Clearly, A = B

$$\left[\begin{array}{l} \because \text{ all the elements of A are in B and} \\ \text{all the element of B are in A.} \end{array}\right]$$

∴the statement is true.

(iii) Here, n(A) = 4, n(B) = 4⇒ A ~ B, therefore, it is a true statement.

(iv) Here, A = {x, x, y, z} = {x, y, z} [As repetition in a set is not allowed]
and B = {x, y, z, w}, so n(A) = 3, n(B) = 4.
⇒ A ~ B, is false because n(A) ≠n(B).

1.4 HIERARCHY OF SETS

For given any two real numbers a and b, you know that either a = b or a < b or a > b. This section will focus on how this type of association is setup in case of sets. Actually, here, we consider the sets contained in some other sets and define them with their appropriate designations.

Subset

Suppose A be the set of all working ladies and B be the set of all ladies then obviously all working ladies are ladies first. That is all the members of the set A are members of the set B. If it is so, then in the terminology of the sets A is known as subset of B.

Now, let us formally define the term subset.

Let A and B be two sets. Then A is said to be subset of B (or B is **super set** of A) if every element of A belongs to B and is denoted by A⊆ B.

"A⊆ B " read as A is contained in B or A is a subset of B.

If we write it as "B⊇A" then we read it as B contains A and we call B is a super set of A.

Remark 4: From above definition of subset, we see that a set A(say) will not be subset of another set B(say) if there is at least one element in A which is not in B. And it is denoted by A⊄B (read as A is not a subset of B or A is not contained in B).

For example,

(i) If N = {1, 2, 3, 4, …}, W = {0, 1, 2, 3, 4, 5, …},

Z = {…, – 3, – 2, – 1, 0, 1, 2, 3 ,…}, $\mathbb{Q} = \{\frac{p}{q} : p, q \in Z, q \neq 0\}$,

then N⊆W, W⊆Z, Z⊆Q, i.e. N⊆W⊆Z⊆Q.

(ii) If A = {1, 2, 4}, B = {1, 2, 4, 7, 9}, then A⊆ B

(iii) If $A = \{a, b, c\}, B = \{b, c, d, e\}$ then A $\not\subset$ B $[\because a \in A \;\; but\; a \notin B]$

Proper Subset

Let A and B be two sets. Then A is said to be proper subset of B if all the elements of A are in B and B has at least one element other than elements of A and is denoted by A⊂ B.

For example, if A = {1, 2, 3} and B = {1, 2, 3, 4, 5},

then A⊂ B. $\left[\begin{array}{l}\because all\ the\ elments\ of\ A\ are\ in\ B\ and\ B \\ has\ two\ extra\ elements, i.e. 4\ and\ 5.\end{array}\right]$

Remark 5:

(i) Empty set is subset of every set, i.e. φ ⊆A for any set A.

(ii) Every set is a subset of itself, i.e. A⊆A for every set A.

Universal Set

Universal Set: For discussion problems relating to educating people in India, we need not take into consideration all human beings, we need to consider *set of only Indians*. While using mathematics for solving problems (like educating Indians) from some domain of human experience, some sets of entities from the domain are considered as and when required. For facilitating arguments, a *sufficiently large set of entities* from the problem domain- *which includes all possible entities* that may be required at any time during the discussion or solution-process of solving the problems of the type under consideration- is taken. For example, *in our present case, the set of all Indians*. ***Such a set is called the Universal set and is generally denoted by U***. *Thus, by definition, every set X under consideration will be a subset of the Universal setU. i. e.* ***X⊆U for all sets x under consideration during solving a particular (type of) problem.***

Let us take IGNOU, there are 21 schools such as school of humanities (SOH), school of social sciences (SOSS), school of sciences (SOS), etc. (source IGNOU dairy 2011). If U is the set of all faculties of IGNOU and $A_1, A_2, A_3, \ldots, A_{21}$ are sets representing the faculties of 21 schools. Then, of course, faculties of all these 21 schools are faculties of IGNOU. That is, all the members of these 21 schools are present in the set U. Here U plays the role of universal set for the sets $A_1, A_2, A_3, \ldots, A_{21}$.

A set U is said to be **universal set** if all the sets under study are subsets of U.

Remark 6:

- It may be noted that for different problem-types or different contexts, there will be different universal sets.

- By definition of universal set U, every set X is a subset of U, i.e. $X \subseteq U$. Also, from the fact that $X \subseteq X$ for every X, we get $U \subseteq U$.
- The concept of *'Universal set'*, since its inception, has been both controversial and problematic. Specially, it has led to a number of well-known paradoxes including the Russell's paradox, which implies that X, the collection of all sets, is not a well-defined collection, and hence, X is not a set.

However, the concept of *Universal Set* has facilitated representation and solution of problems from most of the domains of our concern including from mathematics itself.

For example,

(i) If A = {1, 2, 3, 5, 7, 9}, B = {2, 4, 6, 8, 10}, C = {3, 5, 7}, D = {8, 9, 10}, then U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} can play the role of universal set.

(ii) If in a study, only integers are involved as the elements of the sets, then Z, the set all integers, is the universal set.

Check Your Progress -2

1) Give reasons whether the following statements are true or false:

(i) If A = {2, 9, 7, 7, 5}, B = {5, 2, 2, 9, 7}, then A = B

……
……

(ii) If A = $\{\alpha, \beta, \gamma\}$, B = $\{\lambda, \mu, \nu\}$, then A ~ B

……
……

(iii) If A = {4, – 4, 5, 5}and
B = {x : either $x^2 = 16$ or $x^2 + x - 20 = 0$, x∈Z}, then A = B

……
……

(iv)If A = {a, a, b, b, b, c} and B = {d, e, e, f, g, h}, then A ~B

……
……

2) If A = {a, b, c, d}, then write P(A).

……
……

3) If A = {1, 3, 5, 7, 9}, B = {2, 4, 6, 8}, C = {2, 3, 5, 7, 11, 13, 17, 19, 23}, and D = {5, 10, 15, 20, 25}, then what will be the smallest universal set

……
……

1.5 VENN DIAGRAMS

As we know that examples play an important role to understand the concepts of theory/definitions. Similarly a diagram speaks more than the words that we

may use and they also make the ideas simple and easy to understand even for a fresh reader. Euler (1707-1783), a Swiss mathematician was the first who took the step to represent the sets diagrammatically. Then John Venn (1834-1923), a British mathematician, moving a step ahead simplifying the ideas and made it more user friendly. That is why diagrammatical representation of sets is also known as Venn-Euler diagram. But usually they are known as Venn-diagrams. In Venn diagrams, sets are represented by enclosed areas in a plane as described below:

Notations Used in Venn Diagrams

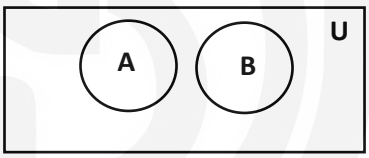
1. **Universal Set**

   Universal set U is represented by the interior of a rectangle as shown in Fig. 1.1

Fig. 1.1: Universal Set

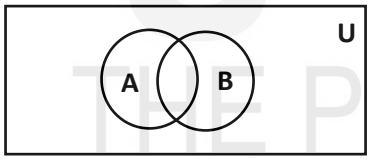
2. **Subsets**

   Subsets of U are described by the interior of closed curves (known as circular discs) within the rectangle, representing the universal set U.
   Fig. 1.2 shows the case when A and B have no common element,
   while Fig.1.3 shows the case when A and B have some common elements.
   Fig. 1.4 shows the case when $A \subseteq B$, and Fig. 1.5 shows the case when $B \subseteq A$.



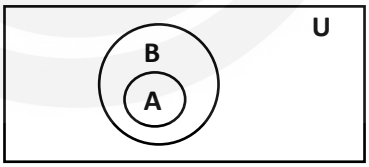
If $A \cap B = \emptyset$

Fig. 1.2: Subset



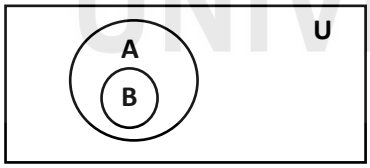
If $A \cap B \neq \emptyset$

Fig. 1.3: Subset



If $A \subseteq B$

Fig. 1.4: Subset



If $B \subseteq A$

Fig. 1.5: Subset

Remark 7:

(i) In general when nothing is mentioned about the common elements of A and B, presentation of Fig. 1.3 is used.

(ii) Sizes that we use to present A and B do not matter.

1.6 SET OPERATIONS

In school days, a child first learns counting numbers and then he/she learns how operations of addition, subtraction, multiplication and division are used on two numbers.

1.6.1 Different Set Operations

A similar type of approach is being used here. In sections 1.2 and 1.3 of this unit you have become familiar with the definition and types of sets respectively. In this section, we will learn about some commonly used operations on sets.

Union

Let A be a set containing the persons getting salary (in Rs) between 10000 and 100000 per month and another set B containing the persons getting salary (in Rs) between 5000 and 20000 per month. For this example, if we are interested in finding those persons who are getting the salary within the range 10000-100000 or 5000-20000 then such persons will be those having salary between 5000 and 100000. The set of such persons is nothing but the union of two sets A and B.

Now, let us formally define the union of two sets.

Let A and B be two sets then union of A and B is denoted by $A \cup B$ and is defined as

$A \cup B = \{x: \text{either } x \in A \text{ or } x \in B\}$.

i.e. $A \cup B$ contains all the elements of A as well as of B (see Fig. 1.6).

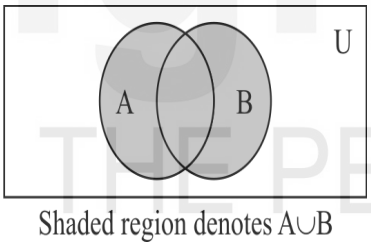


Fig. 1.6: Set Operation $A \cup B$

For example,

(i) If A = {2, 3, 5}, B = {3, 5, 7, 11}, then $A \cup B$ = {2, 3, 5, 7, 11}.

(ii) If A = {a, b, c, d}, B = {d, e, f}, then $A \cup B$ = {a, b, c, d, e, f}.

(iii) If $\mathbb{Q}$ = set of all rational numbers and I = set of all irrational numbers, then $\mathbb{Q} \cup I = \mathbb{R}$ = set of all real numbers.

That is, if all rational and irrational numbers are mixed then that mixture will be the set of real numbers.

Intersection

Let us again consider the example given above. For this example, if we are interested in finding those persons who are getting the salary within the common range then such persons will be those having salary between 10000 and 20000. The set of such persons is nothing but the intersection of two sets A and B.

Now, let us formally define the intersection of two sets.

Let A and B be two sets then intersection of A and B is denoted by $A \cap B$ and is defined as

$A \cap B = \{x: x \in A \text{ and } x \in B\}$.

i.e. $A \cap B$ contains common elements of A and B (see Fig. 1.7).

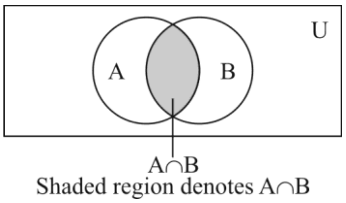


Fig. 1.7 Set Operation $A \cap B$

For example,

(i) If A = {a, b, c}, B = {b, c, d, e}, then $A \cap B$ = {b, c}.

(ii) If A = {5, 7, 9}, B = {10, 11, 18}, then $A \cap B = \{\ \} = \emptyset$ = empty set, as there is no common element in the two sets.

Note: If $A \cap B = \emptyset$ then we say that two sets A and B are **disjoint.**

Complement of a Set

Suppose we have set of persons of a locality having voting right. Then set of those persons of the locality who do not have voting right is its complement, if the set of all persons of that locality is considered as a universal set.

Now, let us formally define complement of a set.

Let U be the universal set, then complement of a set A (where $A \subseteq U$) is denoted by A^c or $\overline{A}$ or A' and is defined as :

$A' = \{x \in U: x \notin A\}$.

That is A' contains those elements of U which are not in A, i.e. A' contains all the elements of U other than A (see Fig. 1.8.).

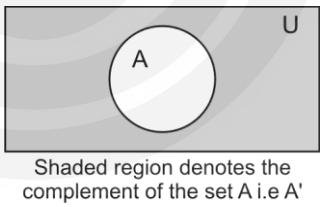


Fig. 1.8: A Complement

For example, if U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} and A = {1, 2, 4, 7, 9, 10}, then A' = {3, 5, 6, 8}, i.e. those elements of U which are not in A.

Difference of Two Sets

Let A and B be two sets then difference of A and B is denoted by A – B and is defined as

$A - B = \{x: x \in A \text{ but } x \notin B\}$. See Fig. 1.9

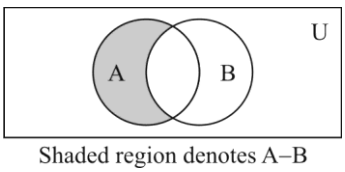


Fig. 1.9: A-B

For example,

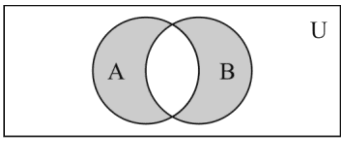
(i) If A = {1, 2, 3, 4, 5}, B = {4, 5, 6, 7}, then A – B = {1, 2, 3}

(ii) If A = {a, b, c}, B = {a, b, c, d, e}, then A – B = { }

Symmetric Difference of Two Sets

Let A and B be two sets, then symmetric difference of A and B is denoted by A⊿B and is defined as

A⊿B = (A– B)∪ (B– A). See Fig. 1.10



Shaded region denotes AΔB

Fig. 1.10: Symmetric Differnece to Two Sets

For example, if A = {1, 2, 3, 4}, B = {3, 4, 5, 6, 7}, then A – B = {1, 2}, B – A = {5, 6, 7}

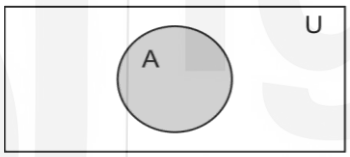
∴ A⊿B = (A – B)∪ (B – A) = {1, 2}∪{5, 6, 7} = {1, 2, 5, 6, 7}.

Example 8: With the help of the Venn diagrams, shade the following sets:

(i) A (ii) A' (iii) A∩B (iv) A – B (v) B – A (vi) A⊿B (vii) A⊆ B (viii) B⊂A

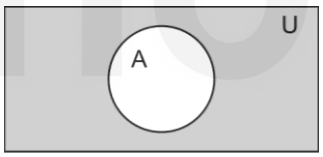
Solution:

(i)



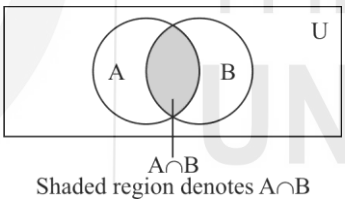
Shaded region denotes the set A

(ii)



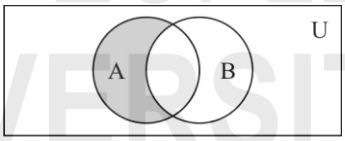
Shaded region denotes the complement of the set A i.e A'

(iii)



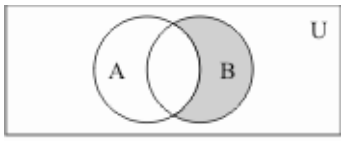
Shaded region denotes A∩B

(iv)



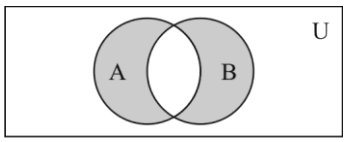
Shaded region denotes A–B

(v)



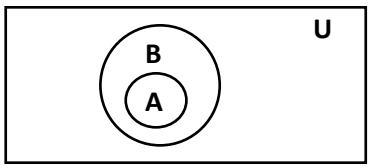
Shaded region denotes B–A

(vi)



Shaded region denotes AΔB

(vii)



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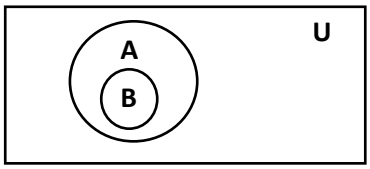

Next, let us we briefly discuss convex set and **Russell's Paradox** .

1.6.2 Convex Set

A convex set in a space (generally 2-dimensional real-space) is a **set** of points such that, given any two points A, B in the **set,** the line AB joining them lies entirely within that **set.**

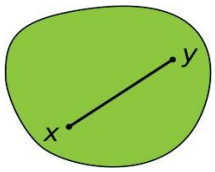

Fig. 1.11: Convex Set Space

Thus, a subset *S* of a space is called *convex* if and only if for any pair of points *p* and *q* in the set *S*, the line section from *p* to *q* is enclosed completely in the set of *S*, as done in the figure on L.H.S. below.

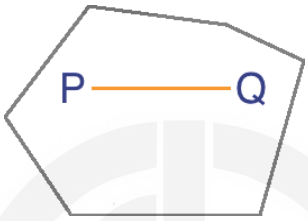

Fig. 1.12: Convex

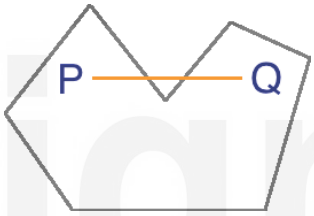

Fig. 1.13: Non-Convex

Algebraically, if x1 and x2 are two number in the set S, then all the points on the straight line x_1 x_2 – which can be represented as $\lambda x_1 + (1-\lambda)\, x_2$, for any λ such that $0 \leq \lambda \leq 1$- also belong to S.

1.6.3 Russell's Paradox

Russell's paradox is a counterexample/ contradiction to naive set theory. In this theory we try to understand working with sets, using natural language rather than formal logic and axioms, which defines a set as any definable collection.

You can construct the Russell's paradox using purely logical statements. This paradox points out the defect in the set theory itself. For example, using the naive set theory, we can construct the set:

$R = \{x | x \notin x\}$;

Here, R is the set of all things x in our mathematical domain (universe) which are not elements of itself. Now the paradox comes here when we try to answer the questions:

Is R itself an element of R? The answer, of question, is either "yes" or "no". We can examine each case separately.

1. If R is an element of R(i.e R. $\in$ R) , then R must satisfy the above membership criterion for R; thus, R $\notin$ R This is of course a contradiction, because by definition R cannot be an element of R.

2. Suppose now that R is not an element of R(i.e R $\notin$ R). Then, R satisfies the membership criterion for R, and hence R $\in$ R. This is again another contradiction, and this implies that R must be an element of R.

   The above two arguments show that: R both can be an element of R and cannot not be an element of R. Which leads to a disastrous and epic contradiction, from which we cannot recover. The same you may infer from the example given below.

A small village has only one (male) barber**,** who shaves all those, and only those males, who do not shave themselves. Let X be the ***collection*** of all the villagers shaven by the barber. Then, it is not possible to tell whether barber belongs to X or not. This is so because, if the *barber as a villager* shaves himself, then he cannot be shaven by the barber. Similarly, if *he, as a villager* does not shave himself, then he must be shaven by the barber. Thus, it is not possible to tell whether the barber is in X or not. Hence, X ***is not a well-defined collection***, and thus, ***not*** a set, this is an example of Russell's paradox.

1.6.4 Intervals

Let R be the set of all real numbers. Then a set $I \subseteq R$ is said to be an interval if whenever $a, b \in I$ and $a < x < b$ then $x \in I$

For example, the set of all real numbers satisfying $2 \le x \le 3$ is an interval where x can take any real value between 2 and 3 including 2 and 3.

Open Interval

An open interval $I \subseteq$ R with end points a and b (a < b) is denoted by (a, b) and is defined by

(a, b) = $\{x \in R : a < x < b\}$

i.e. an open interval contains each value between the end points but does not include the end points.

For example, open interval (2, 5) contains each real number lying between 2 and 5 but does not contain 2 and 5.

Closed Interval

A closed interval $I \subseteq$ R with end points a and b (a < b) is denoted by [a, b] and is defined by

[a, b] = $\{x \in R : a \le x \le b\}$

i.e. a closed interval contains each value between and including extreme values.

For example, closed interval [2, 5] contains each real number lying between 2 and 5. It also contains its end points. i.e.

$x \in [2,5] \;\; \forall \;\; x, \;\; 2 < x < 5$ and

also $2 \in [2,5], \; 5 \in [2, 5]$

more on intervals you will learn in unit 3 of this course.

Now, you can try the following exercises.

Check Your Progress -3

1 **)** If A = {3}, B = {a, b, c}, then write $A \cup B$.

……

……

2) If A = {1, 2, 3}, B = {3, 4, 5, 6}, C = {3, 6, 7}, then write A∪B∪C.

……

……

3) If A = { }, B = {1, 4, 7}, then write A∩B.

……

……

4) If U = {x: x is an English alphabet} and
A = {x : x is an vowel of English alphabet}, then write A'.

……

……

5) If W = set of whole numbers and N = set of natural numbers then write
W – N.

……

……

6) With the help of the Venn diagram justify the following
(i) A⊆ $A \cup B$ (ii) B⊆ $A \cup B$ (iii) A∩B⊆ A (iv) A∩B⊆ B
(v) $(A \cap B)' = A' \cup B'$ (De-Morgan's law) (vi) A⊆ $B \Rightarrow B' \subseteq A'$

……

……

7) Tell which one of the following is a convex set, within set of real numbers.
(i) [4, 5[= {x: 4 ≤ x <5},
(ii) [1, 4] ∪ [7, 11[= {x: 1 ≤ x ≤ 4} ∪ {x: 7≤x<11}

……

……

8) What is the Cardinality of the Power set of the set {0, 1, 2}?

……

……

9) Let $U = \{u, v, w, x, y, z\}$,
i) Find the number of subsets of U.
ii) Find the number of proper non-empty subsets of U.

……

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1.7 ORDERED PAIRS AND CARTESIAN PRODUCTS

Sometimes we need to establish relations between two or more sets. For example, a software development company has a set of specialists in different technology domains, or a company gets some projects to develop. Here the company needs to establish a relation between professionals and the project in which they will participate. To solve this type of problem the following concepts are required.

1.7.1 Cartesian product

Very often we deal with several sets at a time, and we need to study their combined action. For instance, combinations of a set of teachers and a set of students. In such a situation we can take a product of these sets to handle them simultaneously. To understand this product let us first consider the following definitions.

Definition: An **ordered pair,** usually denoted by (x,y), is a pair of elements x and y of some sets. This is ordered in the sense that (x,y) ≠ (y,x) whenever x ≠ y, that is , the order of placing of the element in the pair matters.

Any two ordered pairs (x,y) and (a,b) are equal iff x = a and y = b.

For example if, A= {a,b,c} and B= {x,y,z}, then

A × B = {a,b,c} × {x,y,z} = {(a,x), (a,y),(a,z),(b,x),(b,y),(b,z),(c,x),(c,y),(c,z)}, and

B × A = (x,y,z} × {a,b,c} = {(x,a),(x,b),(x,c),(y,a),(y,b),(y,c),(z,a),(z,b),(z,c)}.

Now let us think about how B × A can be represented geometrically?

For instance,

What is the geometric view of {2} × R ? This is the line x = 2 given in Fig 1.14(a).

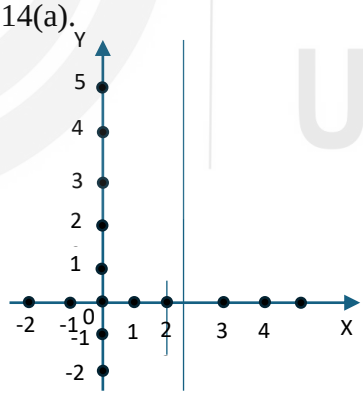

Fig. 1.14:(a) : {2} × R, ie. x=2

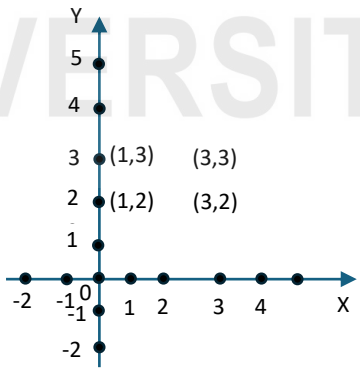

Fig. 1.14: (b): {1,3} × {2,3}

Now, after seeing geometric representation of {2} × **R**, can you tell what {1,3}×{2,3} = {(1,2),(3,2),(1,3),(3,3) } looks like? You will get four points in the first quadrant, as shown in Fig. 1.14(b).

Now, you know that the multiplication of numbers is commutative. Is the Cartesian product of sets also commutative?

For instance, is {1} × {2}= {2} × {1}? No, because (1,2) ≠ (2,1). So, **A×B≠ B×A usually**.

We can extend the definition of A × B to define the Cartesian product of n sets $A_1, A_2, ..., A_n$ as follows.

$A_1 \times A_2 \times A_3 \times ... \times, A_n = \{(x_1, x_2, x_3, ...x_n)\}$:
$x_1 \in A_1, \wedge x_2 \in A_2 \wedge x_3 \in A_3 \wedge ..., \wedge x_n \in A_n\}$.

The element $(x_1, x_2, ..., x_n)$ is called an **n-tuple**. For instance, the 3-tuple (1,1,3) $\in \{1\} \times \{1,2\} \times \{2,3\}$.

1.8 SOME USEFUL AND IMPORTANT LAWS

Some commonly used laws of sets are listed below.

1. Idempotent Laws

For any set A

(i) $A \cup A = A$ (ii) $A \cap A = A$

2. Identity Laws

For any subset A of the universal set U

$A \cup \emptyset = A$, $A \cap U = A$, where $\cup$ is empty set

3. Commutative Laws

For any two sets A and B

(i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$

4. Associative Laws

If A, B, C are any three sets then

(i) $(A \cup B) \cup C = A \cup (B \cup C)$

(ii) $(A \cap B) \cap C = A \cap (B \cap C)$

5. Distributive Laws

If A, B, C are any three sets then

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

6. Complement:

Union: $A \cup A' = U$

Intersection: $A \cap A' = \emptyset$

7. De-Morgan's Laws

For an y two sets A and B the following laws knows as De Morgan's laws., hold

$(i)(A \cup B)' = A' \cap B'$

$(ii)(A \cap B)' = A' \cup B'$

De Morgan's laws can also be expressed as

i) $A \sim (B \cup C) = (A \sim B) \cap (A \sim C)$

ii) $A \sim (B \cap C) = (A \sim B) \cup (A \sim C)$

Example 9: If A = {1, 3, 5}, B = {3, 5, 7, 9}, C = {2, 6, 8, 9} are subsets of the universal set U = {1, 2, 3, 5, 6, 7, 8, 9} then verify

(i) De-Morgan's laws and

(ii) Distributive laws

Solution:

(i) De-Morgan's laws state that

$(a)(A \cup B)' = A' \cap B'$

$(b)(A \cap B)' = A' \cup B'$

To verify (a)

$A \cup B = \{1, 3, 5, 7, 9\}$

$\therefore (A \cup B)' = \{2, 6, 8\}$

$A' = \{2, 6, 7, 8, 9\}$, $B' = \{1, 2, 6, 8\}$

$\therefore A' \cap B' = \{2, 6, 8\}$

We see that here $(A \cup B)' = A' \cap B'$.

Hence verified.

To verify (b)

$A \cap B = \{3, 5\}$

$\therefore (A \cap B)' = \{1, 2, 6, 7, 8, 9\}$ and

$A' \cup B' = \{1, 2, 6, 7, 8, 9\}$

We see that here $(A \cap B)' = A' \cup B'$.

Hence verified.

(ii) Distributive laws state that

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

To verify (a)

$B \cap C = \{9\}$

$\therefore A \cup (B \cap C) = \{1, 3, 5, 9\}$

$A \cup B = \{1, 3, 5, 7, 9\}$, $A \cup C = \{1, 2, 3, 5, 6, 8, 9\}$

$\therefore (A \cup B) \cap (A \cup C) = \{1, 3, 5, 9\}$

We see that here $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Hence verified.

To verify (b)

$B \cup C = \{2, 3, 5, 6, 7, 8, 9\}$

$\therefore A \cap (B \cup C) = \{3, 5\}$

$A \cap B = \{3, 5\}$, $A \cap C = \{\}$

$\therefore (A \cap B) \cup (A \cap C) = \{3, 5\}$

We see that here $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Hence verified.

Duality Principle: The 'duality principle' is very convenient for dealing with theorems about sets. Basically if any theorem is given to you, by applying the duality principle you can get another theorem (the dual of the previous one). In any statement involving the union and intersection of sets, we can get from one of the statements to the other by interchanging $\cap$ with $\cup$ and $\emptyset$ with U.

For example, the dual of $A \cup (B \cap C)$ is $A \cap ((B \cup C)$ and the dual of $U \cup \emptyset = U$ is $U \cap \emptyset = \emptyset$. So, for example what is true for $A \cup (B \cap C)$ will be true for $A \cap (B \cup C)$ too. This is why if the first property in each of the pairs below is proved the second one follows immediately.
For any universal set U and subsets A, B and C of U, the following properties hold.

1.9 APPLICATION OF SETS

Venn diagrams are helpful in establishing many important relations between different sets, some of them are mentioned as under which are helpful in solving many practical problems too.

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

2. $n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$

3. $n(A - B) = n(A) - n(A \cap B)$

4. $n(B - A) = n(B) - n(A \cap B)$

5. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Let us consider an example based on above formulae.

Example 10: In a group of 500 persons, 400 can speak Hindi and 150 can speak English. Then how many can speak

(i) both Hindi and English

(ii) Hindi only

(iii) English only

Solution: Let A and B denote the set of persons who can speak Hindi and English, respectively. Then in usual notations, we are given
$n(A \cup B) = 500$, $n(A) = 400$, $n(B) = 150$

(i) We know that
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow 500 = 400 + 150 - n(A \cap B \Rightarrow n(A \cap B) = 550 - 500 = 50$
$\therefore$ number of persons who can speak both Hindi and English = 50

(ii) We know that
$n(A - B) = n(A) - n(A \cap B) = 400 - 50 = 350$
$\therefore$ number of persons who can speak Hindi only = 350

(iii) We know that
$n(B - A) = n(B) - n(A \cap B) = 150 - 50 = 100$
$\therefore$ number of persons who can speak English only = 100

Check Your Progress -4

1) If X = {a,b,c} and Y = {1,2,3}, find
i) $X \times X$, ii) $X \times Y$, and iii) $X \times \emptyset$.

……

……

2) Under what conditions on A and B is A × B = B × A ?

……

……

3) If the universal set U = {1, 2, 3, 4, 5, 6, 7, 8} and A = {1, 3, 5, 6, 7}, B = {5, 6, 7, 8}, C = {1, 5, 7, 8} are subsets of U, then verify (i) commutative laws and (ii) associative laws

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4) Out of the 50 students in a class, 24 play cricket, 15 play hockey, 18 play football, 6 play cricket and hockey, 8 play cricket and football, 5 play hockey and football and 10 students do not play any of the three games. Then how many play (i) all the three games, (ii) hockey but not football and (iii) cricket and football but not hockey.

……

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1.10 SUMMARY

This Unit has provided a foundational overview of **Set Theory**, beginning with methods for representing collections through roster and set-builder notation, then moving to the structural hierarchy of subsets, power sets, and cardinality. Subsequently, visual logic and abstract operations, using Venn Diagrams to demonstrate set interactions such as unions and intersections, are discussed Also advanced logical concepts such as **Russell's Paradox** and the geometry of **Convex Sets** has been introduced in this unit. The unit concluded with details on Cartesian Products and the application of sets to solve complex real-world problems.

1.11 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress -1

1) (i) It is not a set, because a student may be intelligent according to someone while the same student may not be intelligent according to some other person.

(ii) It is not a set because a hockey player may be good in someone's view while the same player may not be good in view of some other person.

(iii) It is not a set because an actor may be good in someone's point of view while the same actor may not be good in the view of some other person.

(iv) Set of vovels {A,I}

2) (i) x = 2n + 3, n ∈ W = {0, 1, 2, 3, 4, …}

Values are obtained on putting n = 0,1,2,…. In x = 2n+3 e.g. when we put n =0 we get x = 2(0)+3 =3.

∴ x = 3, 5, 7, 9, 11, …

And, therefore, A = {3, 5, 7, 9, 11,…}

(ii) B = $\{7^0, 7^1, 7^2, 7^3\}$ = {1, 7, 49, 343}

(iii) N = {1, 2, 3, 4,…} and W = {0, 1, 2, 3, 4,…}

∴ if x ∈ $N and x \in W$ then the elements which satisfy both are 1, 2, 3, 4, … and hence

C = {1, 2, 3, 4, 5, …}.

(iv) W = {0, 1, 2, 3, …}and

$\mathbb{Q}$ = set of rational numbers,

∴the elements which are common to both are 0, 1, 2, 3, … as no other rational number is a whole number and hence

D = {0, 1, 2, 3,…}.

3) (i) {x∈ N: x = 5n, n∈ N }

(ii) {x : x =$\frac{1}{n}$, n∈ N }

(iii) {x∈ N: x = 2n, n∈ N }

4) (i) X =Y

(ii) X ≠Y as 15 ∈ Y and 15 not ∉ X

(iii) X =Y

5) i) False
ii) False
iii) False
(iv) True
(v) True
(vi) False
(vii) True

6) Set of vowels of English alphabet V={a,e,i,o,u}.
Two subsets of set V are V1={a,e}, and V2={i,o}.
Two supersets of V are V3 = {a,b,c,...,z}, and V4={a,b,c,d,e,i,o,u,...,z}.

7) Powerset of A={a,e,i,o,u} is :
∅,{a},{e},{i},{o},{u},{a,e},{a,i},{a,o},{a,u},{e,i},{e,o},{e,u},{i,o},{i,u},{o,u},{a,e,i}, {a,e,o}, {a,e,u}, {a,i,u}, {a,i,o}, {a,o,u}, {e,i,o}, {e,i,u}, {i,o,u}, {e,o,u}, {a,e,i,u},{a,e,i,o}, {e,i,o,u}, {a,e,o,u}, {a,i,o,u}, {a,e,i,o,u}.

8) For empty set A= { } or ∅, P(A)=1

9) If A ⊂ B, then P(A) ⊂ P(B) because every subset of A is a subset of B.

10) If P(A) = P(B) then A ∈ P(A)= P(B)=A ⊂ B. Similarly, B ⊂ A. Therefore A = B.

Check Your Progress -2

1) (i) A = {2, 9, 7, 5}, B = {5, 2, 9, 7} as repetitions in a set are not allowed.

We see that all the elements of A are in B and all the elements of B are in A.

∴A = B. Hence the statement is true.

(ii) Here, n(A) = 3, n(B) = 3 and so the statement A ~ B is true.

(iii) A = {4, – 4, 5}, B = {x: x^2 = 16, x^2 + x – 20 = 0, x∈ Z}

= {x: x = $\pm 4, (x+5)(x-4) = 0, \rightleftarrows x \in Z$}

= {x: x =± 4, x = – 5, 4, x ∈ Z}= {4}

Here, – 4∈A but – 4∉B

∴A ≠ B, hence the statement is false.

(iv) Here, A = {a, b, c}, B = {d, e, f, g, h}. Thus, n(A) = 3, n(B) = 5.

∴A ~ B is false because n(A) ≠ n(B).

2) P(A) ={φ, {a},{b}, {c},{d}, {a, b}, {a, c}, {a, d}, {b, c},{b, d}, {c, d},{a, b, c}, {a, b, d}, {a, c, d}, {b, c, d}, {a, b, c, d}}

Notice that it has $2^4 = 16$ elements.

3) Smallest universal set is

{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 15, 17, 19, 20, 23, 25}.

Check Your Progress -3

1) A∪B = {3, a, b, c}.

2) A∪B∪ C = {1, 2, 3, 4, 5, 6, 7}.

3) A ∩ B = { } = φ.

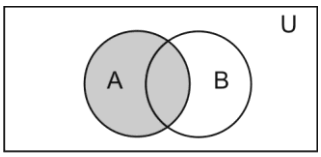
4) A' = {x: x is a consonant of English alphabet}.

5) We know that

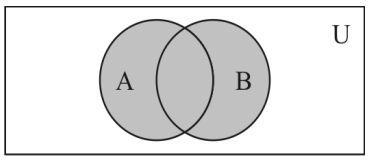
W = {0, 1, 2, 3, 4, …} and N = {1, 2, 3, 4, …}

∴W – N = {0, 1, 2, 3, 4,…} – {1, 2, 3, 4, …}= {0}

6) (i)



Shaded region denotes A

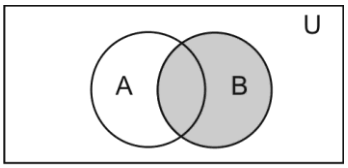


Shaded region denotes A∪B

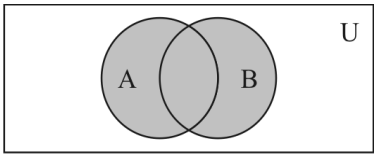
L.H.S. R.H.S.

Two Venn diagrams justify the relationship A⊆ $A \cup B$.

(ii)



Shaded region denotes B

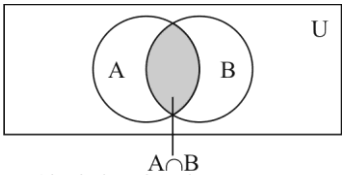


Shaded region denotes A∪B

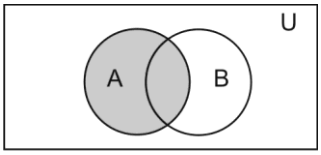
L.H.S. R.H.S.

Two Venn diagrams justify the relationship B⊆ $A \cup B$.

(iii)



Shaded region denotes A∩B

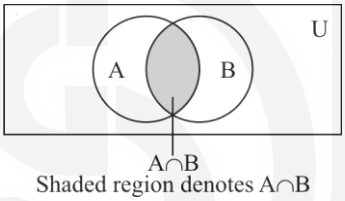


Shaded region denotes A

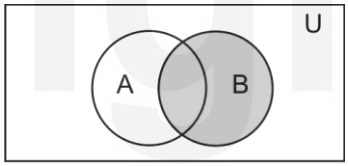
L.H.S. R.H.S.

Two Venn diagrams justify the relationship A∩B⊆ A.

(iv)



Shaded region denotes A∩B

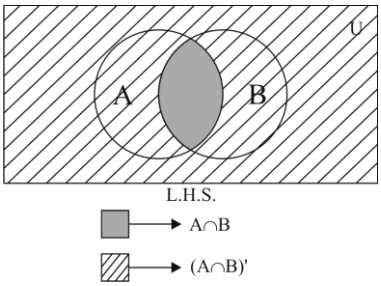


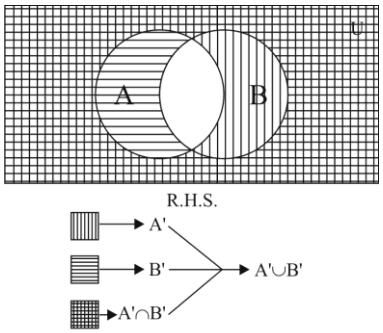
Shaded region denotes B

L.H.S. R.H.S.

Two Venn diagrams justify the relationship A∩B⊆ B.

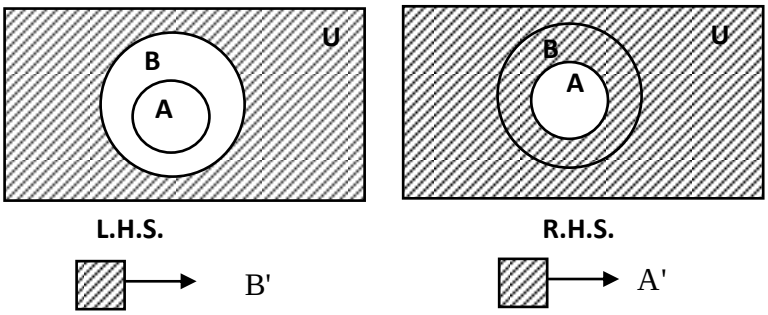
(v)





Two Venn diagrams justify the relationship $(A \cap B)' = A' \cup B'$.

(vi)



Two Venn diagrams justify the relationship $A \subseteq B \Rightarrow B' \subseteq A'$.

7)

i) it is convex
ii) it is not convex

8) $2^3 = 8$

9)

i) There are $2^6 = 64$ subsets
ii) The answer is 62: there are $2^6 = 64$ subsets in U altogether . W exclude two : U itself (because it is not proper subset) and the empty set (because it is not no-empty).

Check Your Progress -4

1) i) X x X = {(a,a),(a,b),(a,c),(b,b),(b,c),(c,c)}.

ii) X x Y = {(a,1), (b,1), (c,1), (a,2), (b,2), (c,2), (a,3), (b,3), (c,3)}.

ii) X xX $\emptyset = \emptyset$.

2) $A \times B = B \times A$ iff A = B or one of A,B is $\emptyset$.

3) (i) Commutative laws state that

(a) $A \cup B = B \cup A$

(b) $A \cap B = B \cap A$

To verify (a)

$A \cup B = \{1, 3, 5, 6, 7\} \cup \{5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$
$\therefore B \cup A = \{5, 6, 7, 8\} \cup \{1, 3, 5, 6, 7\} = \{1, 3, 5, 6, 7, 8\}$
$\therefore$we see that here $A \cup B = B \cup A$.
Hence verified.

To verify (b)

$A \cap B = \{1, 3, 5, 6, 7\} \cap \{5, 6, 7, 8\} = \{5, 6, 7\}$
and $B \cap A = \{5, 6, 7, 8\} \cap \{1, 3, 5, 6, 7\} = \{5, 6, 7\}$
We see that here $A \cap B = B \cap A$
Hence verified.

(ii) Associative laws state that

(a) $(A \cup B) \cup C = A \cup (B \cup C)$

(b) $(A \cap B) \cap C = A \cap (B \cap C)$

To verify (a)

$A \cup B = \{1, 3, 5, 6, 7, 8\}$

$\therefore (A \cup B) \cup C = \{1, 3, 5, 6, 7, 8\}$

$B \cup C = \{5, 6, 7, 8\} \cup \{1, 5, 7, 8\} = \{1, 5, 6, 7, 8\}$

$\therefore A \cup (B \cup C) = \{1, 3, 5, 6, 7\} \cup \{1, 5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$

We see that here $(A \cup B) \cup C = A \cup (B \cup C)$.

Hence verified.

To verify (b)

$A \cap B = \{5, 6, 7\}$

$\therefore (A \cap B) \cap C = \{5, 6, 7\} \cap \{1, 5, 7. 8\} = \{5, 7\}$

$B \cap C = \{5, 6, 7, 8\} \cap \{1, 5, 7, 8\} = \{5, 7, 8\}$

$\therefore A \cap (B \cap C) = \{1, 3, 5, 6, 7\} \cap \{5, 7, 8\} = \{5, 7\}$

We see that here $(A \cap B) \cap C = A \cap (B \cap C)$.

Hence verified.

4) Let C, H, F, denote the set of students who play cricket, hockey and football respectively. Then in usual notations, we are given.

$n(C) = 24$, $n(H) = 15$, $n(F) = 18$, $n(C \cap H) = 6$, $n(C \cap F) = 8$,

$n(H \cap F) = 5$, $n(C' \cap H' \cap F') = 10$

(i) Before finding the required number of students, we are to first obtain the number of students who play at least one of the three games which is given as

$= n(C \cup H \cup F) = 50 - n(C' \cap H' \cap F') = 50 - 10 = 40$

Now, we know that

$n(C \cup H \cup F) = n(C) + n(H) + n(F) - n(C \cap H) - n(C \cap F) - n(H \cap F) + n(C \cap H \cap F)$

$\Rightarrow 40 = 24 + 15 + 18 - 6 - 8 - 5 + n(C \cap H \cap F)$

$\Rightarrow 40 = 57 - 19 + n(C \cap H \cap F) = 38 + n(C \cap H \cap F)$

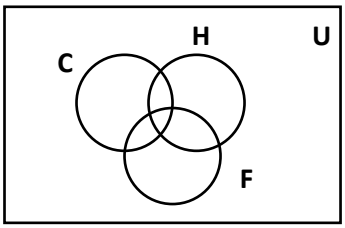
$\Rightarrow n(C \cap H \cap F) = 40 - 38 = 2$

$\therefore$ number of students who play all the three games = 2

(ii) We know that

$n(H - F) = n(H) - n(H \cap F)$

$= 15 - 5 = 10s$



$\therefore$ number of students who play hockey but not football = 10

(iii) n((C$\cap F) - H) = n(C \cap F) - n((C \cap F) \cap H)$ = 8 – 2 = 6

$$\left[\begin{array}{l} \because n(A-B) = n(A) - n(A \cap B) \\ HereA = C \cap F, B = H \end{array}\right]$$

∴ number of students who play cricket and football but not hockey = 6

1.12 REFFERENCES / FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MCS-013: Discrete Mathematics
   Unit-1 Sets, Relations and Functions
   (https://egyankosh.ac.in/handle/123456789/1140)
5. **MEC-203 : Quantitative Methods**
   Unit-1 Fundamental Concepts of Mathematics
   (https://egyankosh.ac.in/handle/123456789/98141)
6. **MST-001: Foundation in Mathematics and Statistics**
   Unit-1 Introduction to Sets
   (https://egyankosh.ac.in/handle/123456789/92122)

UNIT 2 RELATIONS

Structure

2.0 Introduction
2.1 Objectives
2.2 Relation
  2.2.1 Definition, Notation and Illustrations
2.3 Properties of Relations
2.4 Equivalence relations, and Partition of a Set
2.5 Partial Order Relation, Partially Ordered Set
2.5.1 Applications of Equivalence Relation and Partition of a Set
2.5.2 Partial Order Relation, Partially Ordered Sets
2.6Operations *on* Relations
  2.6.1 Set-type Operations on Relations
  2.6.2 Mapping-type operations: inverse of operation, composition
  2.6.3 Closure Operations
2.7 Summery
2.8 Answer to Check Your Progress
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2.0 INTRODUCTION

In Unit 1, we have already talked of set-relations (i. e., relations on sets)—like(subset), and (proper subset). Also, we encounter several situations in day-to- day life to judge number-relations—like < (less than), > (greater than), and = (equal to)—between pair of numbers. In everyday life, we talk of relations like of 'father to daughter', 'mother to son', 'sister to brother', 'teacher to student', and 'employer to employee' to name a few.

In most of such cases, there is a sense of direction/order from the first to the second in the sense that if X is mother of Y, then Y is not mother of X, i.e., X and Y cannot be exchanged in the relation of is-mother-of. Similarly, for the relation of is-less-than (<), we have 4< 5, and we cannot exchange roles of 4 and 5 for the relation to be true. Except in special cases of relations like equal-to (=) in numbers, or the relation of sibling (persons, male or female, having same pair of parents) among human beings, the order of arguments involved in a relation, in general, has to be taken into consideration.

Also, a relation need not be between two similar sets, as discussed above: between two sets of sets, between two sets of numbers, or between two sets of human beings. It may be between two different types of sets. For example, the relation of age-of-person is from a set of human beings to a set of numbers. This leads us to the concepts 'ordered set' and 'cross-product of two sets', which are discussed next.

a) Ordered Set: Earlier, in Unit 1, it was pointed out that in a set, order of occurrence of elements is irrelevant:

For example, {1, 2, 3, 4} = {4, 2, 1, 3} = {3, 2, 4,1} = ….

However, there are situations, as mentioned just above, in which the change of order of occurrence of the elements, changes the situation. For example, in coordinate

geometry (Fig. 3.1), the location of the point (2, 5) is different from the location of the point (5,2).

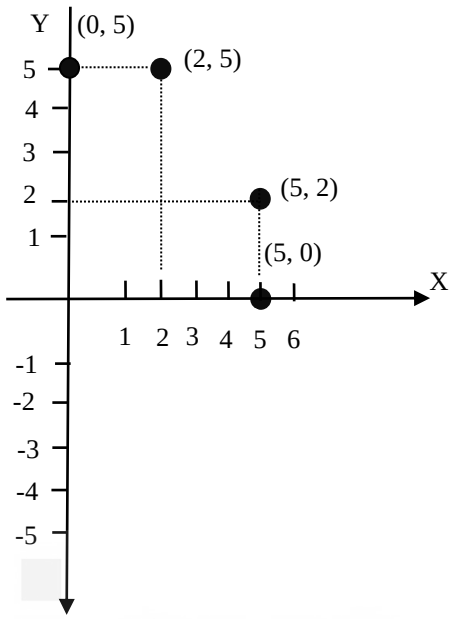


Fig. 2.1: Location of Points in Co-ordinate Geometry

Such a set, in which order of occurrence of elements is significant, is called an **ordered set**. In the notation of ordered sets, the elements are enclosed between parentheses, instead of braces/**curly-brackets.** Thus

$(a, b) \neq (b, a)$, but $\{a, b\} = \{b, a\}$.

b) **Cross-product of two sets X and Y :** for $X \neq \emptyset \neq Y$, denoted by $X \times Y$ is the set of all ordered pairs (x, y) with x belonging to X and y belonging to Y, i. e.

$X \times Y = \{(x, y): x \in X, y \in Y\}$.

If $x = \emptyset$ or $y = \emptyset$ then $X \times Y = \emptyset$

In this unit you learn more on relations.

2.1 OBJECTIVES

After going through this unit, you should be able to:

- define relation,
- explain concepts of relation,
- use the notation for a relation,
- explain different properties like reflexivity, symmetry and transitivity of relations,
- apply the concepts of equivalence relation, equivalence class, and partition of a set,
- identify partial order relation, and partially ordered set,
- draw Hasse diagram for a partially ordered set, and
- apply operations, like composition, inverse and union, to relations.

2.2 RELATION

The concept of 'relation' is one of the few fundamental concepts of mathematics. We begin the discussion with its formal/ mathematical definition and illustrations through some well-known examples.

2.2.1 Definition, Notation and Illustrations

In Mathematics, a relation from a set X to a set Y is defined as a ***subset*** of the cross-product X xY. Earlier mentioned relations like is-mother-of or is-less-than can be discussed as subsets of cross-products of some sets as follows.

Notation (in Mathematics)

If Sheela is the mother of Shivam, then the fact may be expressed as Sheela ***is-mother-of*** *Shivam.*

In mathematics, it is preferred to denote the fact as

(Sheela, Shivam) ∈ ***is-mother-of,***

where the relation "***is mother of"*** is considered as ***the set*** of all the pairs (m, c), for which m "is mother" of c.

In one of the methods of mathematical shorthand, **is-mother-of** is denoted in set notation as:

is-mother-of= {(m, c): m ∈ X, c ∈ Y, and m is mother of c} ⊆ X × Y, ...(i)

where X= Y = set of human beings.

(The above form (i) is called the **set-builder form**, in which the name of the property viz. **is-mother-of** is used)

Then the fact Savitri is not the mother of Mohan, is denoted as

(Savitri, Mohan) ∉ is-mother-of.

Similarly, the relation 'is-less-than' (<) on N, may be considered as a subset of a cross-product by taking X= Y = N = {1, 2, 3,...}—as follows

is-less-than = {(1, 2), (1, 3), (1, 4), ..., (2, 3), (2, 4), (2,5),..., (3, 4), (3, 5), (3, 6), ...}, or more systematically as is-less-than = {(1, 2), (1, 3), (1, 4), (2, 3), (1, 5), (2,4), (1, 6), (2, 5), (3, 4), (1, 7), (2, 6), (3, 5), (1, 8), (2, 7), (3, 6), (4, 5),

(1, 9), (2, 8) ...}⊆ N × N …..... (ii)

Thus, (1, 4)∈ is-less-than, and (4,1)∉ is-less-than.

(The above form (ii) is called **roster form**, in which each pair for which propertyholds is mentioned explicitly instead of the name of the property is-less-than)

The set-builder form of "is-less-than" is given by is-less-than = {(x,y): x,y∈ N, and x < y}

Arguments of a relation: the values supplied to a relation are called arguments of the relation. The relations discussed just above involve exactly two arguments: two human beings, two numbers, or two sets. These relations are known as **binary relations**. However, some relations may involve just one argument, e. g., is-male or is-female: for example,

is-male (Mohan) is true, and **is-female (Mohan)** is false.

A **unary relation** is also called a property of its argument; e.g., is-male is a property of Mohan, is-female is not a property of Mohan; similarly, is-prime-number is a property of 19, is-composite-number is not a property of 19.

Some relations may involve more than two arguments; e. g. the relation is- integer-strictly-between involve three integers, such that 4 is-integer-strictly- between 7 and 2 is True. But 4 is an integer strictly between 7 and 9 is False. Also, 4 is an integer strictly between 1 and 3 is also False. We may express the relation in set-builder form as is-integer-strictly-between = {(x, y, z): either y< x <z, ory> x > z}. This relation involves three arguments, and such relations are called **ternary relations**.

Similarly, the relation **are-parents-of** involves three arguments, and may be defined as **are-parents-of** = {(x, y, z): x, y, z are human beings; and x & y are parents of z}. Similarly, in economics, the (item, cost-production, sales-price) is a ternary relation.

Similarly, we may have relations involving four, five or more arguments. For example, the relation of the timetable of a high school class may involve four arguments as follows.

Time-table = {(class, subject, subject-teacher, room-number): class ∈ {VI, VII, VIII, IX, X},

subject ∈ = {Science, Maths., Hindi, English,...},

subject-teacher ∈ {Ms. Geeta, Mr. John, Mr. Abdul, ...},

room-number = {1, 2, 3, ...}}.

This is an example of **quaternary** relation. In general, a relation R may involve n sets, say X_1, X_2, ... X_n, then $R \subseteq X_1 \times X_2 \times ... \times X_n$, and R is called **n-ary relation**.

However, most of the time, we will be discussing only relations, like is-mother-of or is-less-than, each of which requires exactly two arguments. A relation which involves two arguments is called a **binary relation**.

Unless we explicitly state otherwise, the term 'relation' will stand for '**binary relation**'.

Definition: A relation between two sets A and B is a subset of A × B. Any subset of A × A is **a relation on the set A.**

For instance, if A = {1,2,3} and B = {p,q}, then the subset {(1,p),(2,q),(2,p)} is a relation on A × B. And {(1,1), (2,3)} is a relation on A.

Also, R= {(x,y) ∈ N × N : x > y} is a relation on N, the set of natural numbers, since R ⊆ N × N.

If R ⊆ A × B,. We write xRy if and only if (x,y) ∈ R(xRy is read as 'x is related to y').

Theorem 2: The total number of distinct relations between a finite set A and a finite set B is 2^{mn}, where m and n are the number of elements in A and B, respectively.

For example, R_1 = N × L, where L is the set of straight lines; in this relation, we can give different orderings of the straight lines.

If the relation R_2 = {1,2,3} × {$1_1 1_2$}, then line 1_1 and 1_2 can get three different ordering.

'**Proof:** The number of elements of A × B is mn. Therefore, the number of elements of the power set of A × B is 2^{mn}. Thus, A × B has 2^{mn} different subsets. Now every subset of A × B is a relation from A to B, by definition. Hence the number of different relations from A to B is 2^{mn}.

Some more definitions

- **Domain, Codomain & Range of a relation:** For a binary relation R from X to Y, i. e. R ⊆ X × Y, the set X is called the domain of R and the set Y is called the codomain of R. For example, consider the relation R= (Person, Number), number denotes a 10-digit mobile number. Then the domain of R is the set of all persons, and the codomain of the relation R is the set of all 10-digit natural numbers. The set of all 10-digit numbers, which are used as 10-digit mobile numbers of some persons, constitutes the range of R.
- **Image of an element of the domain:** For the relation R ⊆ X × Y, for x ∈ X (domain), and y ∈ Y (codomain), if (x, y) ∈ R, then an image of x under relation R is y. For example, If Sheela is mother of Shivam, then under the relation is-mother-of, one of the images of Sheela is Shivam.
- **Relation on a set X:** For a binary relation R, if codomain = domain (i. e., R is from X to X), then R is said to be a relation on X.

- **Onto relation:** For a binary relation R from X to Y, we know R ⊆ X x Y. It is not necessary that every element of Y should occur as second component of some ordered pair in R. Consider the relation of (Person, Number), number stands for 10-digit mobile number. Then, it is not necessary that every 10-digit number is a mobile number of someone. Thus, (Person, Number) is not an onto relation. But the relation "is-mother-of" is onto, because, if P is a person, then P must have a mother, say M, Thus, P occurs as second component of (M, P).
- **One-to-one relation:** A binary relation R from X to Y is said to be one-to-one if any two distinct elements of the domain cannot have the same image. One of the examples for one-to-one relation is (Person, Mobile- Number): no

two different persons can have the same mobile number.

- **Empty Relation:** A relation R from a set X to a set Y is called an empty relation if no element of X is related under R to any element of Y, i. e. R = $\emptyset \subset X \times Y$. For example, X = set of decimal digits = {0, 1, 2, 3, ..., 9}, and the relation R is defined on X as R = {(a, b): a - b = 10}. This is an empty relation, as no pair (a, b) of decimal digits satisfies the condition a - b = 10.
- **Universal Relation:** A relation R from a set X to a set Y is called a universal relation if each element of X is related to every element of Y, i.e., R = X × Y.
- The empty relation and the universal relation are sometimes called trivial relations.
- **Notations re-visited:** In general, for R, a relation on a set X, the fact that x is related to y under R may be denoted by either of the notations (i) $(x, y) \in R$, or (ii) x R y. Apart from the above-mentioned set notations for relations, two more forms are popular, (i) Graphs and (ii) Matrices. Fig.3.2 explains how to obtain graphic representation of a relation, say,
- R= {(1, a), (1, b), (2, a), (3, a), (4, b), (4, c), (4, d)} from the set X= {1, 2, 3, 4} to the set Y = {a, b, c, d}

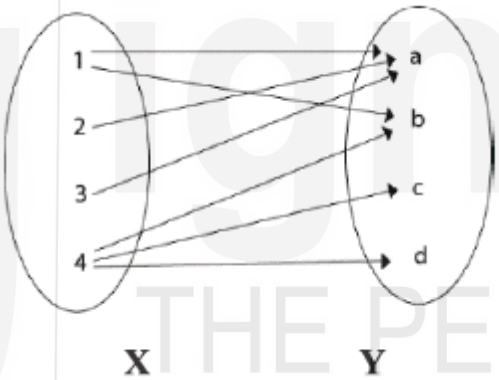


Fig. 2.2: Graphic Representation of a Relation

Fig. 2.3 given below explains how to obtain a matrix representation of a relation, say,

R= {(1, a), (1, b), (1, c), (2, b), (2, c), (2, d)} from the set X= {1, 2, 3, 4} to the set Y={a, b, c, d}

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{Bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix} \end{array}$$

Fig. 2.3: Matrix Representation of a Relation

For representing a relation R on a set X, i. e. when domain of R = codomain of R, a directed graph may also be used. For example, the relation R = {(1, 1), (1, 2), (2, 2), (3, 1), (3, 3)} on the set X= {1, 2, 3}, may be represented by the following directed graph (Fig. 3.4).

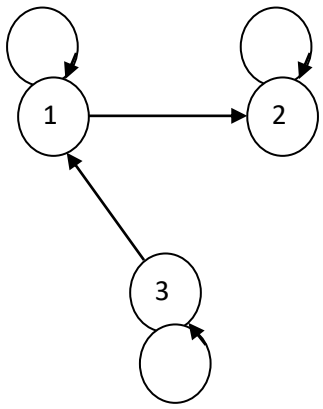


Fig. 2.4: Directed Graph Representing a Relation R on a Set X

In the next subsection, we discuss relations on (set of) relations (rather, we will discuss mainly properties (i.e. binary relations) of relations. In the subsequent subsection, we discuss operations on (set of) relations.

2.3 PROPERTIES OF RELATIONS

We will discuss properties of only for those relations R for which the domain and range is the same set, say X, i.e., the relations on a set, say X. Unless mentioned otherwise, all relations are on the set X (fixed for all the relations under consideration). The discussion revolves around three important properties of relations on a set, viz. (i) reflexive, (ii) symmetric, and (iii) transitive. Other relations to be discussed are directly or indirectly modifications of these properties.

1) Reflexive, Not-Reflexive, Anti-Reflexive/Irreflexive

a) A relation R on set X is said to be reflexive, if for every $a \in X$, then $(a, a) \in R$. For example, each of the following is a reflexive relation

i) the relation of equality (=) on N = {1, 2, 3,...}

ii) The universal relation on X, is reflexive.

b) by definition under (a) above, R is not reflexive on X, if for at least one $a \in X$, we have $(a, a) \notin R$. For example,

i) Let X= N = {1, 2, 3, ...}. Define $(a, b) \in R$, if and only if the product $(a \times b)$ is an even number. Then $1 \times 1 = 1$ is not even, therefore $(1, 1) \notin R$. Therefore, R is not reflexive on N. Note that there are other elements like 3, for which $(3, 3) \notin R$, but we need not discuss other cases, as just one example of $(1, 1) \notin R$ is sufficient to establish R as not reflexive.

ii) The empty relation on a set $X \neq \emptyset$ is not reflexive, because for $a \in X$, then $(a, a) \notin \emptyset$ (because $\emptyset$ has no element).

iv) We may construct our own simple counter-examples, e. g. Let $X = \{1, 2, 3\}$, and $R= \{(1, 1), (1, 2), (1,3), (3,3)\}$. Then as $(2, 2) \notin R$., R is not reflexive.

c) **Anti-reflexive/irreflexive:** A relation R on set X is said to be anti-reflexive (or irreflexive), if for every $a \in X$, we have $(a, a) \notin R$. For example, the relation

i) is-less-than (<) on numbers is anti-reflexive, as $x < x$ is false for every number x, i. e. $(x, x) \notin$ is-less-than

ii) is-mother-of on the set of human beings is anti-reflexive, for every human x, $(x, x) \notin$ is-mother-of

iii) $R= \{(2, 3), (3,1), (2, 1)\}$ on the set $\{1, 2, 3\}$ is anti-reflexive.

Remarks 1:

a) Difference between not-reflexive and anti-reflexive

For a relation R to be not-reflexive, we require for some $a \in X$, (a, a) fails to be in R. For a relation R to be anti-reflexive, we require for every $a \in X$, (a, a) fails to be in R.

b) Every anti-reflexive relation is a not-reflexive relation, but a not-reflexive relation may not be anti-reflexive.

If R is anti-reflexive on set X, then, by definition, for every $a \in X$, $(a, a) \notin R$ implies for at least one a, $(a, a) \notin R$ implies R is not reflexive.

Some examples:

i) $R= \{(2, 2), (3, 3), (2, 1)\}$ on the set $\{1, 2, 3\}$ is a not-reflexive relation, as $(1, 1) \notin R$. But it is not anti-reflexive, as $(2, 2) \in R$.

ii) The relation is-brother-of on the set of all human beings is not reflexive, because, if x is a female human being, then $(x, x) \notin$ is-brother-of. But the relation is not anti-reflexive; also, as for each male x, we have $(x, x) \in$ is- brother-of, contradicting the requirement of anti-reflexivity.

c) Just by changing the underlying set of a relation, its property may change.

For example,

i) For the relation "is-brother-of ", if

a) Underlying set X= set of all males, then the relation is reflexive

b) Underlying set X= set of all females, then the relation is anti-reflexive

ii) For the relation R on integers, such that $(x, y) \in R$ if and only the product $x \times y$ is even, then if

a) Underlying set X= set of **all even integers**, then the relation is reflexive

b) Underlying set X= set of **all integers**, then the relation is not reflexive, and also not anti-reflexive

c) Underlying set X= set of **all odd integers**, then the relation is anti- reflexive

2) Symmetric, Not-Symmetric, Anti-Symmetric, Asymmetric

a) A relation R on a set X is **symmetric** for a, b ∈ X, if (a, b) ∈R, then ((b, a) ∈R.

Note: ∅ relation is a symmetric relation.

For example, each of the following is a symmetric relation:

i) the relation of equality (=) on N= {1, 2, 3, ...}.

ii) The relation of is-congruent-to on the set X of all triangles

iii) the relation of is-sibling-of on the set of all human beings.

But the relation less-than-or-equal-to (≤) is NOT symmetric, though ≤ is reflexive on numbers.

b) By definition (a) above, a relation R on a set X is **not symmetric** if there is some pair a, b ∈ X, with (a, b) ∈R, but (b, a) ∉R. For showing not-symmetric, we need to show just one instance violating the condition of symmetry.

Some examples are:

i) The relation R of less-than-or-equal-to (≤) on integers is NOT symmetric, as (3, 4) ∈ R, but (4,3) ∉R

ii) Relation R of is-mother-of on the set of all human beings is not symmetric

iii) Relation R= {(1, 1), (2, 2), (3, 3), (1,2)} on the set {1, 2, 3} is NOT symmetric, as (1,2)∈ R, but (2, 1) ∉R. We may note that R is a reflexive relation

c) **Antisymmetric:** a relation R on a set X is anti-symmetric, if we have (a, b) ∈R, and (b, a) ∈ R—for two elements a, b ∈ X-then we must have a = b.

Examples:

i) The relation of "is less than or equal to" on integers is antisymmetric

ii) On the set X = {1, 2, 3}:

a) Each of the relations {(1, 1)}, and {(1, 1), (2, 2)} is both symmetric and antisymmetric.

b) The relation {(1, 2), (1, 3)} is antisymmetric, but not symmetric.

Remarks 2:

a) Difference between not-symmetric and anti-symmetric: For a relation R to be not-symmetric, we require for some pair a, b ∈ X, if (a, b) ∈ R, then (b, a) fails to be in R. For a relation R to be anti-symmetric, we require for every pair a, b ∈ X, if (a, b) ∈ R, and a ≠ b then (b, a) fails to be in R.

b) Every anti-symmetric relation R is a not-symmetric relation, but a not- symmetric relation need not be anti-symmetric. If R is anti-symmetric on set X, then, by definition, for every pair a, b ∈ X, and a ≠ b, if (a, b)∈ R, then (b, a) fails to be in R. It implies for at least one pair a, b ∈ X, and a ≠ b, if (a, b)∈R, then (b, a) fails to be in R, which implies R is a not-symmetric relation. For the converse, only an example is required: The relation R = {(a, b), (a, c), (c, a)}is not-symmetric as (a, b) ∈ R, but (b, a) fails to be in R. However, R is not anti-symmetric as (a, c) ∈ R, and also (c, a) ∈ R.

Check Your Progress-1

1) Let R be a relation from X= {1, 2, 3, 4}, to Y = {1, 2, 3, 4}, with

R = {(1, 2), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 3)}

i) Give Graphic representation of R

ii) Give Matrix representation of R

---

2) For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) The relation of "is-congruent-to" on the set X of all triangles

ii) the relation "less-than-or-equal-to (≤)" on numbers

iii) The relation "less-than (<)" on N

---

3) For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) R={(1, 1),(2,2),(1,3),(2,3)}on X={1,2,3}

ii) R = {(1,2),(2,1),(1,3),(3,1)}on X={1,2,3}

iii) R = {(1, 2), (2,3), (1,3)}on X={1,2,3}

.....

.....

3) Transitive, Not-Transitive, Anti-Transitive

a) A relation R on a set X is **transitive** if whenever (a, b) ∈ R and also (b, c) ∈R-for elements a, b, c ∈X ; then (a, c) ∈R.

b) From (a) above, a relation R on a set X is **not transitive** if—for some a, b, c ∈ X-whenever (a, b) ∈R and also (b, c) ∈R ; then (a, c) fails to be in R.

Remarks 3:

If (a, b) ∈ R and there is no c ∈ X such that (a, c) ∈ R, then R is transitive

Some examples:

i) Each of the relations=,and<is transitive on set of numbers.

ii) The relation is-congruent-with on the set of all triangles is transitive.

iii) The relation is-mother-of is not transitive on the set of all human beings (because, if a is mother of b, and b is mother of c, then a is grand-mother of c, and a is not mother of c.)

iv) The relation is-brother-of is transitive on the set of all human beings.

For (v) & (vi), on the set X = {1, 2, 3},

v) Each of the relations R_1= {(1, 2), (3, 3), (2, 3), (1, 3)}, and R_2= {(1, 1), (3, 3)} is transitive,

vi) The relation R_3 = {(1, 2), (3, 3), (2, 3), (2, 2), (2, 1)}is not transitive, because, for (1, 2) ∈R and (2, 3) ∈ R, (1, 3) ∉ R. The proof is completed. However, the claim could also be established in another way as follows: (1, 2) ∈ R, (2, 1) ∈R, but (1, 1) ∉ R.

c) A relation R on a set X is **anti-transitive** if for each triple a, b, c ∈X ; whenever (a, b) ∈R and also (b, c) ∈R, then (a, c) fails to be in R.

Examples of anti-transitive:

i) In the plane geometry, the relation in the set of all straight lines of being perpendicular (⊥) is anti-transitive: because, if for any three straight lines a, b, and c, with a ⊥ b, and b ⊥ c, then a $\not\perp$ c (a is not perpendicular to c)

ii) By the same argument discussed earlier, the relation of is-mother-of on the set of human beings is anti-transitive.

iii) On the set X = {1, 2, 3}, the relation R= {(1, 2), (2,3), (3, 1)} is anti- transitive, because only pairs of the form (a, b) and (b, c) are:

a) (1,2), (2,3), for which (1, 3) fails to be in R

b) (2,3), (3, 1), for which (2, 1) fails to be in R

c) (3, 1), (1, 2), for which (3, 2) fails to be in R.

Remarks 4:

a) Difference between not-transitive and anti-transitive: For a relation R to be not-transitive, we require for some triple a, b, c ∈ X-if (a, b) ∈R and also, (b, c) ∈R then (a, c) fails to be in R. For a relation R to be anti- transitive, we require for every triple a, b, c ∈ X—if (a, b) ∈R and also (b, c) ∈R then (a, c) fails to be in R.

b) Every anti-transitive relation is a not-transitive relation, but the converse may not be true. From (a) above, if R is anti-transitive, then for every triple a, b, c ∈ X-if (a, b) ∈R and also (b, c) ∈R then (a, c) fails to be in R. Hence, for relation R, for at least one triple a, b, c ∈ X—if (a, b) ∈Rand also (b, c) ∈R then (a, c) fails to be in R. And hence R is not-transitive. For the opposite direction, just one example is sufficient. Example: (i) R= {(1, 2), (2, 3), (1, 3), (3, 2))} on the set {1, 2, 3} is a not-transitive relation as (2, 3)∈R, (3,2)∈R, but (2, 2)∉ R. But it is not anti- transitive, as (1, 2)∈R, (2, 3)∈R, and also (1, 3)∈R.

4) A Simple Method of Generating Examples and Counter-Examples

Start with a small set say, X= {a, b, c}, and for the required relation R, take pairs, one by one, of elements of X, which satisfy, or violate the conditions as required for the relation R. For example, if R is required to be (i) reflexive & (ii) transitive, but not symmetric. Take pairs which make the relation reflexive, by including in R the pairs (a, a), (b, b) and (c, c). So far, the relation R is transitive and symmetric. But we require R to be nonsymmetric. Include, say, only (a, b), but not (b, a), so that{(a, a), (b, b), (c, c), (a, b)} ⊆R.

As L. H. S. satisfies the required conditions, the process may be terminated.

Care must be taken that while satisfying one condition, some earlier satisfied condition may be violated. Then the process needs to be continued. For example, we wish to have a relation which is reflexive, but not symmetric and not transitive. Then we may continue from {(a, a), (b, b), (c, c), (a, b)} ⊆R, and add a pair so that transitivity is disturbed, e. g.{(a, a), (b, b), (c, c), (a, b), (b, c)} ⊆R, which satisfies the required conditions. If R is required not to be reflexive, exclude, say (a, a) from R. And so on.

Various combinations of the above-mentioned properties lead to some mathematically interesting and practically useful properties of relations. These interesting and useful combinations of properties include

i) Equivalence of relations, and

ii) Partial ordering of relations.

These two form the subject matter of next two sections.

2.4 EQUIVALENCE RELATIONS, AND PARTITION OF A SET

A relation R on a set X is said to be an equivalence relation, if R is reflexive, symmetric and transitive. In more detailed form, R is an equivalence relation on set X, if and only if

(i) R is reflexive on X, i. e. for every $x \in X$, we have $(x, x) \in R$, and

(ii) R is symmetric on X, i. e. for $x, y \in X$, if $(x, y) \in R$, then $(y, x) \in R$, and

(iii) R is transitive on X, i. e. for $x, y, z \in X$, if $(x, y) \in R$, and $(y, z) \in R$, then $(x, z) \in R$

Some examples:

i) The relation of equality (=) on the set of numbers, say Q, the set of rational numbers, is an equivalence relation

ii) The relation of 'is-parallel-to' on the set of all straight lines in a plane, is an equivalence relation

iii) The relation of 'is-congruent-to' on the set of all triangles in a plane, is an equivalence relation

iv) The relation of 'is-sibling-of (male or female having same parents) on the set of human beings, is an equivalence relation.

Not-equivalent relation: A relation R on a set X is **not an equivalence relation** if R

i) fails to be reflexive on X: i. e., even for one particular $a \in X$, if $(a, a) \notin R$. For example, for set $X = \{1, 2, 3\}$, the relation $R= \{(2, 2), (3, 3), (2, 3), (3, 2)\}$ is not equivalence relation, as it is not reflexive, because $(1, 1) \notin R$, despite $(2, 2), (3, 3) \in R$(this is an example of a relation which is symmetric, and transitive, but which is not reflexive)

Or

ii) fails to be symmetric on X, i. e. even for one pair of elements $a, b \in X$,we have $(a, b) \in R$, but $(b, a) \notin R$. For example, $X = \{1, 2, 3\}$, the relation $R= \{(1, 1), (2, 2), (3, 3), (2, 3)\}$ is not symmetric, and hence not equivalence relation on X, as $(2, 3) \in R$, but $(3, 2) \notin R$. (this is an example of a relation which is reflexive, and transitive, but which is not symmetric)

Or

iii) fails to be transitive on X, i. e. for $a, b, c \in X$, if for pair of elements $(a, b) \in R$, and $(b, c) \in R$ but $(a, c) \notin R$. For example, $X = \{1, 2, 3\}$, the relationR= $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ is not transitive, and hence not equivalence relation on X, as $(1, 2)$, and $(2, 3) \in R$, but $(1, 3) \notin R$. (this is an example of a relation which is reflexive, and symmetric, but which is not transitive)

Next, we discuss a concept which is quite closely associated with equivalence relation, viz Partition of a set. The close relation (which we will prove) is that for each equivalence relation R on set X, there is a partition of X, and for every partition of a set, say X there is an equivalence relation on X.

2.4.1 Partition of A Set

For a given set X, a partition of X is a non-empty, finite or infinite, collection $P = \{X_i\}$, where Xi's are mutually disjoint non-empty subsets of X, with their union equal to whole X. More explicitly, a collection $P=\{X_i\}$ is a partition of X, if

i) $X_i \neq \emptyset$, for each $X_i \in P$,

ii) $X_i \cap X_j = \emptyset$, for every distinct pair of X_i and X_j in P

iii) $\cup_{i \in p} X_i = X$

A non-empty set may have several distinct partitions. Examples: Let X = {a, b, c}. Then each of the following is a partition of X:

i) {{a}, {b}, {c}}; (ii){{a}, {b, c}} (iii) {{a, b}, {c}} (iv) {{b}, {a, c}} (v) {{a, b, c}}

2.5 EQUIVALENCE CLASS OF AN EQUIVALENCE RELATION

For an equivalence relation R on a set X, the subset [a] of X defined as [a] = $\{x \in$ X: (x, a) ∈R} is called an equivalence class of a under the equivalence relation R.

Example: Consider the relation R= {(a, a), (b. b), (b, c), (c, b), (c, c)} on set X= {a, b, c}. The relation is an equivalence relation. Two equivalence classes in X under R are {a} and {b, c}.

Every equivalence relation on a set X determines a partition of X. Conversely, if we are given a partition of a non-empty set, then it determines an equivalence relation on X.

Let $P = \{A_i\}$ be a partition of X. Then we define a relation R on X such that for x, y∈ X, (x, y) ∈ R if and only if x and y belong to the same class A_i of the partition. Then, to show R is an equivalence relation on X: (I) Reflexive: for each x ∈ X, the elements x, and x belong to the same class, hence (x, x) ∈ R. (II) Symmetry: if (x, y) ∈ R, i. e., if x and y belong to the same class of partition, then y and x also belong to the same class. Hence, (y, x) ∈ R. (III) Transitivity: if (x, y) ∈ R, and (y, z) ∈ R. Then, by definition of R, x and y are in the same class A_i and y and z are in the same class A_j . By definition of partition, A_i and A_j are either equal or are disjoint. But y belongs to both A_i and Aj. Hence, $A_i = A_j$ Implying x and z belong to the same class of the same class of the partition; implying (x, z) ∈R.

2.5.1 Applications of Equivalence Relation and Partition of a Set

Motivation for equivalence relation is that for some specific purpose, many apparently distinct elements behave identically, and hence need not be distinguished from each other. By clubbing all such elements, we reduce complexity for designing solutions of problems concerning the purpose. For example, for designing a timetable for a school with 1000 students, 5 classes (say, VIII to XII), each class having 4 sections (say, A, B, C & D), and each section having 50 students. Now the timetable need not be designed independently for 1000 distinct students, but only for 20 class sections (5 x 4). For the purpose of designing a timetable, all the 50 students of a class- section are treated as a single entity, which is an equivalence class under the equivalence relation R with $(x, y) \in R$ if and only if students x and y belong to the same class-section. These 20 class sections determine a partition of the set of all students in the school.

The equivalence relation has useful applications in the design of digital computers and other sequential machines.

2.5.2 Partial Order Relation, Partially Ordered Sets

A relation R on a set X is said to be a partial order, if R is reflexive, antisymmetric and transitive. In more detailed form, R is an partial order on set X, if and only if

i) R is reflexive on X, i. e. for every $x \in X$, we have $(x, x) \in R$, and

ii) R is anti-symmetric on X, i. e. for $x, y \in X$, if $(x, y) \in R$, and $(y, x) \in R$, then $x=y$, and

iii) R is transitive on X, i. e. for $x, y, z \in X$,if $(x, y) \in R$, and$(y, z) \in R$, then $(x, z) \in R$

Some examples:

i) The relation of less-than-or-equal-to ($\leq$) on the set of numbers, say Q, the set of rational numbers, is a partial order relation

ii) The relation of subset-of($\subseteq$) on a set of subsets of a given set, is a partial order.

iii) The relation of is-divisor-of on N, the set of natural numbers is a partial order.

The relation is called partial order because, some pairs of elements of domain under a partial order may not be related. For example, in (ii) just above, the two subsets {a, b, c} and {b, c, d} are not related to each other under subset-of($\subseteq$) relation. Similarly, for (iii) just above, the number 5 does not divide 9, nor 8 divides 5. Hence, 5 and 9 are not related to each other under divisibility relation.

Partially Ordered Set (Poset): An ordered pair (X, R)- where X is non-empty set and R, a partial order relation on X - is called a partially ordered set (for short, Poset). For example, (N, is-divisor-of) is a partially ordered set.

Total order: For a partial order R on a set X, if for every pair $x,y \in X$, either $(x, y) \in R$, or $(y, x) \in R$, i. e. every pair of elements is related, then the relation R is called Total order, and (X, R) is called a totally ordered set. For example, If for the relation

is-divisor-of is defined on the set $\{2^0, 2^1, 2^2, 2^3, ...\}$, of all powers of 2, then is-divisor-of is a total order on the set. In general, the relation is-divisor-of defined on the set $\{p^0 = 1, p^1, p^2, p^3, ...)$, of all powers of p, for a prime number p, is a total order.

Hasse Diagram

In partial order theory, Hasse Diagram is a graphical representation of a finite partially ordered set obtained as follows. Let (X, R) be a finite partially ordered set. The Hasse diagram is obtained by drawing an upward line from x to y whenever for x, y ∈X, we have x R y, (i.e., (x, y)∈ R) as shown and explained in the following examples.

Example: Let X = divisors of 12= {1, 2, 3, 4, 6, 12}, and for x R y if x is a proper divisor of y, for x, y ∈ R. Then the corresponding Hasse diagram is shown in Fig. 2.5.

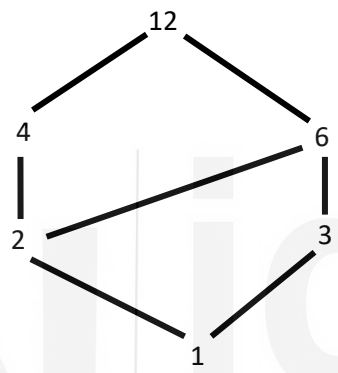


Fig. 2.5: Hasse Diagram - 1

Example: Let X= Power set of {x, y, z}, i. e. X = {∅, {x}, {y}, {z}, {x, y}, {x, z}, {y, z}, {x, y, z}}. Define R on X such that for A, B ∈ X, then (A, B) ∈R, if A⊂ B. The Hasse diagram is as shown in Fig 2.6.

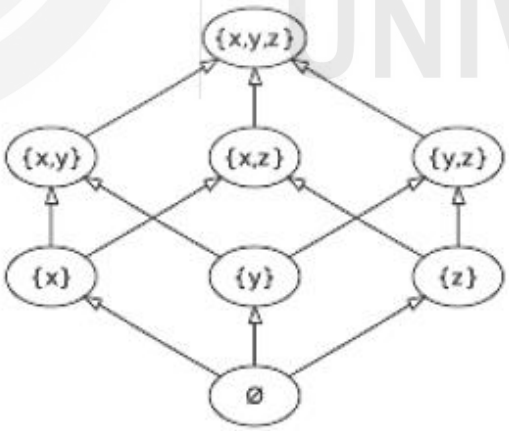


Fig. 2.6: Hasse Diagram - 2

Example: Let X = {3, 4, 12, 24, 48, 72}, and under the relation R, for x, y ∈X, (x, y) ∈R, if x is proper divisor of y. The Hasse diagram is shown in Fig 2.7.

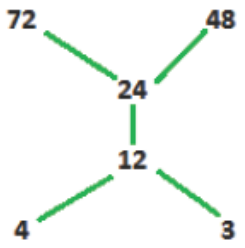


Fig. 2.7: Hasse Diagram -3

Following is an another example of a totally ordered set ({1,2,3,4,8},/) , where " /" is division symbol.

As we know a set is "totally ordered" if **every pair** of elements in the set is comparable. This means for any two numbers a and b, above either a divides b or b divides a.

Hasse diagram for this set will be a strait line.

Figure 2.8 Hasse Diagram - 4

Some Definitions

Here we provide definitions of maximal element and minimal element, of a set under a partial order.

In the previous Fig. 2.7 there are two elements, viz. 4 and 3, at the lowest level in the sense that x∈X does not divide 3, for x ≠ 3, and x∈X does not divide 4 for x ≠ 4. Such elements are called minimal elements.

Thus, in a poset (X, R), an element a of X is a minimal element under R, if there does not exist x in X, such that x is related to element a. A Poset may have one, two or more minimal elements. We have shown in a previous example that there are cases with two minimal elements. When we have only one minimal element, we also call it the minimum element; in this case, it is called the **least element**.

Maximal Element: In Figure 2.7, there are two elements, viz. 48 and 72, at the highest level in the sense that 48 does not divide x∈X, for x ≠ 48, and also 72 does not divide x∈X for x ≠ 72. Such elements are called Maximal elements.

Thus, in a Poset (X, R), an element a of X is a maximal element under R, if there does not exist y in X, such that a is related to y. A Poset may have, one, two or more maximal elements. In the example corresponding to Fig.3.7, we have two maximal elements, viz. 48 and 72.

In the example corresponding to Fig.2.5 we have just one maximal element, viz. 12. In such cases, the maximal element is also called greatest element of the Poset.

Note: Point to remember that **least and greatest element are unique**.

Check Your Progress -2

1) For each of the following relations, tell whether it is transitive, not-transitive, anti-transitive.

   i) In the plane geometry, the relation R—in the set of all straight lines—is defined as R (a, b) ∈ R, for straight lines a and b, if and only angle between a and b is 30 degrees.

   ii) The relation less-than-or-equal-to (≤) on numbers

   iii) The relation is-brother-of, if underlying set X= set of all human beings

   .....
   .....
   .....
   .....

2) Show that the three properties of relations, viz. reflexive, symmetric and transitive are independent of each other, i. e. a relation may be (i) reflexive, but neither symmetric, nor transitive; (ii) symmetric, but neither reflexive, nor transitive; and (iii) transitive, but neither reflexive, nor symmetric.

   .....
   .....
   .....
   .....

3) Find out partition of the set X = {1, 2, 3, 4} given by the equivalence relation

   R = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (3, 1), (2, 4), (4, 2)}

   .....
   .....
   .....
   .....

2.6 OPERATIONS ON RELATIONS

A relation on input of required number of arguments returns true/false, e. g. is- less-that (3, 5) is true, and is-less-that (5,3) is false. On the other hand, an operation on input of required number of arguments returns a new element of the domain, e. g. Sum (3, 5) returns 8.

In this section, we discuss operations on relations, i. e., operations for which inputs are relations, and the output is again a relation. Unless otherwise mentioned, all operations are binary relations on (a single) set, say, X.

The operations which we discuss may be classified as:

a) Set-type operations: union, intersection and difference of relations

b) Mapping-type operations: inverse of operation, composition

c) Closure operations: reflexive-closure, symmetric-closure, transitive closure.

2.6.1 Set-type Operations on Relations

The set-type operations on relations include union of relations, intersection of relations etc. The definitions of these operations are exactly on the lines of the definitions of operations on sets given in Unit 1 of this course. We explain this type of operation on relations only through examples. For the discussion of this type of operations on relations, let R and S be two relations from X = {1, 2, 3, 4} to Y= {a, b, c}, with

R= {(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)}

S= {(1, b), (2, a), (3, a), (3, b), (4, a)}

Then, Universal relation U has 4 x 3 =12 elements of the form (x, y) with all x ∈ X, and all y ∈ Y.

i) Union of R and S denoted by RUS, is given by:

RUS = {(1, a), (2, a), (2, c), (3, b), (4, b), (4, c), (1, b), (3, a), (4, a)}

ii) Intersection of R and S denoted by R∩S, is given by R∩S = {(2, a), 3,b)}

iii) Difference R-S = {(1, a), (2, c), (4, b), (4, c)}, and

Difference S-R = {(1, b), (3, a), (4, a)}

iv) Complement of relation R= R'= {(1, b), (1, c), (2, b), (3, a), (3, c), (4, a)}.

2.6.2 Mapping-type operations: inverse of operation, composition

i) For a relation R from set X to Y, the inverse relation of R, denoted by R^{-1} is the relation from Y to X, such that (y, x) $\in R^{-1}$, if and only if (x, y) ∈R. For example, if R= {(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)} from X = {1, 2, 3, 4} to Y= {a, b, c}then R^{-1}= {(a, 1), (a, 2), (c, 2), (b, 3), (b, 4), (c, 4)} is from Y= {a, b, c} to X = {1, 2, 3, 4}.Note the difference between R' and R^{-1}.

ii) Composition of relations denoted by R∘S.

Here, R and S are defined in a slightly different way than these were defined just above.

Let R be a relation from X = {1, 2, 3, 4} to Y= {a, b, c}, say, R= {(1, a), (2, a), (2, c), (4, c)};

And, S be a relation from Y= {a, b, c} to Z = {triangle, square}, say S= {(a, triangle), (b, square), (b, triangle), (c, square)}. Note the Codomain of R = Domain of S, is essential for the definition of composition.

Then the relation R∘S is from X = {1, 2, 3, 4} to Z = {triangle, square}, given by

R∘S= {(1, triangle), (2, triangle), (2, square), (4, square)}

2.6.3 Closure Operations

The reflexive-closure, symmetric-closure, and transitive closure operations are relevant only for a relation R on a set, say X. The operations are illustrated through examples:

Let X= {1, 2, 3}, and R = {(2, 2), (3, 3), (2, 3), (3, 1)}

i) Reflexive-closure of a relation R is a relation S such that R⊆S, and S is obtained from R by adding minimum number of ordered pairs (x, x) so that S is reflexive. For example, for R = {(2, 2), (3, 3), (2, 3), (3, 1)}, Reflexive-closure (R) = {(2, 2), (3, 3), (2, 3), (3, 1), (1, 1)}

ii) Symmetric-closure of a relation R is a relation S such that R⊆S, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is symmetric. For example, for R = {(2, 2), (3, 3), (2, 3), (3, 1)}, Symmetric-closure (R) = {(2, 2), (3, 3), (2, 3), (3, 1), (3, 2), (1, 3)}

iii) Transitive-closure of a relation R is a relation S such that R⊆S, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is transitive. For example, for R = {(2, 2), (3, 3), (2, 3), (3, 1)}, Transitive-closure (R) = {(2, 2), (3, 3), (2, 3), (3, 1), (2, 1)}.

Check Your Progress - 3

1) Under the relation of is-proper-divisor-of, draw Hasse diagram for set X, where X= all divisors of

(i) 30 (ii) 24 and for each case, write all the minimal elements, and maximal elements.

.....

.....

.....

2) Under the relation of *is-proper-divisor-of,* draw Hasse diagram for set X = {1, 2, 3, ..., 10}, and tell all the minimal elements, and maximal elements.

.....

.....

.....

2.7 SUMMERY

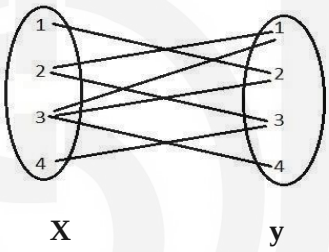
Relation one of the most fundamental concepts of Mathematics. In this unit, we introduced the concepts through respective definitions and illustrative examples. Also, we discussed three main properties of relations, viz. reflexivity, symmetry, and transitivity, along with some variants of these properties and some interesting combinations of these properties, viz. equivalence relations and partial orders. Finally, we discussed some operations on relations.

2.8 ANSWER TO CHECK YOUR PROGRESS

Check Your Progress – 1

1.

i) Graphic Representation



ii) Matrix Representation

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 0 & 0 & 1 & 0 \end{array}$$

2. i) The relation of is-congruent -to on set X of all triangles is **symmetric.**

ii) The relation less-than-equal to (<=) on numbers is not symmetric , as 2<=3 is true but 3<= 2 is not true. . Also,<= is anti-symmetric as x<=y and y<=x then x = y.

iii) The relation < on N is asymmetric.

3. i) {(1,1),(2,2),(3,3)}is both symmetric and antisymmetric.

ii) {(1, 2), (2,3), (1,3)}is antisymmetric, but not symmetric.

iii) {(1,2),(2,1),(1,3),(3,1)}is symmetric, but not anti-symmetric ,as (1,2), (1,2) , but 1 ≠2.

Check Your Progress – 2

1.

i) The relation R defined as R(a, b), for straight lines a and b, if and only angle between a and b is 30 degrees, is not *transitive* ,because for R(a, b) and R (b, c), the angle between (lines a and b) and also (lines b and c) may be 30 degrees, but angle between a and c may be 60 degree also.

ii) The relation less-than-or-equal-to (<=) on numbers *is transitive*

iii) The relation *is-brother-of*, if underlying set X= set of all human beings *is transitive.*

2. The three properties of relations on a set X, are independent of each other. Examples of relation which is:

i) *reflexive, but neither symmetric, nor transitive:*

R on Z, the set of integers, (a, b) ∈ R, if and only if 0 <=a-b <=1

Reflexive: 0 = a-a<= 1; **Non-symmetric: (4,3) ∈R, (3,4) ∉R;**

nor transitive (4,3) ∈R, (3,2) ∈ *R*, (4,2) not ∈ *R*

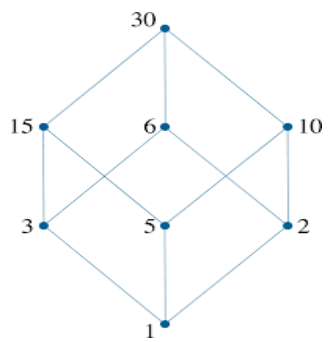
ii) symmetric but neither reflexive, nor transitive: R, the relation of being perpendicular to each other in the set of all strait lines in a plane.

iii) R is less than (<) on the set of numbers, say, rational numbers

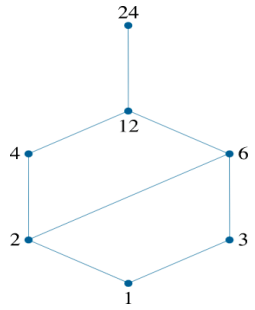
3. Partition of X = { (1,3}}, {2,4}}

Check Your Progress – 3

1. i)



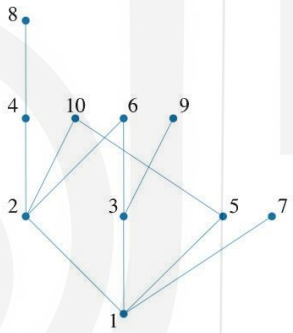
For(i) above, Minimal element = 1, Maximal element =30;



ii).

For(ii) above, Minimal element = 1, Maximal element = 24

2. Minimal element=1, Maximal elements=5,6,7,8,9,10.



2.9 REFERENCES AND FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MCS-013: Discrete Mathematics
   Unit-1 Sets, Relations and Functions
   (https://egyankosh.ac.in/handle/123456789/1140)
5. MEC-203 : Quantitative Methods
   Unit-3 Relations and Functions
   (https://egyankosh.ac.in/handle/123456789/98141)
6. MST-001: Foundation in Mathematics and Statistics
   Unit-2 Functions
   (https://egyankosh.ac.in/handle/123456789/92122)

UNIT 3 FUNCTIONS

Structure

3.0 Introduction
3.1 Objectives
3.2 Quantity
3.3 Interval
3.4 Function
  3.4.1 Mapping
  3.4.2 Pictorial Presentation of Functions
  3.4.3 Operations on Functions
  3.4.4 Some Commonly Used Functions
  3.4.5 Monotonic Functions
3.5 Types of Functions Based on Mapping
  3.5.1 Injective Function
  3.5.2 Surjective Function
  3.5.3 Bijective Function
  3.5.4 Geometrical Meaning of Injective, Surjective and Bijective Functions
3.6 Inverse of Functions
3.7 Summary
3.8 Answer to Check Your Progress
3.9 Refferences and Further Readings

3.0 INTRODUCTION

Many times, we observe the association of the elements of one set with the elements of another set. For example, roll numbers of students in an examination of a university are associated with their corresponding marks. Such type of associations is discussed under the topic “Function”. In this unit, we will define function, explain its classification and various types. Function is a fundamental concept of Mathematics, viz. that of a ***function,*** which is a ***special case of relation***, but m***ore general than that of 'operation'*** on a set. While discussions in this s unit, we will be using some concepts related to sets discussed in the unit -1 of this block.

3.1 OBJECTIVES

After completing this unit, you should be able to:

- define constant quantity, variable, interval, give some examples of each;
- define function, some particular functions;
- evaluate the value of some particular functions at given points;
- get an idea of one-one, onto and one to one correspondence and their geometrical interpretation; and

- finding inverse of a function.

3.2 QUANTITY

Before defining function, let us first explain what we mean by quantity, constant, variable and intervals.

Quantity

Here, by quantity we mean those things on which four basic mathematical operations addition, subtraction, multiplication and division can be applied.
For example, temperature, height, weight, and time all these are quantities but they are continuous in nature, whereas books in a library, number of trees, number of balls is discrete in nature.

Note 1: If the nature of the quantity is such that it can take any possible value between two certain limits, then such a quantity is known as continuous in nature.

Let us consider the following example:

Suppose at the time of birth, height of a baby was 1.5 ft and after 15 years, the height of the same baby is 5 ft, then we know that height of this baby took all possible values between 1.5ft. to 5ft. That is, this is not the case that at the time of birth the height was 1.5 ft and in the next moment it reached to1.6 ft and in successive moment to 1.7 ft. In fact, there are infinitely many values between 1.5 ft and 1.6 ft and all these values were taken by the height of that baby.

Note 2: Generally in mathematics and statistics, a variable is considered **discrete** if its possible values form a set of isolated points on a number line. If you can count the possible values—even if that counting goes on forever (countably infinite)—the variable is discrete.

Let us consider an example:

Number of children per family in a locality is an example of discrete quantity.

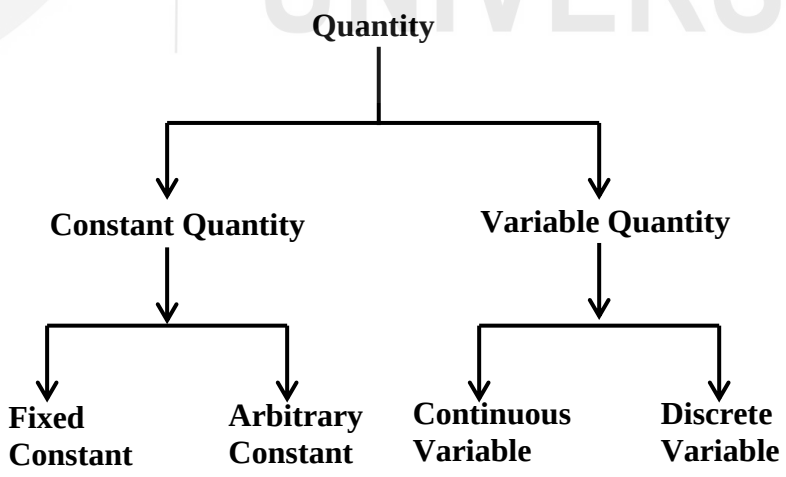


Fig. 3.1 : Representation Quantity

Constant Quantity

A quantity which remains same (unchanged) throughout a particular investigation is known as a constant quantity.

Fixed Constant

Those types of constants which always remain same (unchanged) independent of the purpose of user, place and time are known as fixed constants.

For example,

(i) 2, 5, $\frac{3}{2}$, $\sqrt{13}, -\sqrt{17},$, etc.

(ii) quotient of circumference of a circle and its diameter which is always equal to π.

Arbitrary Constants

Those types of constants which remain same in one problem but may vary from problem to problem are known as arbitrary constants and are generally denoted by a, b, c, l, m, n, $\alpha, \beta, \gamma,$ etc.

For example, heights of the houses constructed by the different owners of the plots in a locality as per their own choice, is an example of arbitrary constant because heights of the houses vary from house to house as it depends on the choice of the owners.

Variable

A quantity which may change its value even in a particular problem is known as variable.

For example, blood pressures of a person as it varies time to time.

3.3 INTERVAL

Let R be the set of all real numbers. Then a set I⊆R is said to be an interval if whenever $a, b \in I$ *and* a < $x < b$ then $x \in$ I

For example, the set of all real numbers satisfying $2 \leq x \leq 3$ is an interval where x can take any real value between 2 and 3 including 2 and 3.

In unit 1 of this course, you have learned about Open Interval and Closed Interval, now let us learn some more intervals

Left Open and Right Closed Interval

A left open and right closed interval I⊆ R with end points a and b (a < b) is denoted by (a, b] and is defined as

(a, b] = $\{x \in R: a < x \leq b\}$

In this case $x \in (a, b], \forall x,$ a < x $\leq b$ and a $\notin (a, b]$, but b $\in$ (a, b]

Left Closed and Right Open Interval

A left closed and right open interval I⊆ R with end points a and b (a < b) is denoted by [a, b) and is defined as

[a, b) = $\{x \in R: a \leq x < b\}$

In this case $x \in [a, b), \forall x,$ a $\leq x < b$ and a $\in [a, b)$, but b $\notin$ [a, b)

Length of an Interval

Length of each of the intervals (a, b), [a, b], (a, b], [a, b) is defined as
$b - a, a < b$
i.e. length of the interval = difference of the end points

For example, if I = (2, 7) then l (I) = 7 – 2 = 5, where l (I) denotes the length of the interval I.

Finite Interval

An interval is said to be finite if its length is finite.

For example, if I = (– 3, 5) then l (I) = 5 – (– 3) = 5 + 3 = 8 which is finite.
∴interval I is finite.

Infinite Interval

An interval is said to be infinite interval if its length is not finite.

For example,

(i) The set :
$\{x \in R : x > a, a \in R\}$
is an infinite interval and is denoted by (a,∞)
(ii) The set $\{x \in R : x < a, a \in R\}$ is an infinite interval and is denoted by $(-\infty, a)$

Similarly, infinite intervals [a,∞), $(-\infty, a]$ are defined as
[a, ∞) = $\{x \in R : x \geq a\}$, where a is a fixed real number
$(-\infty, a] = \{x \in R : x \leq a\}$,where a is a fixed real number

Remarks 1:

(i) Each interval contains infinitely many elements.

(ii) Each interval is an infinite set but an infinite set may or may not be an interval. For example, N, W, Z, Q are infinite sets but are not intervals.

(iii) A set may or may not be an interval.

(iv) R, set of real numbers, is an infinite interval given by R = $(-\infty, \infty)$

(v) Remember that $-\infty$ and ∞ are not included in the set of real numbers. Also, if extreme value is $\infty \; or - \infty$ then open bracket is used on the side having extreme value.
(vi) When we say that x is a finite number/real number it means that
$-\infty < x < \infty$

Now we are in a position to define function.

3.4 FUNCTION

A function is a *mathematical* term, which is a tool used to describe some aspects of the real world, including the following:

a) Given that a vehicle moves at a constant average speed *(say 50 km./hour)*, then the *distance (say d)* covered depends on the *time (say t hours)* over which the vehicle moves,

b) If a person takes a loan L (say, of Rs. 50,000/-) on simple interest *R* (say, at 6% per annum), the amount *(say A)* payable depends on the time *T*, after which the payment is made, *in addition to L and R.*

c) The area *(say Area)* of a circle depends on the radius *(say r)* of the circle.

The statements (a), (b) and (c) above express some *real* life situations, and each of which can be *mathematically* expressed respectively as (a) d = 50* t, (a) A = 50000 $(1+6/100)^t$, and Area = $\pi.r^2$.

In each of these cases, there are two *variable quantities* involved: one on LHS and the other on RHS.

For example,

in (a) the quantity *d* occurs on LHS and quantity *t* occurs on RHS;

in (b) the quantity *A* occurs on LHS and quantity *t* occurs on RHS; and

in (c) the quantity *Area* occurs on LHS and quantity *r occurs* on RHS.

For further discussion, first, we consider only (a) above. The value of d, occurring on L.H.S., is given by RHS, viz. 50*t, and *50*t* depends on the value of t *only*. For example, if t = 3(hours), the distance covered, i. e. d=50* t=50*3= 150 km.; if t= 5 hours, then distance covered, i. e. d = 50*5=250 km.

Further, the dependence of 50*t on t *is unique* in the sense that for t=3, we get the *unique* value 150 or 50*3, and for t=5, we get the *unique* value 250 or 50*5. This unique *dependence* of 50*t, and hence of variable quantity d is, in mathematical terms, called ***functional*** dependence, and 50*t is denoted as ***f(t),*** for some f, and t inside shows that it depends on variable quantity t.

In view of the discussion above, ***f*** is called the ***function name***, the variable quantity ***t*** is called the ***independent variable*** and the variable quantity d is called the ***dependent variable, and the relation between d and t is called d's functional dependence on t.***

Going back briefly to (b), A = 50000 $(1+6/100)^t$. Let g(t) = 50000 $(1+6/100)^t$, then A = g(t). Here, g is a function name, t is independent variable, and A is the dependent variable, ***and the relation between A and t is called A's functional dependence on t.***

In case of (c), let h(r) = πr^2, then Area = h(r), and h is a function name, r is the independent variable and *Area* is the dependent variable, ***and the relation between Area and r is called Area's functional dependence on r.***

Further, let X and Y be two sets. Then a rule which associates each element of X to a unique element of Y is called a function.

X is called domain of the function.
Y is called co-domain of the function and the set of only those values of Y for which function is defined is called range of the function.
That is, subset $\{y \in Y : y = f(x) for\ some\ x \in X\}$ of Y is called range of the function.

Notation:

(i) A function is generally denoted by f, g, h, etc., in the case of above definition we write f: X→Y and read as f is a function from X to Y.

(ii) A function f: X→Y is generally described by writing $y = f(x), x \in X,$ where $f(x)$ is an expression in terms of x.

A function is a special kind of relation. Functions are not restricted to sets of numbers only. For instance, let A be the set of mothers and B be the set of human beings. Then the rule that assigns to every mother her eldest child is a function. But the rule that assigns to **each mother her children** is **not a function** because it does not relate a unique element of B to each element of A.

In other words we can say that a function from a set, say A, to a set, say B, is a special type of relation where (i) each element a ∈ A is in relationship with some element b ∈ B, and (ii) there is exactly one element b ∈ B in relationship with each element a ∈ A.

More explicitly,

For two given *non-empty* sets X and Y, ***a function f from X to Y,*** denoted as

f: X→ Y is a *rule* which associates to each element x of X a *unique* element, say, y of Y. The unique element y associated with x of X is denoted as f(x), i. e. y = f(x).

For the function f: X→Y, *the set X is called the domain of function f and Y is called the codomain of function f.*

3.4.1 Mapping

If f(x) = y, then y is called image of x under f, and x is called *pre-image* of y. Also, the y, or f(x), is called the *value* of x under the function f. *The names 'map' and 'mapping' are also used instead of the name 'function'*. The subset of Y which consists of all images f(x) of elements of X is called the *range* of function f (in Y)

Remarks 2:

Important to note that the role/meaning of '→' is just a part of notation for a function. It just indicates that the symbol X, immediately on left of '→' is the *domain,* and the symbol Y, immediately on the right of '→' is the *codomain* of the function-symbol f, the symbol which precedes ':'.

Definitions: Let f be a function from A to B. The **set A is called the domain** of the function f and **B is called the co-domain** of function f. The set {f(x)|x ∈A} is called the range of f, and is also denoted by f(A).

Given an element x ∈A, the unique element of B to which the function f associates, it is denoted by f(x) and is called the **f-image (or image) of x** or the value of the function f for x. We also say that f maps x to f(x). The element x is referred to as the **pre-image of f (x).**

For example, if A = {1,2,3,4}, B = {1,8,27,64,125}, and the rule f assigns to each member in A its cube, then f is a function from A to B. The domain of function f is A, its codomain is B and its range is {1,8,27,64}.

Can you tell what will be the domain and codomain for rule f: f(x) = x/(1-x)?

As you can see that 1-x =0, if x = 1, in this case f(x) will be undefined. Here domain of f can be taken as R~{1} and codomain can be R.

Definition: A function from a non-empty set A to a non-empty set B is a subset R of A × B such that for each a ∈A ∃ a unique b ∈B such that (a,b) ∈ R. So, this relation satisfies the following two conditions:

for each a ∈A, there is some b ∈B such that (a,b) ∈ R

if (a,b) ∈ R and (a,b´) ∈ R then b = b´.

In previous examples we have seen that, we usually present functions as a rule associating elements of one set with another. So, let us present the definition again, with this view.

Definition: Let A and B be non-empty sets. A function (or a mapping) f from A to B is a rule that assigns to each element x in A exactly one element y in B. We write this as f: A →B, read it as 'f is a function from A to B'.

Note that :

to each a ∈A, f assigns an element of B; and

to each a ∈A, f assigns only one element of B.

So, for example, suppose A = {1,2,3}, B = {1,4,9,11} and f assigns to each member in A its square values. Then f is a function from A to B. But if A={1,2,3,4}, B={1,4,9,10} and f is the same rule, then **f is not a function** from A to B since no member of B is assigned to the element 4 in A.

There are two conditions for a rule to be a function.
(i) Each element of X must be associated to some element of Y.
(ii) There is unique element of Y corresponding to each element of X.

Similarly, consider an example where A={1,2,3}, B={1,4,5,6,7} and the rule f which associate 1→1, 2→1, 2→4. **Here f is not a function from A to B** since no member of B is assigned to the element 3 ∈ A.

Note that the former example, 11 ∈ B, but there is no element in A which is assigned to 11. **This does not matter**. It is not necessary that every element of B be related to some element of A.

Remarks 3: Each element of A has a unique image, and each element of B need not appear as the image of an element in A. Further, more than one element of A can have the same image in B.

Let us look at some examples of functions, and non-functions now.

If b is be a fixed element of B, then f: A→B : f(x) = b ∀ x ∈ A is called a **constant function.**

Note that if b=0, then f is called the **zero map**, and is denoted by 0.

f : A → A : f(x) = x ∀ x ∈ A is called the **identity function**, and is denoted by **I**.

Also, in our discussion so far, in each case there is only ONE independent variable associated with each dependent variable. However, for a dependent variable, there may be ***more than one independent variable***.

For example, the area of a rectangle of length *l* and breadth *b* is given by Area_Rectangle = *l*b*. The dependent variable *Area_Rectangle* depends on two independent variables, viz. *l* and b. It may be written as Area_Rectangle = f1(*l*, *b*), where f (l, b) = *l*b*. Note the comma in Area_Rectangle =f1(*l*, b).

Similarly, the volume v of a railway container of length *l*, breadth b and height h is given by *v = l*b*h*. Then the volume v may be written as a function f_2 *(l,b, h)* of three independent variables *l, b, and h.*

Let us look at some more examples for illustrating the concept of function.

i) Let X Y = N, the set of natural numbers, then

f: N→N, such that f(x)=2x+1, ∀ x ∈ N.

Clearly, for each x ∈ N, 2x+1 exists in N and is unique for each x.

Hence, f: N→ N is a function.

ii) Let f: Z→N, such that f(x) = x^2+3,

then for each x ∈ Z, x^2+3 ∈ *N and is unique, therefore, f is a function*

iii) Let f: Z→N, such that f(x) = x^2-3,

then for x=0, x^2-3=-3∉ N, Hence, f: Z→ N is ***not* a function** from Z to N.

iv) Let f: Z→ Z, such that f(x) =x^2 -3 ∈ Z

then x^2- 3 ∈ Z (*codomain),* ∀ *x* ∈ *Z (domain).* Hence, f: Z → Z is a function from Z to Z.
Let us see some examples:

Example:

Function Name: It is indeed called the **greatest integer function** or, more commonly in modern mathematics, the **floor function**.

Domain and Codomain: The function correctly maps real numbers (R) to integers (Z).

We denoted this function's action by f(x)=[x], where [x] is the greatest integer < = x.

Some more examples:

- For x=0.6, the greatest integer less than or equal to 0.6 is **0**.
- For x=2.3, the greatest integer less than or equal to 2.3 is **2**.
- For x=−5, the greatest integer less than or equal to -5 is **-5**.

Example:

Function f: R→R: f(x)= | x | is known as the modulus (or absolute value) function, where | x | is the absolute value of x.

For example, if x=10 then f(x)= | x | =10 and if x= -10, then f(x) = | x |=10.

Now let us take sets, A={a,b,c} and B={1,2,3,4,5}. Consider the rule f which associates a→1, a→3, b→2, c→3. This is **not a function** from A to B because, elements 1 and 3 ∈ B are assigned to the same element a ∈ A.

3.4.2 Pictorial Presentation of Functions

A function can be presented diagrammatically. Here we will see few examples of that. In next unit of this block, you will learn in more detail about diagrammatical representation of different types of functions.

Consider A = {1,2,3,4}, B= {1,4,5} and the rule f which associates 1→1, 2→4, 3→5, 4→5. **Then f is a function from A to B**.

Pictorial representation of above function f :

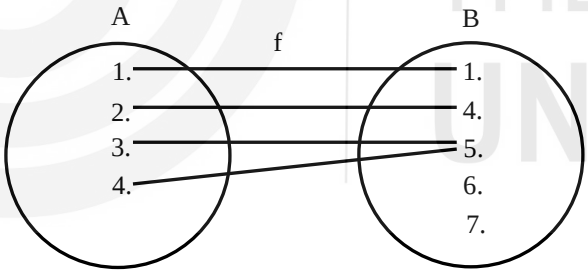


Fig. 3.2: The rule f is a function

Let us take another example of function f where f(x) = x+5 . If we take domain for this function {1,2,3,4} then diagrammatic representation will be :

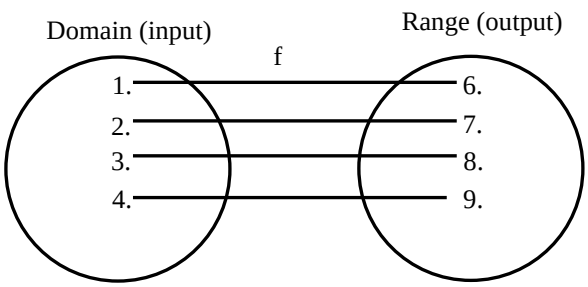


Fig. 3.3: Diagramatic representation function f(x) = x+5

Check Your Progress-1

1) . Let $f : N \to N$ defined by f(n) = 3n, $n \in N$
Express the function diagrammatically. Also, write domain, range and co-domain of the function.

.....
.....

2). If $f : R \to R$ be a function defined by
$f(x) = 3^x$, $x \in R$, then obtain (i) Domain of f (ii) Range of f

.....
.....

3). Find the domain and range of the following real-valued real functions:
(i) f(x)= - |x| (ii) f(x) = + $\sqrt{9 - x^2}$

.....
.....

4). For each of the following, tell whether f is a function or not. Justify , your answer.
i) f: N→N, such that f(x) = x -3
ii) f: N→ Z, such that f(x) = x – 3

.....
.....

5). Find the domain of the function ,$f : R \to R$ defined by
$f(x) = \sqrt{(x-3)(5-x)}$, $x \in R$
Also evaluate f(3), f(4), f(5).

.....
.....

6). Find the domain and range of the function f: R→R given by f(x) = $\frac{1}{x+3}$, $x \in R$.
Also, evaluate f(0), f(2), f(-3) if possible.

.....
.....

7). Every function is a relation. Is every relation a function? Why?

.....
.....

8). Consider the following pseudocode:

```
read(n)
while n > 1 do
begin
if n is even then n := n div 2
else n := 2n + 1;
end
```

Write a function of n that describes the operations performed.

……
……

Some more Concepts Related to Function are given as under

(i) If $f: X \to Y$ is a function given by $y = f(x)$ then x is known a pre image of y and y is known as image of x.

(ii) If $y = f(x)$ is a function then values of y depend on values of x. So, y is known as dependent variable and x is known as independent variable.

Let us now take up some examples which will enable you to distinguish as to whether a rule is a function or not.

For example,

(a)

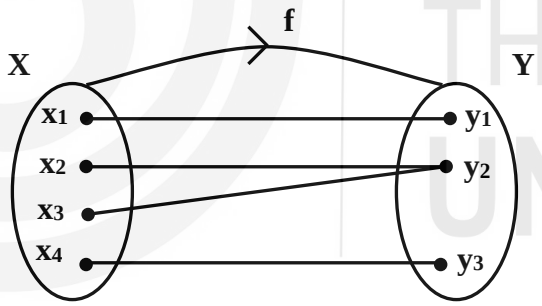


Fig. 3.4: A function

The rule f shown in Fig. 3.4 is a function because it satisfies both conditions i.e.

(i) Each element of X is associated to some element of Y

(ii) There is unique element of Y corresponding to each element of X.
i.e. y1 is unique element of Y corresponding to $x_1 \in X$
y_2 is unique element of Y corresponding to both $x_2, x_3 \in X$
y_3 is unique element of Y corresponding to $x_4 \in X$
Here domain of $f = \{x_1, x_2, x_3, x_4\}$
Co-domain of f = { y_1, y_2, y_3} and range of f = { y_1, y_2, y_3}

(b)

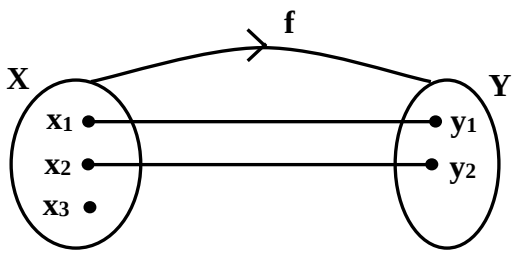


Fig. 3.5: Not a function

The rule f shown in Fig. 3.5 is not a function because $x_3 \in X$, but it is not associated to the element of Y.

Out of two restrictions for a rule to be a function, first is that each element of X must be associated to some element of Y.

(c)

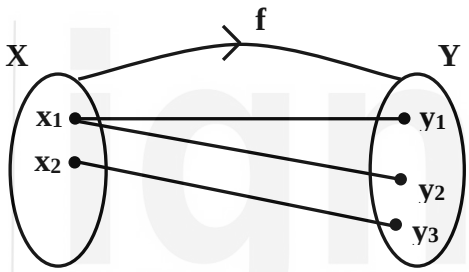


Fig. 3.6: Not a function

The rule f shown in Fig. 3.6 is also not a function because $x_1 \in X$ is not associated to unique element of Y.

For a rule to be a function it is must that each element of X is associated to a unique element of Y.

(d)

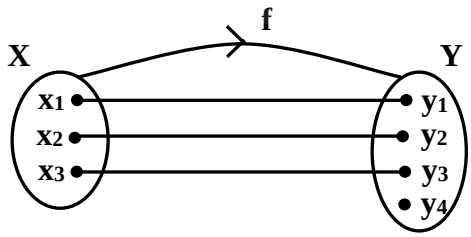


Fig. 3.7: A function

The rule f shown in Fig. 3.7 is a function because it satisfies both the conditions for a rule to be a function, i.e.

(i) Each element of X is associated to some element of Y.

(ii) There is unique element of Y corresponding to each element of X.

Here domain of the function $f = \{x_1, x_2, x_3\} = X$
Co-domain of the function $f = \{y_1, y_2, y_3, y_4\} = Y$
Range of the function $f = \{y_1, y_2, y_3\} \subset Y$

Real valued function

A **real-valued function** is a function where the **output (the "value")** is always a real number.
In mathematical notation, if we have a function f: X →Y, it is a real-valued function if the codomain (Y) is the set of real numbers (ℝ) or a subset of it.

Remark :

1. **Real-Valued Function:** Only the **output** must be a real number. The input (domain) could be anything—points on a map, people, or complex numbers.
2. **Real Function:** Both the **input** and the **output** must be real numbers.

Now let us see how different operations like addition, subtraction, multiplication and division can not only be applied only on real valued functions.

Here it is important to note that the "Algebra of Functions" is commonly taught using real-valued functions, these operations can be applied to many other types of functions as well.

The ability to add, subtract, multiply, or divide functions depends on whether the **outputs** of those functions belong to a mathematical structure that allows those **operations (like a Field, Ring, or Vector Space).**

For real-valued functions f and g, the operations are defined **pointwise**. This means we perform the math on the outputs for every shared input x.

3.4.3 Operations on Functions

The *binary operation* '+' on Natural numbers may be considered for the function ***Plus:*** N × N→N, such that

Plus (x, y) = (x+y), for each pair of Natural numbers.

The union operation **X U Y** on sets can be considered as a function

Union: ***P×P→P,*** where P is a set of all subsets of universal set U, such that ***Union (X, Y) = X U Y***, for each pair X, Y of sets from P.

In general, an operation is a function in which the domain of function is either the codomain or cross-product of codomain of function, Cross-product may involve the codomain finitely many times: Codomain × Codomain ... X Codomain.

For example, the square operation on N is the function SQ: N →N, such that SQ (n) = n^2. The domain of function SQ is the codomain of the function SQ. In the case of Plus and Union, just sown above, the Domain is a cross- product of codomain.

If given whose domain ranges are subsets of the real numbers, we define the function f+g by (f + g) (x) to be the function whose value at x is the sum of f(x) and g(x). Symbolically,

(f + g) (x) = f(x) + g(x). This is called pointwise addition of f and g.

The domain of f+g is the intersection of the domains of f and g since to compute (f + g) (x) it is necessary and sufficient to compute both f(x) and g(x).

Other operations on functions are defined similarly:

- (fg) (x) = f(x)g(x) (pointwise multiplication)
- f^p (x) = $(f(x))^p$ for any real exponent p with the domain of f^p consisting of those points for which the p-th power of f(x) makes sense.
- (f/g)(x) = f(x)/g(x), for g(x) ≠ 0 (pointwise multiplication)

For example, if f(x) = 3 sin (x) and g(x) = x^2, then

(f+g)(x) = 3 sin(x) + x^2

(fg)(x) = 3 sin(x)* x^2

(f-g)(x) = 3 sin(x) - x^2

(f/g)(x) = 3 sin(x) / x^2

The domains of both f and g are all real numbers, but the domain of f/g is {x | x ≠ 0}.

Now let us consider two functions f and g from A = {1,2} to, B = {1,2,3,4}, where f={1,1),(2,4)}. Let g be defined by the rule g(x)=x^2 where the domain of g is the set {1,2}. Here both have the same domain. Since f and g assign the same image to each element in the domain, they have the same effect throughout. This is why we treat them as the same, or equal.

Definition: If f and g are two functions defined on the same domain A and if f(a)=g(a) for every a ∈ A, then the functions f and g are equal, i.e., f = g.

Let us take another example f(x) = x^2 +5, where x is a real number, and g(x) = x^2 +5, where x is a complex number. Then the function f is not equal to the function g since they have different domains although f(x)= x^2 + 5= g(x) ∀ x ∈ R. By this example we can conclude that even if f(a)= g(a), f and g may not be the same.

So far, the operations you have seen are the same as those for member systems.

However, there is yet another operation on functions which we now define.

Definition: Let f and g be the operation of combining two functions by applying them one after the other. That is, the composition of f(x) and g(x), denoted by, fog.

For example, consider f : R→R: $f(x) = (x^3 + 2x)^3$. We can write it as the composition of g and h, where the value of f(x) can be obtained by first calculating $x^3 + 2x$ and then taking its third power. We can write g for first or inside function $g(x) = x^3 + 2x$. We write h for the second function: $h(x) = x^3$. The use of the variable x is irrelevant, we could as well write $h(y) = y^3$ for $y \in R$. We can see that go $h(x) = g(x^3 + 2x) = (x^3 +2x)^3 = f(x)$.

In general $(f \circ g) \neq (g \circ f)$.

For example, if $f(x) = x^2$ and $g(x) = x+1$, then $(f \circ g)(x) = (x + 1)^2$ and

$(g \circ f)(x) = x^2 + 1$.

Here we can see that $f \circ g \neq g \circ f$.

Function Composition is Not Commutative. This property means that the **order** in which you compose functions matters.

Function Composition is Associative:

Let us see another example, where $f(x) = x^2$, $g(x) = x+1$, $h(x) = x^3$

Here, $(f \circ (g \circ h))(x) = (x^3 + 1)^2$ and $((f \circ g) \circ h)(x) = (x^3 + 1)^2$. Here we can see $f \circ (g \circ h) = (f \circ g) \circ h$.

This property means that when you are composing three or more functions, the **grouping** of the functions does not matter.

Let us break down the calculation for $f(x)=x^2$, $g(x)=x+1$, and $h(x)=x^3$:

1. Calculating $f\circ(g\circ h)$

- First, we find the inner function $(g\circ h)(x)$: $(g\circ h)(x)=g(h(x))=g(x^3)=x^3+1$
- Now, we compose f with that result: $(f\circ(g\circ h))(x)=f(x^3+1)=(x^3+1)^2$

2. Calculating $(f\circ g)\circ h$

- First, we find the inner function $(f\circ g)(x)$:
  $(f\circ g)(x)=f(g(x))=f(x+1)=(x+1)^2$
- Now, we compose that result with h(x) (by substituting h(x) for x):
  $((f\circ g)\circ h)(x) = (h(x)+1)^2 = (x^3+1)^2$

Both methods produce the exact same result, confirming that the composition is **associative**.

3.4.4 Some Commonly Used Functions

Based on the nature of classification a function may be given some particular names. In this section you will see some commonly used function names, their definitions and their graphical representations. More on graphical representations of different types of functions you will learn in next unit of this Block.

Constant Function

$Let\ X, Y \subseteq R$, then a function $f: X \to Y$ is said to be a constant function if it is defined as: $f(x) = a, \forall\ x \in R$, where a is a real constant.
i.e. a function is constant if range is a singleton set.
i.e. all elements of the domain are associated to a single element of the co-domain of the function.

For example,

(i) $f: N \to N$ defined by $f(x) = 3,\ n \in N$

is a constant function because all elements of the domain are associated to the single element 3 as shown in the Fig. 3.8

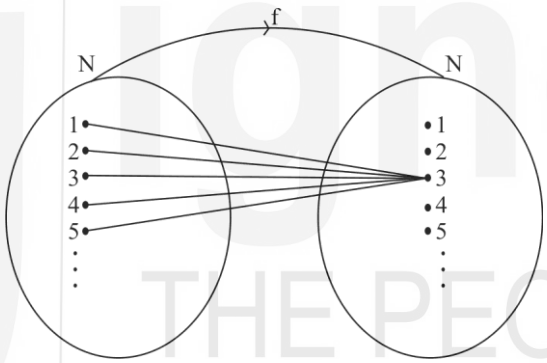


Fig. 3.8: Constant Function

(ii) $f: R \to R$ defined by $f(x) = 2, x \in R$ is also a constant function and its graph is given below in Fig. 3.9.

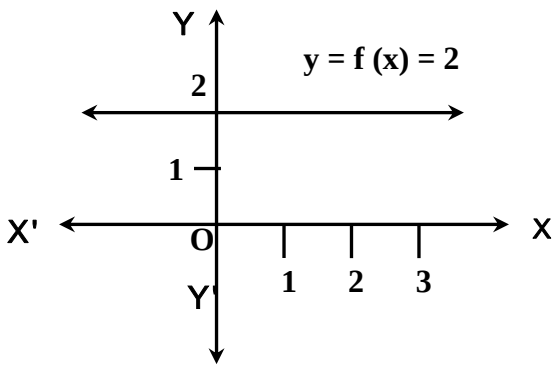


Fig. 3.9: Constant Function

Identity Function

$Let\ X \subseteq R,$ a function $f: X \to X$ is said to be an identity function if it is defined as :

$f(x) = x, \quad x \in X$

i.e. a function is said to be identity function if each element is associated to itself.

For example,

(i) $f: N \to N$ defined by $f(n) = n, \quad n \in N$
is an identity function as shown in the Fig. 3.10.

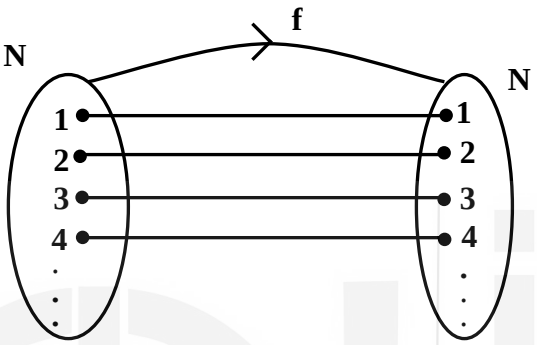


Fig. 3.10: Identity Function

(ii) Function f : $R \to R$ defined by $f(x) = x, \quad x \in R$
is also an identity function and its graph is given in Fig. 3.11.

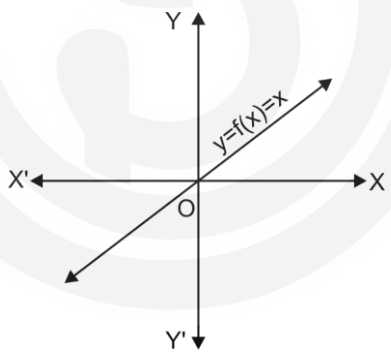


Fig. 3.11: Identity Function

3.4.5 Monotonic Functions

When a function that is either entirely non-increasing or non-decreasing it is called monotonic function.

For Example,

Increasing: If x ≤ y, then f(x) ≤ f(y).

Any straight line with a positive slope is an increasing function. The general form is f(x) = mx+c, where the slope m is greater than 0.

Strictly Increasing: If $x < y$, then $f(x) < f(y)$.

Note that strictly increasing functions are always injective.

It is commonly found that some functions to be increasing only on specific portions of their domain.

For Example $f(x) = x^2$: This parabola is increasing for all $x > 0$. (It is decreasing for $x < 0$).

Linear Function : Any line with a positive slope, like $f(x) = 2x+1$. For every step you take to the right on the x-axis, you go up on the y-axis.

Exponential Function: The natural exponential function $f(x) = e^x$. This function not only increases but does so at an ever-faster rate.

Logarithmic Function: The natural logarithm $f(x) = \ln(x)$, $x > 0$. This function increases for all positive x values, though its rate of increase slows down as x gets larger.

Check Your Progress -2

1. Let $f(x) = 1/x$ and $g(x) = x^3 + 2$. Find the following functions, where $x \in R$.

$(f + g)(x)$

$(f – g)(x)$

$(fg)(x)$

$(f/g)(x)$

……
……..

2. Let $f(x) = \sqrt{(x+1)}$ $\forall\ x \geq -1$ and $g(x) = x^3$ $\forall\ x \in R$. Define the following functions. Also give their domains.

$(f + g)\ (x)$

$(f – g)(x)$

$(fg)(x)$

$(f/g)(x)$

$(f \circ g)(x)$

……
……

3. If $f(x) = x^2$, $g(x) = x+2$ and $h(x) = x^3 + x+2$ find $g \circ f$ and $f \circ g \circ h$

……
……....

3.5 TYPES OF FUNCTIONS BASED ON MAPPING

Now let us discuss some types of functions based on how inputs are mapped to outputs. In other words it describes how describe how the inputs in the domain relate to the outputs in the codomain.

By definition of function f: X→ Y, for each x, its image f(x) must belong to Y and f(x) should not have more than one value in Y, as happens in the case of f(x) = square-root of x. ***However, there is no restriction on elements of X, for f to be a function***. For example, for the function Mod, and for both values 5 &-5, Mod (5) = Mod (-5) = 5. So mapping of function helps in deciding its type. Let us look at different types of mappings.

3.5.1 Injective Function (One-One Function)

A function f: X → Y is called **injective,** if for x_1, x_2 in X with $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

The definition can be equivalently stated as:

A function f: X → Y is called **injective**, if $f(x_1) = f(x_2)$ then $x_1 = x_2$

Note: *The term 1-1 is also used instead of* ***injective****. However, there is a similar term 1-to-1* used for a distinct concept viz. ***bijective. For this reason, the terms 1-1 and 1-to-1 may be avoided.***

Some examples:

i) The function f: N→N, such that f(x) = 2x+1, ∀ x ∈ N is injective. Let f(x) = f(y), then 2x+1 = 2y+1 implying x = y

ii) The function f: N→ R, defined by the rule f(x) = *positive* square-root of x is injective. Let f(x) =f(y), i. e. √x = √y squaring both sides, we get x =y.

iii) The function f: N→ R, defined by the rule $f(x)=x^2$ is injective, as, for x, y from N, if f(x) = f(y), i. e. $x^2 = y^2$ then x = y or x = *-y, but as y is in N, the number -y cannot be in N*. Hence, *x = y*.

Procedure (generally adopted) to establish a function as injective:

For function f: X→ Y,

i) assume that f(x) = f(y) and then show x = y (must) for x, y ∈ X or alternatively, in order to establish ***f is NOT injective***, show for ***some*** pair of elements x, y, with x ≠ y of X, that f(x) = f(y).

The latter method was used in showing f: ***Z***→ R, defined by the rule $f(x) = x^2$, is not injective by using f (2) = f (-2) = 4. ***We have already used the first one to show*** the function f: ***N***→ R, defined by the rule $f(x) = x^2$ as injective.

Injective Mapping: A mapping f A→ B is said to be injective (or one-one) if the images of distinct elements of A under f are distinct, i.e., if $x_1 \neq x_2$ in A, then $f(x_1) \neq f(x_2)$. So, f is 1-1.

In other words a function $f: X \to Y$ is said to be 1-1 or **injective** function if distinct elements of X are associated to distinct elements of Y under f.
i.e. if $x_1, x_2 \in X$ be s.t.
$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

Or if $x_1, x_2 \in X$ and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

If we compare this definition with example of daughters and mothers then one-one function means each daughter must have different mother, i.e. there cannot be two daughters having same mother for a function to be one-one.

For example,

(i)

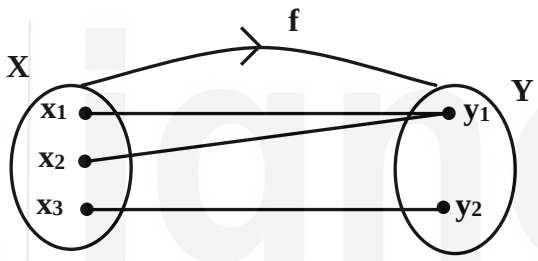


Fig. 3.12: Not a 1-1 function

The function f shown in Fig. 3.12 is **not one-one** functions because two different elements x_1, x_2 of X have the same image y_1.

(ii)

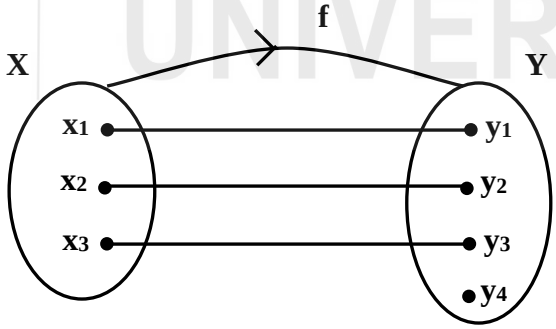


Fig. 3.13: A 1-1 function

The function f shown in Fig. 3.13 is one-one function because all the three elements of X have distinct images in Y.

Example:

Show that the function $f: R \to R$ defined by $f(x) = 7x + 5$ is 1-1 function.
Solution: Let x_1, x_2 be s.t.
$f(x_1) = f(x_2) \Rightarrow 7x_1 + 5 = 7x_2 + 5 \Rightarrow 7x_1 = 7x_2 \Rightarrow x_1 = x_2$
$\Rightarrow f$ is 1-1 function

3.5.2 Surjective (Onto) Function

For a function f: X→Y, it is essential that every element of X(***domain***) corresponds to some element of Y. But, for f to be a function, it is not compulsory that for every y of Y (***codomain***), there be a corresponding element x of X such that f(x) = y.

Onto Mapping: A mapping f: A → B is said to an onto (or surjective) mapping if f(A) = B, that is, the range and co-domain coincide. In this case we say that f maps A onto B.

For example, f: Z → Z : f(x) = x+1, x ∈ Z, then every element y in the co-domain Z has a pre-image y-1 in the domain Z. Therefore, f(Z) = Z, and f is an onto mapping.

If we take another example, for the function ***f: N→ N, with f(x) = 2x+1*** ∀ x ∈ N, for the element 4 of N (the codomain) an *x* in N (the domain), if exists, satisfies 2x+1 = 4, i. e. x = 3/2, which is not in N. ***Hence f is not surjective.***

A function f: X → Y is called **surjective**, if for an arbitrary element y of Y (***the codomain***), we are able to find an x in X (***the domain***) such that f(x) = y.

Procedure for establishing a function as surjective/onto:

Take an arbitrary element y of the ***codomain*** Y, and show the existence of some element x of domain X with f(x) = y. If it is possible to find such an element x of ***domain*** X, then the function is surjective, *else*, it is not surjective.

Example: For the function Mod: Z→ N such that:

$$\text{Mod}(x) = |x| = \begin{cases} -x, x < 0 \\ x, x \geq 0 \end{cases}$$

Let x be in the codomain N, then this x belongs to Z, the domain (*because* $N \subseteq Z$) with |x|=x, hence, Mod is surjective.

Next, consider f: Z → C, defined by the rule f(x) = *positive* square-root of x.

Let, for x in Z, with f(x) = 3.75, i. e. positive-square-root of x = 3.75. Therefore, x = $(3.75)^2$, which is not in Z. Hence, **f: Z→ C** *is not surjective*

A function $f: X \to Y$ is said to be onto or **surjective** if each element of Y has at least one pre image in X.
i.e. for each y ∈ Y, there exists at least one $x \in X$ such that
$f(x) = y$

If we compare this definition with example of daughters and mothers then onto function means each mother must have at least one daughter.

For example,

(i)

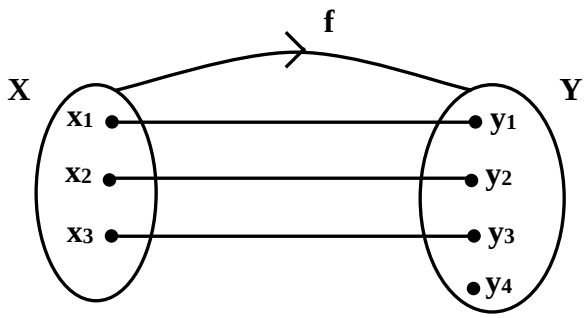


Fig. 3.14: Not an onto function

The function f shown in Fig. 3.14 is not onto function because $y_4 \in Y$ but there is no $x \in X$ such that: $y_4 = f(x)$

(ii)

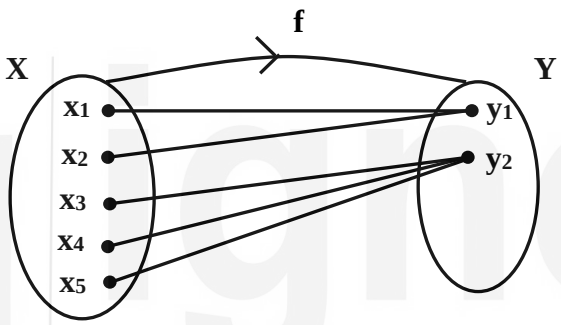


Fig. 3.15: An onto function

The function f shown in Fig. 3.15 is onto function because each element of Y has at least one pre image, i.e. y_1 has two pre images and y_2 has three pre images.

Remark 4: If we want to show that a function $f(x)$ is onto then first we take an element y in Y and we have to show that there exists an element x is X such that $f(x) = y$

For example, show that the function $f: R \to R$ defined by $f(x) = 7x + 5$ is onto function.

Solution: Here X = R, Y = R

Let $y \in Y = R$, then $\frac{y-5}{7} \in X = R$ s.t.

$f\left(\frac{y-5}{7}\right) = 7\left(\frac{y-5}{7}\right) + 5 = (y-5) + 5 = y$

$\therefore f$ is an onto function.

3.5.3 Bijective (One-One and Onto) Function

A function f: X → Y is called Bijective if and only if it is both injective and surjective. A bijective function from X to Y is also called ***1-to-1 correspondence*** between X and Y.

Bijective Mapping: A Member f: A → B is said to be bijective (or one-one onto,) if f is both injective and surjective, i.e., one-one as well as onto.

For example, f: Z→Z : f(x) = x+2, x ∈ Z is both injective and surjective. So, f is bijective.

Let us take another example: f(x)=x−4.

This function is one-to-one (different inputs give different outputs) and onto (any real number output y can be achieved with the input x=y+4), so, f is bijective.

A function $f: X \to Y$ is said to be one-one and onto or **bijective** or **one-one correspondence** if:

(i) f is one-one

(ii) f is onto

If we compare this definition with example of daughters and mothers then one-one and onto function means each mother have exactly one daughter and there is no mother who does not have any daughter. The function f shown in Fig.3.16 represents a situation of one-one and onto function.

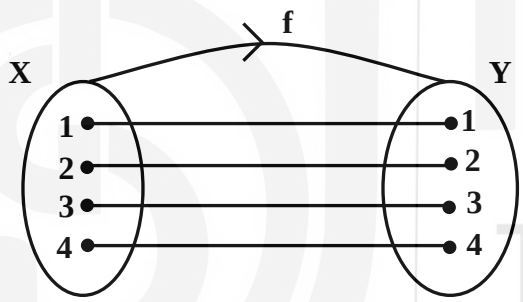


Fig. 3.16: one – one and onto function

3.5.4 Geometrical Meaning of Injective, Surjective and Bijective Functions

One-One Function

If a function is 1-1 then geometrically it will satisfy the following condition. Each horizontal line either does not intersect the graph of the function or if it intersects it will intersect exactly at one point.

For example,

(i) Graph shown in Fig. 3.17 (b) is the graph of a one-one function because each horizontal line either does not intersect the graph or if it intersects, it will intersect exactly at one point, i.e. each horizontal line above x-axis will intersect the graph exactly at one point and each horizontal line below x- axis not intersect the graph at all.

(ii) Graph shown in the Fig. 3.17 (c) is also the graph of a one-one function because each horizontal line intersects the graph exactly at one point.

(i) But the graph shown in the Fig. 3.17 (a) is not the graph of a one-one function because if we draw any horizontal line below the x-axis, then it will intersect the graph at two points. Similarly graph shown in Fig. 3.17 (d) is also not a one-one function.

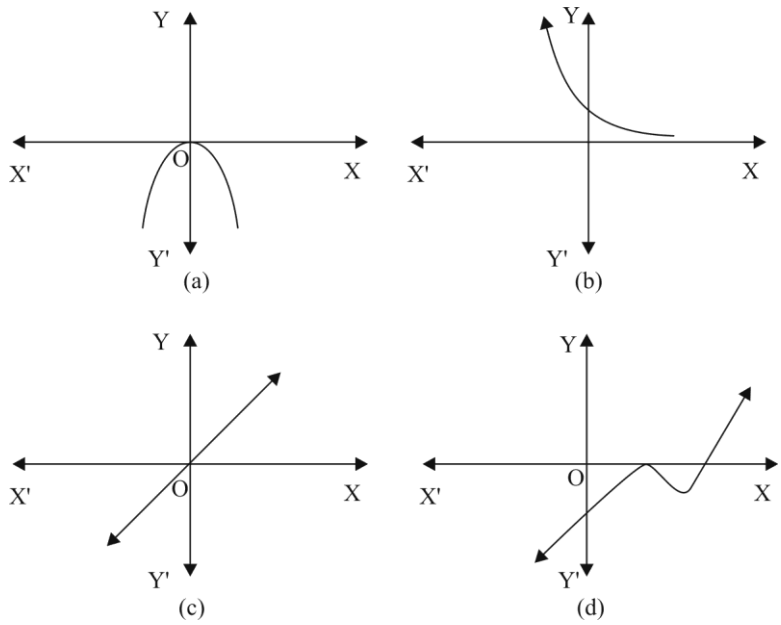


Fig. 3.17: Some functions

- **Onto or Surjective Function**

If a function is onto then geometrically it will satisfy the following condition. Each horizontal line must intersect the graph of the function at least at one point.

For example,

(i) Graph shown in the Fig. 3.17 (d) is the graph of an onto function because each horizontal line intersect the graph of the function at least at one point. Graph shown in Fig. 3.17 (c) is also onto function because each horizontal line intersects the graph exactly at one point.

(ii) But the graph shown in the Fig. 3.17 (b) is not the graph of a surjective function because if we draw any horizontal line below the x-axis, then it will not intersect the graph.

(iii) Similarly, the graph shown in the Fig. 3.17 (a) is not the graph of an onto function because if we draw any horizontal line above the x-axis, then it will not intersect the graph.

- **Bijective Function**

If a function is bijective or one-one and onto or one-one correspondence, then each horizontal line must intersect the graph of the function exactly at one point.

For example,

(i) Graph shown in the Fig. 3.17 (c) is the graph of a bijective function because each horizontal line intersects the graph of the function exactly at one point.

(ii) But the graphs shown in the Fig. 3.17 (a) and (b) are not the graphs of a bijective function because they are not onto.

(iii) Similarly, the graph shown in the Fig. 3.17 (d) is not the graph of a bijective function because it is not one-one.

3.6 INVERSE OF FUNCTIONS

Purpose of an **inverse of a function** is to **"reverses" or "undo "** the action of another function. For example if you take an input, apply a rule/function (f(x)) to it, and then apply the inverse function ($f^{-1}(x)$) to the output, you get your original input back. Here function f(x), its inverse is denoted as $f^{-1}(x)$. Here it is very important to note that the "-1" is not an exponent but it is just notation for the inverse.

The key idea while using inverse function is that if f(a) = b, then $f^{-1}(b) = a$.

Example:

Let us say the function f(x)=3x+5 takes an input, triple it, and adds 5.

Then for x = 2; f(2)=3(2)+5=11

Then no need to say that inverse function, $f^{-1}(x)$, must undo this.

Let us check

It would subtract 5 and then divide by 3.

$f^{-1}(x)= (x-5)/3$

Now let us check it with the output 11:

$f^{-1}(11) = (11-5)/3 = 2$ which is our original input.

Now let us see the formal definition of inverse function.

Definition: Let f: A→B be a bijective mapping. Then the mapping g: B→ A which associates to each element b ∈ B the unique element a ∈ A, such that f(a) = b, is called the inverse mapping of the mapping f: A→B.

Remember, by definition a function to have an inverse, it must be **bijective**, meaning it is both **one-to-one** (each output comes from only one input) and **onto** (it covers all possible outputs).

Hence, if f: A→B is a one-one onto mapping, then f^{-1} : B→A exists, and is also 1-to-1.

For example, A= R~{3} and B= R(1}, and the function f: A→B is defined by f(x) = (x-2)/(x-3) .

We can see that f is a one-to-one and onto function.

∴ f inverse exists.

To find $f^{-1}(x)$ the following steps are required:

1. Replace f(x) by y in the equation describing the function.

Here you will get $y = (x-2)/(x-3)$.

2. Interchange x and y. In other words, replace every x by y, and vice versa.

Here you will get $x = (y-2)/(y-3)$.

3. Solve for y.

4 Replace y by $f^{-1}(x)$

By applying these steps, we get $f^{-1}(x) = (3x-2)/(x-1)$.

Example:
Let's find the inverse of the function f(x)=3x+5 (which we have used above for verification the concept)

Step 1: Replace f(x) with y.
$y = 3x+5$
Step 2: Swap x and y.
$x = 3y+5$
Step 3: Solve for y. Subtract 5 from both sides:
$x-5 = 3y$
Divide both sides by 3:
$(x-5)/3 = (3y)/3$
$y = (x-5)/3$
Step 4: Replace y with $f^{-1}(x)$.
$f^{-1}(x) = (x-5)/3$
The inverse of $f(x) = 3x+5$ is $f^{-1}(x) = (x-5)/3$.

The concept of inverse function is used in cryptography & security (encryption and decryption), solving mathematical equations, undo functionality in general computer applications (command (Ctrl+Z) in software is an application of an inverse function), computer graphics etc.

Check Your Progress -3

1. Which of the following kinds of function would you use to provide photo identity numbers and why?

i) Constant function, ii) one-to-one function, and iii) identity function.

……
……

2. Explain why $f : Z \rightarrow Z$: $f(x) = x^2$ is onto? Domain and range of f is Z.

……
……....

3. Explain whether function defined by $f(x) = x^2 - 2x + 1$ is onto function or not?

.....
.....

4. Find f^{-1} of rule $f : f(x) = x^3 - 3$.

.....
.....

5. If $f: R \rightarrow R$ be defined by $f(x) = x^2 + 1$ then find $f^{-1}(17)$

.....
.....

3.7 SUMMARY

This unit provides a foundation of Functions, beginning with basic definitions like quantities and intervals before moving into the mechanics of how sets interact. This unit explains various ways to represent functions—whether through mapping diagrams, pictorial graphs, or algebraic operations and also introduces monotonicity, which describes whether a function consistently increases or decreases. Further, this unit focuses on the classification and reversibility of functions based on their mapping properties. It distinguishes between injective (one-to-one), surjective (onto), and bijective functions, using both algebraic definitions and geometric interpretations. Finally, the unit concludes by explaining how to find the inverse of a function.

3.8 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

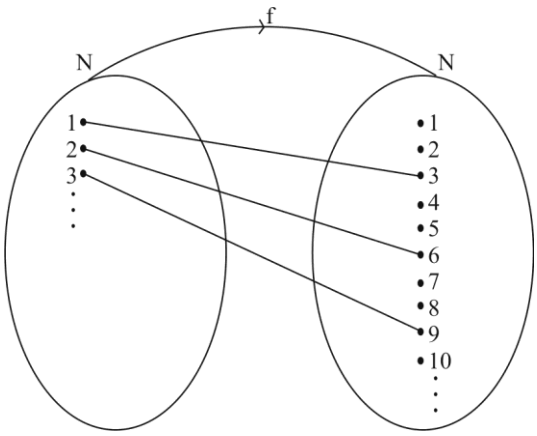
Answer to Check Your Progress- 1

1.

$f : N \rightarrow N$ defined by

$f(x) = 3n, \quad n \in N$

$\therefore f(1) = 3, \; f(2) = 6, \; f(3) = 9$, and so on. See Fig. 3.6



Domain of the function $f = \{1, 2, 3, ...\} = N$
Range of the function f = Set of only those values for which function is define = {3, 6, 9, …}
Co-domain = Set of all values of Y = {1, 2, 3, …}= N

2. $f : R \to R$ is defined by

$$f(x) = 3^x, \qquad x \in R$$

(ii)Since $f(x)$ is defined for all $x \in R$

$\therefore$ domain of $f = R =$ set of all real numbers

(iii) We know that

$$3^x > 0 \qquad \forall\, x \in R$$

i.e. $f(x) > 0 \qquad \forall\, x \in R$

$\therefore$ range of f = $(0, \infty)$

3.

(i) Domain = R, Range = $\{x \in R \text{ and } x \le 0\}$.
(ii) Domain = [0,3], Range = [0,3].

4. i) f: N→N is not a function, as f(0) = -3 ∉ N, the codomain of f.
ii) f: N→Z, is a function.

5. Given function is

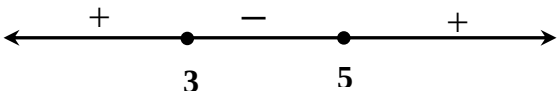
$$f(x) = \sqrt{(x-3)(5-x)}, \qquad x \in R$$

For $f(x)$ to be real, the quantity under the square root should be non negative and hence

$(x-3)(5-x) \ge 0$

$\Rightarrow -(x-3)(x-5) \ge 0$

$\Rightarrow (x-3)(x-5) \le 0$



Now, when we take x less than 3, the L.H.S. comes out to be (–ve) (–ve) = + ve, hence does not satisfy the inequality.
Also, if we take x greater than 5, the L.H.S. comes out to be (+ ve) (+ ve) = + ve, and hence this value does not satisfy the inequality.
But, if we take 3 < x < 5, the L.H.S. becomes (+ ve) (–ve) = –ve and hence satisfies the inequality.
$\Rightarrow x \in [3, 5]$

$\therefore$ domain of $f = [3, 5]$

Also,

$f(3) = \sqrt{(3-3)(5-3)} = \sqrt{0 \times 2} = \sqrt{0} = 0$

$f(4) = \sqrt{(4-3)(5-4)} = \sqrt{1 \times 1} = 1$

$f(5) = \sqrt{(5-3)(5-5)} = \sqrt{2 \times 0} = \sqrt{0} = 0$

6. Here f(x) takes real values for all $x \in R$, except at x = –3

$\therefore$ domain of f = R – {– 3}

and f(x) cannot takes zero values at any real number,

$\therefore$ range of f = R– {0}

Also, $f(0) = \frac{1}{0+3} = \frac{1}{3}$, $f(2) = \frac{1}{2+3} = \frac{1}{5}$

f(– 3) is not defined because x = -3 is not a point of the domain of f.

7. Not every relation is a function. For example, this relation does not satisfy the property that:

Each element of A must have been assigned to one element in B.

If a ∈ A is assigned b ∈ B and a ∈ A is assigned b´ ∈ B, then b = b´

That is why relations that don't satisfy the above properties are not a function

8. We can see that the code has no effect on the value of n < 0. In the While loop, the value of n is halved whenever it is even. If n becomes odd before reaching 1, the second part of the while loop is invoked, and n remains odd and increases forever.

This shows that f : N → N is the function defined by :

f(n) = n/2 if n is even and 2n+1 if n is odd

Note : for f(1) the loop will not be executive , infect n > 1 is the condition for loop execution .

Answer to Check Your Progress- 2

1.

i) (f+g) (x)= $1/x + x^3 + 2$

ii) (f – g) (x)= $1/x - (x^3 + 2)$

iii) (f.g) (x)= $(1/x)(x^3 + 2)$

iv) (f/g) (x)= $1/(x(x^3+2))$ ∀x ∈R.

2.

i) (f+g) (x) = $\sqrt{(x+1)} + x^3$ ∀x $\geq$ -1

ii) (f-g) (x) = $\sqrt{(x+1)} - x^3$ ∀x $\geq$ -1

iii) (f.g) (x) = $(\sqrt{(x+1)}).\ x^3$ ∀x $\geq$ -1

iv) (f/g) (x) = $\sqrt{(x+1)}/x^3$ ∀x $\geq$ -1, x $\neq$ 0

v) (f o g)(x) = $f(x^3) = \sqrt{(x^3+1)}$ ∀x $\geq$ -1

3. Let us find the composite functions $(f \circ g \circ h)(x)$ and then $(g \circ f)(x)$.

Given:

$f(x)=x^2$

$g(x)=x+2$

$h(x)=x^3+x+2$

A. Calculating $(g \circ f)(x)$:

The composition $(g \circ f)(x)$ means we apply function f first, and then apply function g to the result.

We will start with the outer function: $g(x)=x+2$

1. Substitute f(x) into g(x) in place of x: $g(f(x))=f(x)+2$
2. Replace f(x) with its definition, x^2: $g(f(x))=x^2+2$
3. So, the final result is: $(g \circ f)(x)=x^2+2$

B. Calculation of $(f \circ g \circ h)(x)$

This composition means we work from the inside out: first h(x), then g(x), and finally f(x). It's calculated as f(g(h(x))).

i) First, find the inner composition, g(h(x)): $g(h(x)) = h(x)+2$

Substitute the definition of h(x): $g(h(x)) = (x^3+x+2)+2$

$g(h(x)) = x^3+x+4$

ii) Now, substitute this result into the outer function, f(x):

$f(g(h(x))) = (g(h(x)))^2$

Replace g(h(x)) with the expression we just found:

$f(g(h(x))) = (x^3+x+4)^2$

So, the composite function is: $(f \circ g \circ h)(x) = (x^3+x+4)^2$

Answer to Check Your Progress- 3

1. One-to-one function will be used for providing identity card number, because each person must have unique identity numbers.

2. Function $f(x) = x^2$ is one-to-one because for every value of x, x^2 will be a number that is different for different x. Hence, $f(x) = x^2$ is one-one mapping.

3. The function $f(x)=x2-2x+1$ is **not** an onto function. Because a function is **onto** (or surjective) if its **range** is equal to its **codomain**. For this function, the codomain is typically assumed to be all real numbers, R. However, the range of this function is restricted.

This function can be simplified by recognising that it is a perfect square:

$f(x)=x^2-2x+1=(x-1)^2$

Since the square of any real number **R** is always non-negative, the output of this function can never be a negative number. The minimum value of f(x) is 0, which occurs when x=1.

Therefore, the **range** of the function is $[0,\infty)$, which is not equal to the codomain of **R.** For instance, there is no real number x for which $f(x) = -9$

4. Step 1: $y = x^3 - 3$

Step 2: $x = y^3 - 3$

Step 3: $y = \sqrt[3]{(x+3)}$

Step 4: $f^{-1}(x) = \sqrt[3]{(x+3)}$

5. To find the value of $f^{-1}(17)$, we need to determine which input value, x, results in an output of 17 for the function f(x)=x2+1.
In other words, we need to solve the equation: f(x)=17
Step-by-Step Calculation

1. Set up the equation: Substitute the function definition into the equation: x2+1 = 17
2. Isolate the x2 term: Subtract 1 from both sides of the

   equation: $x^2 = 17-1 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$

   Hence $x = 4$ or $x = -4$

The values of x for which f(x)=17 are 4 and -4.Therefore, $f-1(17)=\{4,-4\}$.
It's important to note that since two different input values (4 and -4) produce the same output (17), the function $f(x)=x^2+1$ is not a one-to-one function over the real numbers, and therefore **does not have a true inverse function** unless its domain is restricted. However, the set of values that map to 17 is $\{4,-4\}$.

3.9 REFFERENCES AND FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MCS-013: Discrete Mathematics
   Unit-1 Sets, Relations and Functions
   (https://egyankosh.ac.in/handle/123456789/1140)
5. MEC-203 : Quantitative Methods
   Unit-3 Relations and Functions
   (https://egyankosh.ac.in/handle/123456789/98141)
6. MST-001: Foundation in Mathematics and Statistics
   Unit-2 Functions
   (https://egyankosh.ac.in/handle/123456789/92122)

UNIT 4 GRAPHICAL REPRESENTATION OF FUNCTIONS

Structure

4.0 Introduction
4.1 Objectives
4.2 Vertical Line Test of Function
4.2.1 Relationship between Equation and Function
4.2.2 Co-ordinate Geometry
4.2.3 Cartesian Co-ordinate System
4.3 Translating Geometric Figures to Algebraic Equations
4.4 Graphing Linear Functions
4.4.1 Absolute Value Function
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4.5 Graphing Non-Linear Functions
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4.5.2 Quadratic Functions
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4.10 Answer to Check Your Progress
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4.0 INTRODUCTION

From the previous unit, we know that a function is a relationship between its domain (X, in Fig. 4.1a) and co-domain (Y, in fig. 4.1a) such that every element of the domain is related to exactly one element of the co-domain/range. On this ground, Fig. 4.1a does not represent a function (or functional relationship) on each of the following two grounds:

i) The element 5 of the domain (X) is related to three elements, viz. 4, 5, 7 of the co-domain.

ii) each of the elements 3 and 4 of domain X is not related to any element of the co-domain Y.

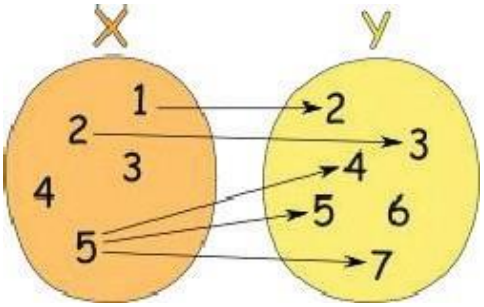


Fig.4.l a: A Non-Functional Relationship

However, for a relationship to be a function, there is *no such restriction* on the elements of *co-domain* Y. For example, Fig. 4.lb represents a function or functional relationship, even when

i) element C of co-domain Y corresponds to two elements viz. 2 and 3 of domain X, and

ii) element A (or even B) of co-domain Y is not related to any element of domain X.

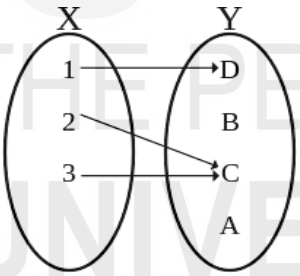


Fig.4.l b: A Functional Relationship

4.1 OBJECTIVES

After going through this unit, you should be able to

- Apply vertical line test to identify anon-function;
- explain how co-ordinate geometry differs from the conventional/synthetic geometry;
- obtain algebraic equations corresponding to simple geometric figures;
- graphically represent various types of functions of a single variable; and
- handle the graphs of two variable functions at a preliminary level.

4.2 VERTICAL LINE TEST OF FUNCTION

The **vertical line test** for graphically testing***(only) failure*** of a relationship f(x) to be a function of x indicates that the vertical line crosses the graph of f(x) at two distinct points (Fig 4.2).

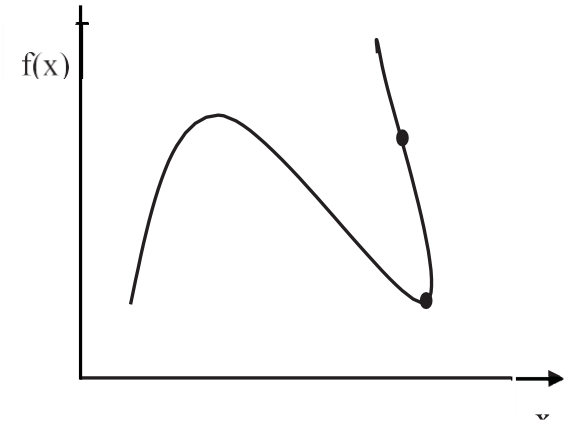


Fig. 4.2: Graph that does not Represent a Function

Thus, using vertical-line text we can identify the graphs that are not functions.

For example, you can show that each of the following graphs is not a function.

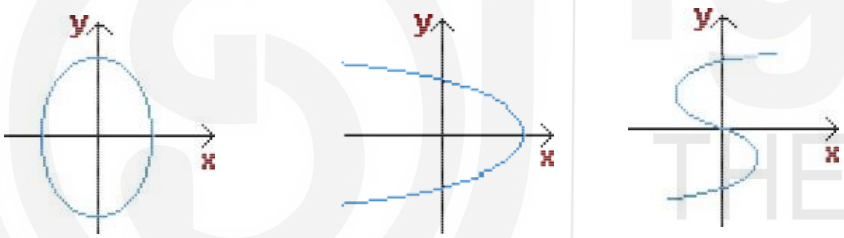


Fig.4.3: Graph of Not-Function

But the graph shown in Fig 4.4 below represents a function. Thus, some of the important equations such as ellipse, circle and parabola may not represent a function.

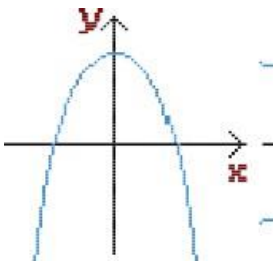


Fig.4.4: Graph of a Function

One must note that the above test can be used *to test whether a relationship is* functional. It cannot be used to test (or confirm) whether a relationship is functional. There are other additional requirements for a relationship to be functional.

4.2.1 Relationship between Equation and Function

Before we discuss functions in detail, we need to be clear about the possible relationship between an equation and a relation/function.

Consider the equation

$X^2+Y^2=4$ …… (1)

Then to talk about mathematical relation/function, we rewrite the above equation so that one of the variables, say Y is written on L.H.S. and everything else is shifted to R.H.S. as follows:

$Y^2 = 4 - X^2$ ……. (2)

Next step to talk about a relation/ function corresponding to Equation (1) is to rewrite (2), so that only Y (and not its power, or square-root etc.) is written on L.H.S. as follows:

$Y = \pm\sqrt{(4 - X^2)}$ ……. (3)

At this stage, after reducing the L.H.S. to a single variable, with the power of the variable as exactly 1, we can talk of a (mathematical) relation/function. As (3) does not satisfy single-value rule, hence, equation (1) does not correspond to a function. Of course, it corresponds to a relation. Equation (3) is obtained to make Y as dependent variable, and X as independent variable. By interchanging the roles of X and Y in the above process, we can make X as dependent variable, and Y as independent variable. Similarly, by taking the following equation in three variables

$X^2 + Y^2 + Z^2 = 4$ …… (4)

Through the similar process, we may rewrite the Equation (4)

as: $Z = \pm\sqrt{(4 - X^2 - Y^2)}$ …….. (5)

At this stage, we may talk of Z being a function or not, and this time, function of two independent variables, viz. X and Y. Again (5) does not represent a function.

4.2.2 Co-ordinate Geometry

Co-ordinate Geometry is an *approach* to investigating and solving problems in the domain of geometric objects such as straight lines, triangles, planes, cubes, etc. The ***approach consists in representing geometric objects as Algebraic expressions and equations*** like $a x + b y + c = 0$ for a straight line, and $x^2 + y^2 = c$ for a circle etc. And then use algebraic tools for making inferences and solving problems. The process of translating geometric objects to algebraic expressions are achieved using ***co-ordinate systems.*** There are several different co-ordinate systems, each more useful than the others for some specific purposes. We will discuss some of these briefly. However, before that we mention some synonymous and related terms, and a little bit of history.

Co-ordinate geometry is also called ***Analytical geometry,*** or ***Cartesian geometry***

(in honor of *Rene Descartes*,17th century French philosopher and mathematician). Though evidence of the use of co-ordinates in geometry is found in the work of earlier mathematicians, particularly some Greeks and the 11th-century Persian mathematician Omar Khayyam, yet the discipline in its current form was invented independently by two 17th-century mathematicians, viz. Rene Descartes and Pierre de Fermat.

Earlier forms of geometry, which have been practised for thousands of years, were retrospectively called, since the 17th century, synthetic geometry, which uses, instead of algebraic tools, the axiomatic method (based on proof techniques) to prove results from a few basic properties called postulates/axioms.

From our experience with school mathematics, we know that in 2-dimensional space, the Cartesian coordinate system uses two intersecting, mutually perpendicular real lines, as shown in Fig. 4.5.

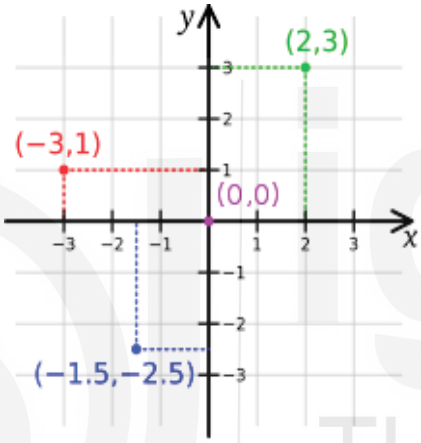


Fig 4.5: Two-Dimensional *Cartesian* Co-ordinate System

Though, we will be using mainly cartesian co-ordinate system, yet we should know the names of other co-ordinate systems which include ***polar co-ordinate system, cylindrical & spherical co-ordinate system, homogeneous co-ordinate system and Curvilinear co-ordinate system.***

4.2.3 Cartesian Co-ordinate System

In the rest of the discussion in the unit, unless mentioned otherwise, the term ' ***co-ordinate system****' will be used for* ***cartesian co-ordinate system,*** which is also called ***rectilinear co-ordinate system.***

In this co-ordinate system for 2-dimensional space, two mutually perpendicular intersecting lines ***(called axes),*** as shown in the Fig. 4.6 below, form the basis of *rectangular co-ordinate system for 2-dimension*

1) The horizontal number line is called the ***x*- axis.**

2) The vertical number line is called the ***y*-axis.**

The **origin** is where the two lines/axes intersect. (See Fig.4.6). It has four **Quadrants** marked with Roman numerals.

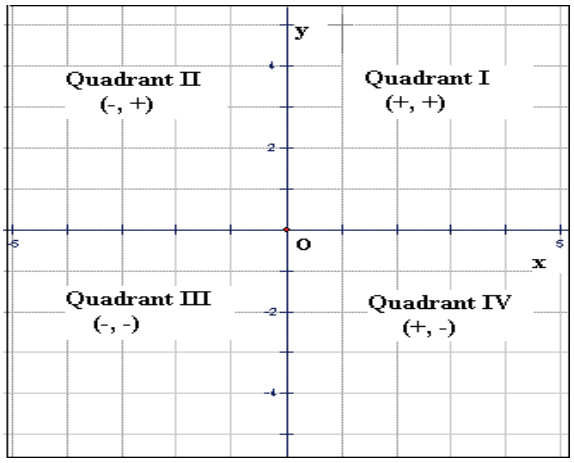


Fig. 4.6: Rectangular Co-ordinate System

Each point on the graph corresponds to an **ordered pair**. *At the origin*, it is assumed that value of x = 0 = value of y. when dealing with an (x, y) graph, the ***x* co-ordinate is** always written first and the ***y* co-ordinate** as second in the ordered pair ***(x, y)***.

Normally, the values of the **independent** variable (generally the ***x*-values**) are placed on the horizontal axis. In contrast, the values of the **dependent** variable (generally the ***y*-values**) are placed on the vertical axis.

For a point P (x, y), the *x*-value, called the **abscissa,** is the perpendicular distance of *P* from the *y*-axis, and the *y*-value, called the **ordinate,** is the perpendicular distance of *P* from the *x*-axis.

The values of *x* and *y* together, written as (*x*, *y*) are called the **co-ordinates** of the point *P*. The centre of the co-ordinate system (where the two lines intersect) is called the **origin.**

Such a formulation is called the ***"rectangular" co-ordinate system*** because the scale used along the *x*-axis is evenly spaced, as is the scale along the *y*-axis.

4.3 TRANSLATING GEOMETRIC FIGURES TO ALGEBRAIC EQUATIONS

For translating a geometric figure, ***say, a straight line,*** we create a co-ordinate system around the object, or, equivalently, the geometric figure is embedded in a co-ordinate system as shown in Fig. 4.7. How and where to place the geometric figure in the co-ordinate system depends on the problem under consideration. Each point in Fig 4.7, has co-ordinates, as one of the points, say A, in the figure above, has co-ordinates (2, 0), with 2 as abscissa and 0 as ordinate, and another point, say B, has co-ordinates (0, -2).

To represent a geometric figure by an algebraic equation, we take an arbitrary point in Fig 4.7, assuming its co-ordinates as (x, y) with x as abscissa and y as ordinate. Then an ***algebraic equation*** representing the figure is some relationship (to be determined) between x and y.

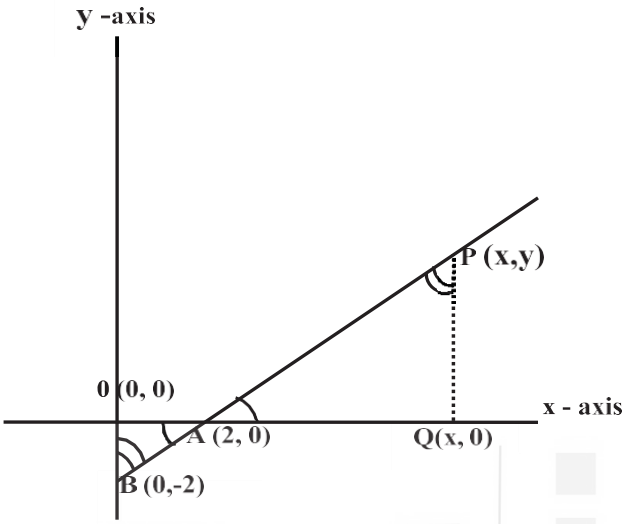


Fig. 4.7: Coordinate System

To **derive the algebraic equation** corresponding to Fig. 4.7, let us take P (x, y) outside the segment A (0, -2)and B (2, 0), and drop a perpendicular PQ on x-axis. Then triangles AOB and AQP are similar. From conventional/synthetic geometry, we know:

$|AO| / |AQ| = |BO| / |PQ|$, ... (i)

Where, $|X|$ denotes length of line segment X.

Now
$|AO| = 2$, $|AQ| = x-2$, $|BO| = 2$, $|PQ| = y$ ...(ii)

From (i) and (ii) we get $2/(x-2) = 2/y$, implying

$y = x-2$ is the algebraic translation of the Figure 4.7.

Using similar method, we may obtain algebraic equations corresponding to the well-known curves, including the following standard algebraic forms

- **$x^2+y^2=a^2$ for Circle**

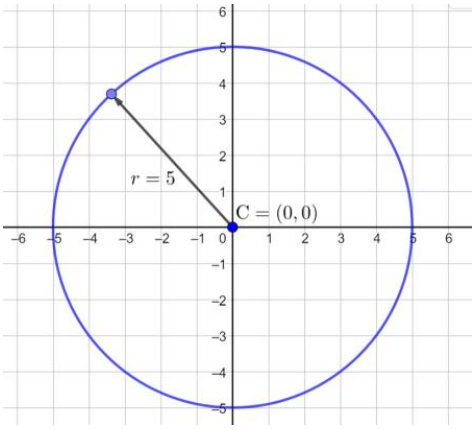


Fig. 4.8: Circle : $x^2+y^2=5^2$

- **$(x/a)^2 + (y/b)^2 = 1$ for Ellipse**

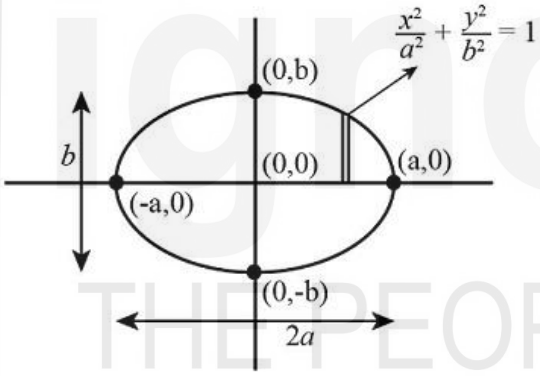


Fig. 4.9: Ellipse : $(x/a)^2 + (y/b)^2 = 1$

- **$y^2 = 4ax$, with a> 0 for Parabola**

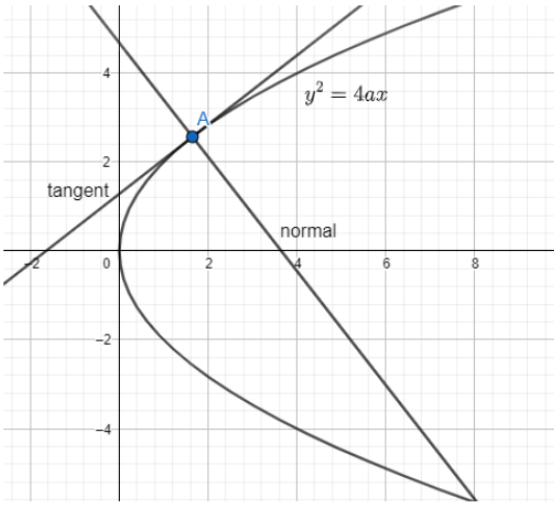


Fig. 4.10: Parabola : $y^2 = 4ax$, with a> 0

- **$(x)^2 - (y)^2 = 1$ for Hyperbola**

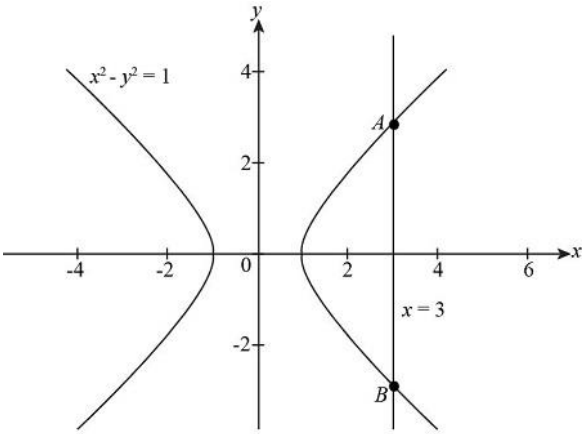


Fig. 4.11 : Hyperbola : $(x)^2 - (y)^2 = 1$

- **$xy=c^2$ for Rectangular Hyperbola.**

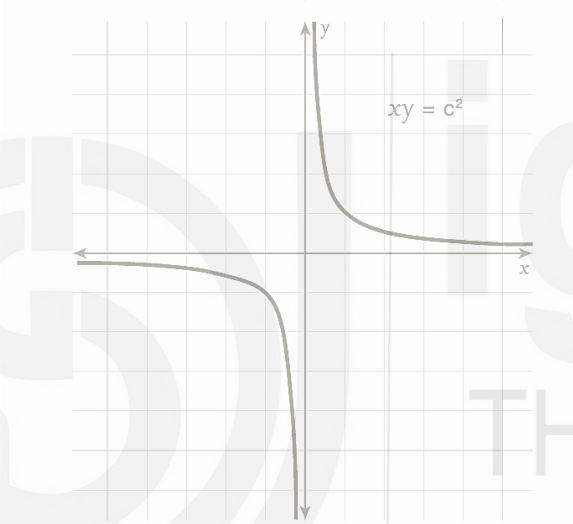


Fig. 4.12 : Rectangular Hyperbola : $xy=c^2$

4.4 GRAPHINGLINEARFUNCTIONS

A function *y =f(x)*, is called ***linear*** if the power of x in f(x) is not more than 1. A linear *function* corresponds to a linear *equation* A x + B y + C = 0, in which degrees of x and y do not exceed 1, and A, B, C are constants. Where at least one of A, B ≠ 0.

To plot a graph of a function, we take, one by one, values in the domain (x_i) as the independent variable x, and then find the corresponding values in the range y_i of the dependent variable y. It helps record the x and y values while plotting the graphs. For example, consider a linear function depicting the relationship between price (*x*) and market demand (y) for a commodity. To plot the graph for evaluating the relationship, it is convenient to first prepare a table, called a t-chart, comprising the values of x and y. If a function takes the form of *y=7-5*x, then *t*-chartis as under:

Table 4.1: t-chart of y=7-5x

x	**y=7-5x**
-1	12
0	7
1	2
2	-3
3	-8

In general, we may need to take many values of the independent variable x and find the corresponding values of y. But in the *case of functions*, we need to take ***only two*** values of the independent variable x and calculate two corresponding values of the dependent variable y. Thus, to draw the required graph, we ***need only*** the following part of the table :

X	Y = 7-5X
-1	12
0	7

The graph of such a function is as follows (refer Fig. 4.13):

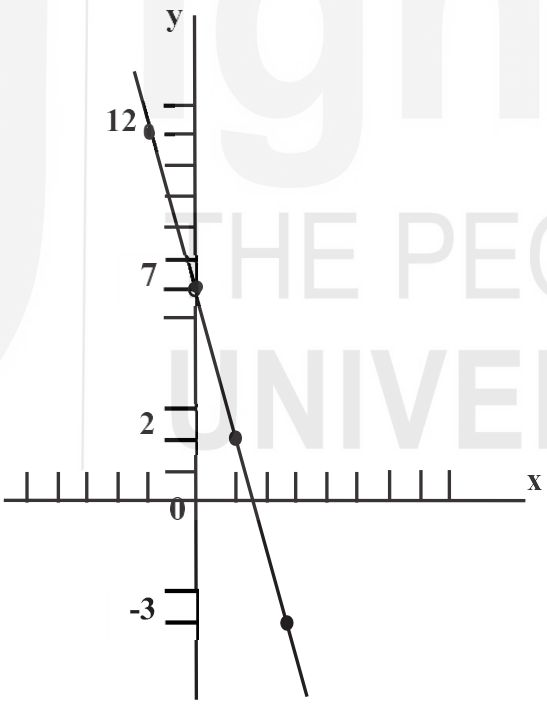


Fig.4.13:Graph of Function *y*=7 - 5x

The graph has ***x-intercept*** on a point where y is zero and ***y-intercept*** on a point where x is zero. For example, to find the x and y intercepts of the equation $3x+4y=12$ proceed as follows:

To find the x-intercept, set y=0 and solve for x. That is, $3x+4(0) = 12$ *or*, $x = 12/3=4$. Similarly, to find the y-intercept, set x =0 and solve for y.

That is, 3(0) + 4y=12 *or*, y=12/4=3.Thus, the x-intercept is (4,0) and the y-

intercept is (0,3).

As a linear equation represents a straight line, and a straight line is specified by only two of its point. Hence, the graph of the line is as shown in Fig. 4 .14.

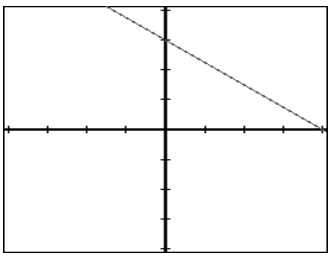


Fig.4.14: Graph of 3x+4y=12

4.4.1 Absolute Value Function

Absolute Value Function is given as f(x) = | x | indicating that we need to consider modulus | x | of a real number keeping it as the non-negative value without regard to its sign. That is, $|x| = x$ for a positive x, $|x| = -x$ for a negative x (in which case $-x$ is positive), and $|0| = 0$.

Thus, the function $y = |x|$, is defined by the following two equations

$y = |x| = x$ for $x \geq 0$, …..(1)

and $y = |x| = -x$ for $x < 0$ …..(2)

As each of (1) and (2) is a linear equation, each is represented by a straight line as shown in Fig. 4.15 below:

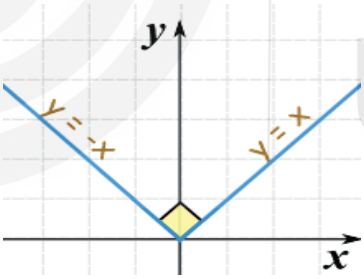


Fig.4.15: Graph of Absolute Value

However, for some functions, the x-intercept & y-intercept method may not yield the required result, because the required curve may not intersect x-axis or y-axis.

Example: f is a function given by f (x) = $|x|$ + 2. The y interceptis given by (0, $f(0)$) = (0, |-2|) =(0, 2); x intercept is at the point (2,0)since we solve for $|x-2|$ =0.But 0 =y=f(x)= $|x-2|$=0 has no solution.

Since $|x-2|$ is either positive or zero for x=2; the domain of f is the set of all real numbers and the range of f is given by the interval $[0,+\infty)$.The corresponding graph is shown in Fig 4.16.

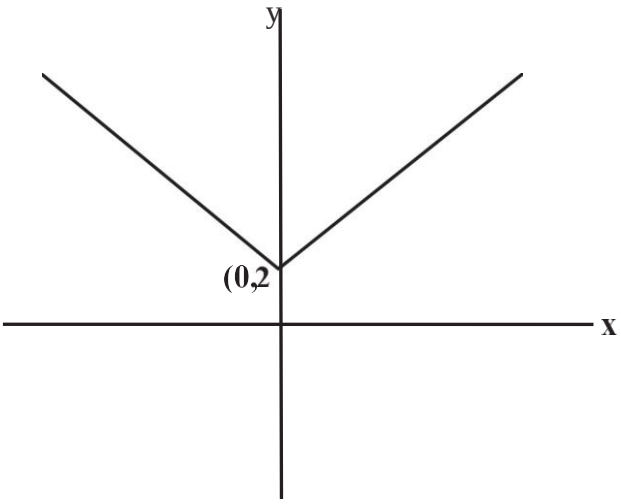


Fig.4.16: Graph of f(*x*)=|*x*|+2.

In general, for drawing a graph for a linear function, we need ***just two*** *pairs* (x_i, y_i) i =1, 2 of values of (x, y).

4.4.2 Step Function

A step function (or staircase function) is a piecewise function containing all constant "pieces" . The constant pieces are observed across the adjacent intervals of the function, as they change value from one interval to the next. A step function is discontinuous (not continuous) as you can see in the Fig. 4.17. You cannot draw a step function without removing your pencil from your paper.

Example of a step function: For definition of our particular step function y=f(x),

First, we write every x = Int - x + Fraction-x, where Fraction-$x \geq 0$.

E.g., for x = 5.46, Int-x =5, and Fraction-x = .46

For x = - 3.87, Int-x = -4, and Fraction-x = .13

Then we define y = f(x) = Int-x

Thus, f (5.46) = 5, and f (-3.87) = -4.

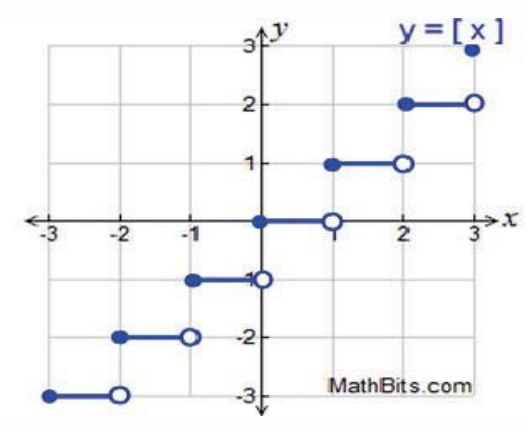


Fig.4.17: Graph of Step Function

Note: The *domain* of a function is the set of all inputs for which the function is defined. For piece wise functions, the domain is union of all the individual cases. The *range* of a function is the set of all the possible function outputs. For piecewise functions, this is the union of the ranges, so fall back to the individual cases.

Check your Progress -1

1) Draw the graph of the equation $y^2 = x+5$ and comment if it is a function.

……

……

……

……

2) Find algebraic equation corresponding to a circle of radius r.

……

……

……

3) Draw the graph of $y = (-5/3)\,x-2$.

……

……

……

4) Find when $f(x)=2|3x+2|-12$ crosses the x-axis.

……

……

…….

5) A truck rental company, charges a rental fee of Rs. 25 plus Rs. 0.30 per mile for driving between 0 and 500 miles for the standard –sized truck. For the same truck, the company charges Rs. 100 plus Rs. 0.15 per mile for driving more than 500 miles, but less than 1000 miles. Write a step function that represents the situation.

…….

…….

…….

4.5 GRAPHING NON-LINEAR (CURVE TYPE) FUNCTIONS

Determining the nature of the functions that produces non-linear type curves requires additional considerations compared to those of linear types. When you graphed straight lines, you only needed two points to graph the line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly *not* enough/sufficient for graphing curves of quadratic or cubic functions. We discuss the caution to be exercised for sketching the graphs of such function.

4.5.1 Even and Odd Functions

A function ***f*** is **even** if its graph is symmetric with respect to the *y - axis*. This criterion can be stated algebraically as follows: *f* is even if $f(-x)=f(x)$ for all x in the domain off. For example, if you evaluate at 3 and at-3, then you will get the same value, if *f* is even. Geometrically, such functions are symmetric about the *y-axis*.

We have seen that for an even function $f(x)=f(-x)$ for all x ,there is symmetry about y-axis. The graph, in this case is shown in Fig 4.18.

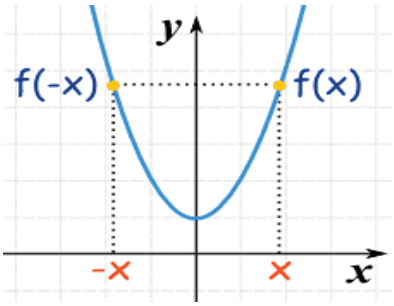


Fig.4.18: Graph of Even Function

A function is **"odd"** when f(x) = - f (-x) for all x and symmetry is observed from the point of origin (Fig. 4.19).

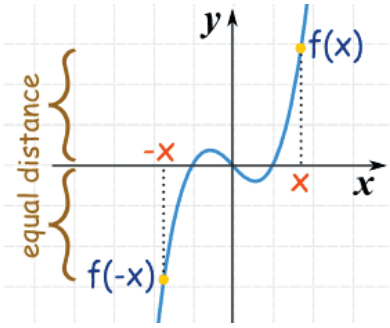


Fig. 4.19: Graph of Odd Function

Note: The graph based on $f(x)=x^3-x$ is an odd function. It is symmetrical about the origin.

4.5.2 Quadratic Functions

Quadratic functions help describing demand, cost, revenue and profit which you will find while covering microeconomic analysis. The general technique for graphing quadratics is the same as **for graphing linear functions.** However, since quadratic graphs are represented by curvy lines (for example, parabolas, ellipses etc.), rather than the straight lines generated by line are equations, there are some additional considerations.

The most basic quadratic is $y = x^2$. When you graphed straight lines, you only needed two points to graph your line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly not be enough points for graphing a quadratic. For example, compute these three points:

Table 4.2: T-chart of $y=x^2$

X	$y=x^2$
0	0
1	1
2	4

Then, based only on this experiment, the plotted graph will produce a straight line. Thus, the graph is not the correct presentation of the function. More points need to be considered. By extending the above table with more points we construct Table 4.3.

Table 4.3: Extension with more points

X	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

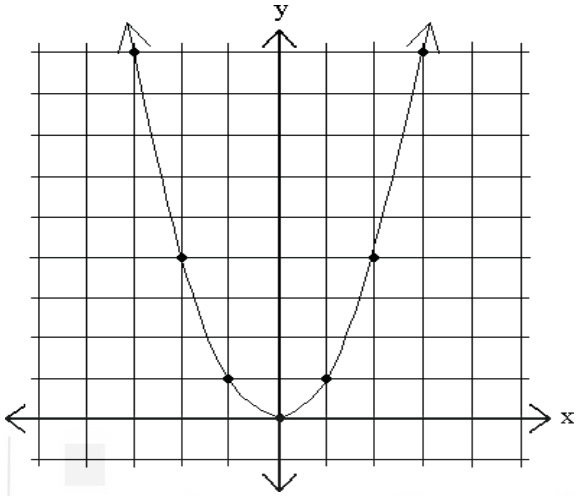


Fig. 4.20: Graph of Equation $y=x^2$

We observe that the graph in Fig 4.20 is a parabola. Note that the parabola does not have a constant slope. In fact, as x increases by 1, starting with x=0, y increases by 1,3,5,7,,.As x decreases by 1, starting with x=0, y again, increases by 1,3,5,7,….

Consider a quadratic function of the form: $f(x) = ax^2 + bx + c$ with $a \neq 0$. The graph of such a function takes one of the two general forms shown in Fig. 4.21, depending on the sign of a.

A quadratic function with $a > 0$ A quadratic function with $a < 0$

Fig.4.21: Graph of Quadratic Function

Example: Sketch the graph of function $y = 2x^2 - 8x + 6$. As the coefficient of x^2 is positive, so the graph is *U-Shaped* (Fig.4.22)

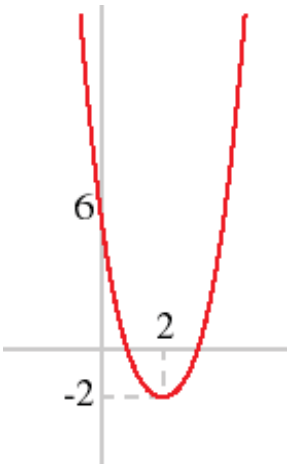


Fig.4.22: Graph of Function $y=2x^2-8x+6$

In the graph of $y = x^2$, the point (0, 0) is called the vertex. The vertex is the lowest point in a parabola that opens upward. In a parabola that opens down ward, the vertex is the highest point. We can also graph a parabola with a different vertex. For example, see the graph of $y=x^2+3$ (Fig 4.23). The graph is shifted up 3 units from the graph of $y = x^2$, and the vertex is (0, 3).

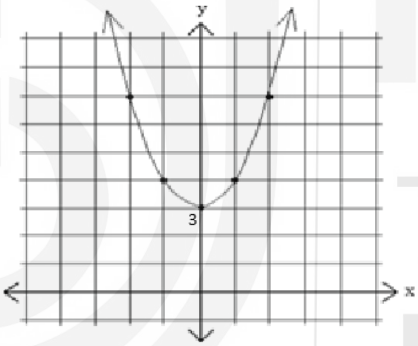


Fig. 4.23: Graph of $y=x^2 + 3$

We can also shift the vertex left and right. Observe the graph of $y=(x+3)^2$
In Fig 4.24.

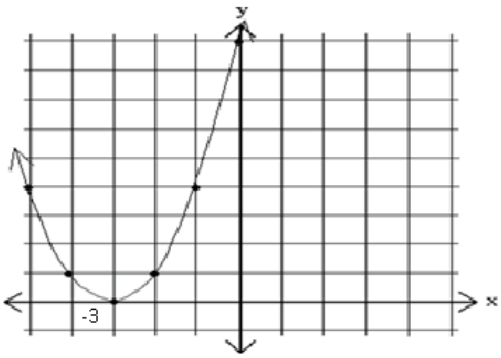


Fig.4.24: Graph of $y=(x+3)^2$

The graph is shifted *left* 3 units from the graph of $y=x^2$, and the vertex is

(-3, 0). Next, see Fig. 4.25, the graph of $y=(x-3)^2$, which is obtained by

shifting to the right.

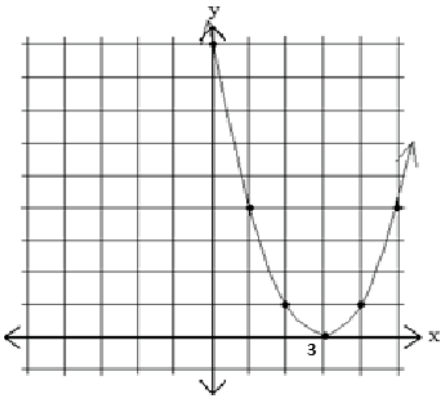


Fig. 25 : Graph of y = (x-3)^2

The graph is shifted to the *right* 3 units from the graph of $y=x^2$, and the vertex is (3, 0). Remember that the ***axis of symmetry*** is the line which divides the parabola into two symmetrical halves. It is given by the equation

$x=h$. For example, the line of symmetry in the graph (Fig. 4.26) of $y = (x-2)^2 + 1$ is $x = 2$.

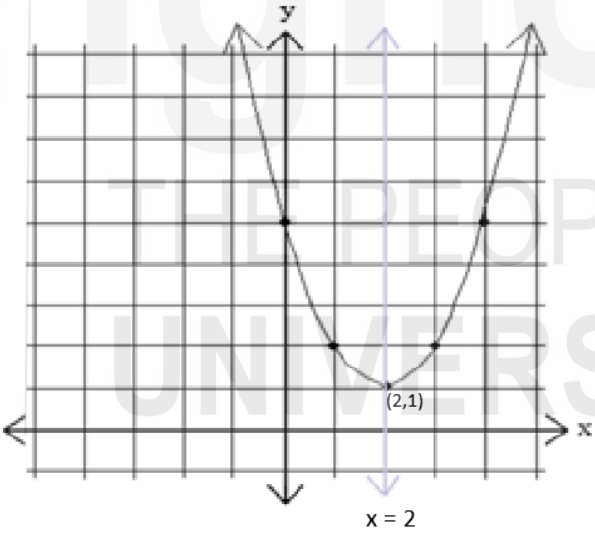


Fig.4.26: Axis of Symmetry

4.5.3 Computation of the co-ordinates of Vertex

Consider a generic quadratic expression: $y=ax^2+bx+c$. Start with **completing the square** on the equation to have $y = a\left[x^2 + \frac{bx}{a} + \frac{c}{a}\right]$ or $y=a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a}\right]$

The expression $-\left(\frac{b}{2a}\right)^2 + \frac{c}{a}$ is a constant and it does not depend on x. So, we can replace it with k.

Thus, we write :

$$y = a\left[\left(x + \frac{b}{2a}\right)^2 + k\right]$$

Now, depending on whether *a* is positive or negative, the parabola given by *y* will either have either a maximum value or a minimum value. Since *a* and *k* are fixed, this must occur when $\left(x + \frac{b}{2a}\right)^2 = 0$ Hence, *x* = - *b/2a* which implies that the function *y* has either a minimum value or a maximum value when *x*= - *b/2a.*

Thus, x-coordinate of the vertex is -*b/2a.* They-coordinate can be obtained by replacing, in the equation $y=ax^2+bx+c$, the variable x by *(-b/2a).*

Example: Find the vertex of $y=3x^2+x-2$ and graph the parabola.

To find the vertex, we look at the coefficients a, b and c. Then *k* is obtained by evaluating *y* at *h* = - $\frac{1}{6}$.

x-coordinate of the vertex $= \left(-\frac{b}{2a}\right) = \left(-\frac{1}{2x3}\right) = -\frac{1}{6}$

The y-coordinate is given by replacing x by $-\frac{1}{6}$ in the equation

$y=3x^2+x-2.$

$=3(-1/6)^2+(-1/6)-2 = 3/36-1/6-2$

$=1/12-2/12-24/12 = -25/12$

We get y-coordinate as $-\frac{25}{12}$. Thus, the vertex is at $(-\frac{1}{6}, -\frac{25}{12})$. T-chart can be prepared as shown in Table 4.4.

Table 4.4: T-chart of y = 3x^2 + x-2

x	**y = 3x^2 +x-2**
-2	8
-1	0
0	-2
1	2
2	12

The graph (Fig. 4.27) can be drawn with the specification vertex:

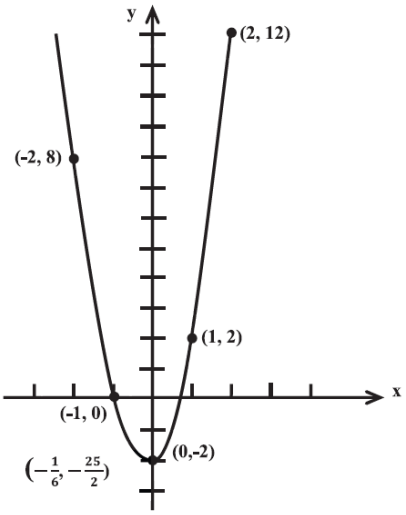


Fig.4.27: Graph of $y=3x^2+x-2$

4.5.4 Cubic Functions

Cubic functions are encountered in the discussion of total cost in microeconomic analysis. Consider a cubic function of the form $f(x)= ax^3+bx^2+cx+d$. Its graph (Fig 4.28) is obtained as follows:

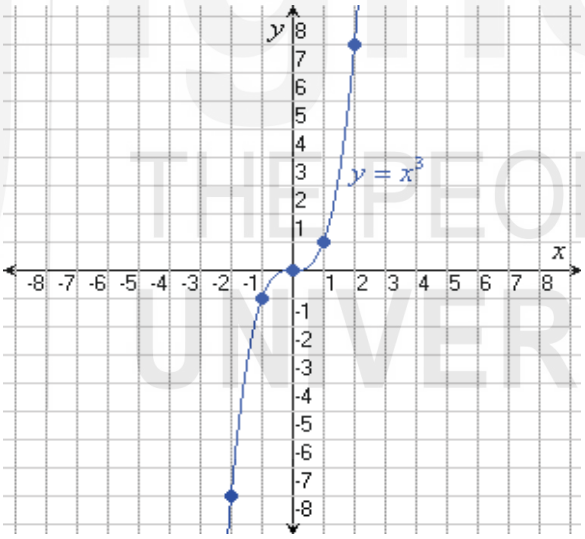


Fig. 4.28: Graph of Cubic Function $f(x) = x^3$

To trace the cubic function given by $f(x)=x^3$, find the x and y intercepts of the graph of f, find the domain and range of f and then sketch the graph of f. The y intercept is given by $(0, f(0)) = (0, 0)$. The x co-ordinates of the x-intercepts are the solutions to $x^3 =0$. The one x-intercept is at the points (0,0). The domain of $f(x)$ is the set of all real numbers. Since the leading coefficient of x^3 is positive, the graph of: f is up on the right and down on the left and hence the range of : f is the set of all real numbers. Table of values is as follows:

x	-2	-1	0	1	8
$f(x)=x^3$	-8	-1	0	1	8

Since f(-x) = - f(x), function f is odd and its graph is symmetric with respect to the origin (0,0).

4.6 GRAPHING FUNCTIONS ASYMPTOTICTYPES

We are familiar with exponential growth models such as population growth and compound interest. Such models are represented by functions, which are called asymptotic functions. Next, we examine behavior of such functions through their graphical representations.

4.6.1 Graph of Asymptote

An asymptote is, essentially, a line that a graph approaches, but does not intersect (Fig. 4.29). For example, in the following graph of $y=1/x$, the line approaches the x-axis (y=0), but never touches it. No matter how far we go towards infinity, the line will not actually reach y=0, but will always get closer and closer to it.

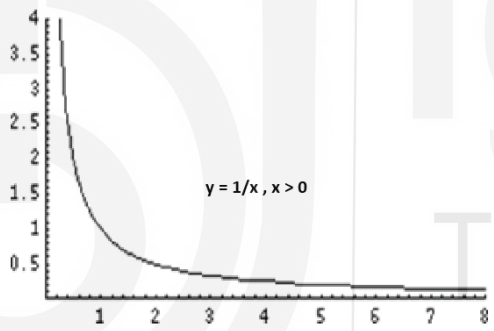


Fig. 4.29: Graph of Asympote

This means that the line y=0 is a horizontal asymptote. Horizontal asymptotes occur most often when the function is a fraction where the top remains positive,

but the bottom goes to infinity. Going back to the previous example, y=1/x is a fraction: When we go out to infinity on the x-axis, the top of the fraction remains 1, but the bottom gets bigger and bigger. As a result, the entire fraction gets smaller, although it will not approach zero. As we increase the values of x, the values of the function will gradually have values 1/2, then 1/3, then 1/10, even 1/10000, but never quite 0. Thus, y=0 is a horizontal asymptote for ***the function y=1/x.***

The function also serves as an example of ***vertical asymptote, because as y becomes larger and larger in absolute value, x becomes smaller and smaller in magnitude, but never becomes 0.***

Similarly, the equation $y = \frac{4}{x-2}$, gives one of the horizontal asymptotes, viz. y= 0, and one of the vertical asymptotes, viz. x=2 (Fig.4.30).

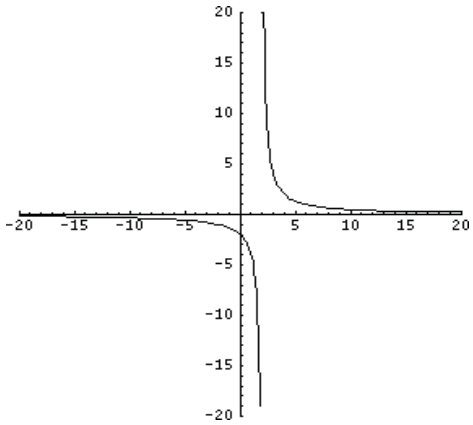


Fig. 4.30: Graph of the Function $y = \frac{4}{x-2}$

Note that as the graph approaches x=2 from the left, the curve drops rapidly towards negative infinity. This is because the **number or** is staying at 4, and the denominator is approaching to 0. That means that the fraction itself is getting very big and negative. When x is exactly 2, the function does not exist because you cannot divide by 0. Immediately after 2 it resumes at positive infinity, because the numerator is 4 and the denominator is again very tiny, but this time it is positive.

Next, we consider two examples of functions one of which gives only

(i) horizontal asymptote, but not a vertical asymptote, and (ii) vertical asymptote, which is not horizontal.

i) The function y=1/(x-2), for $0 \leq x < 2$, is asymptotic with x=2, being a vertical asymptote. Similarly,

ii) The function x=1/ (y - 2), for $0 \leq y < 2$ is asymptotic with y=2, being a horizontal asymptote

4.6.2 Square Root Function

As the name indicates we take a variable x to write the function in the form of f(x)= √x, with Domain/Range: [0,∞), and intercept (0,0), and which is neither even nor odd and is increasing on(0, ∞). Plotting its graph yields Fig. 4.31.

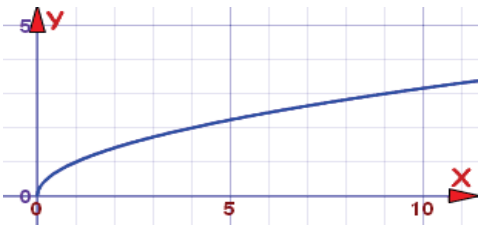


Fig.4.31: Graph of Square-root Function

You may see that square root function is used to depict an *is o cline* which provides information on the way different is o quant are related to each other. In fact, this is o cline based on square root function converges to a point where output is maximized and marginal product is zero.

4.6.3 Exponential Function

A general exponential function is of form $y=a^x$, where $a > 0$, $a \neq 1$, constant, and x any real variable. The discussion of the general exponential function requires substantial space, and hence, will not be discussed here. We discuss a special such function which is found useful in several applications, viz. the natural exponential, which is of the for my= e^x where *e* is Euler's number (a number approximately equal to 2.718281828). If we take its domain as the set of all real numbers i.e., $(-\infty, \infty)$, then its range is $(0, \infty)$. Since the graph never intersects or goes below the x-axis, they-values are not zero or negative (Fig. 4.32).

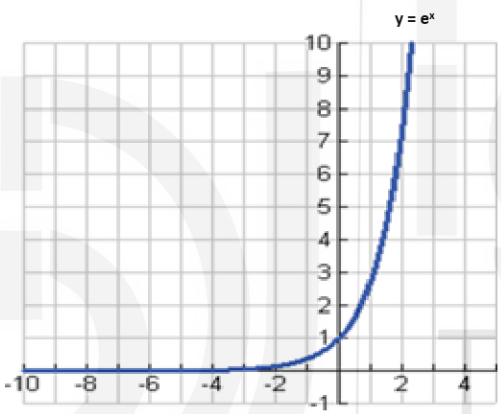


Fig. 4.32: graph of Exponential Function

Three of the most common application of exponential and logarithmic functions (to be discussed soon) have to do with interest earned on an investment, population growth, and carbon dating. For example, exponential functions are used to describe growth or decay processes with constant growth (or decay rate), population growth, growth of GDP, inflation etc.

4.6.4 Logarithmic Function

A logarithm function is a sort of inverse function of exponential function. Thus, if $b > 0$, $b \neq 1$, $y = b^x = f(x)$ is an exponential function, then $log_b y$ = g(y) (read as log of y to the base b) is such that, with some restrictions, f(g(y)) =y, and g(f(x)) =x. A function given by $y=log_b x$, where *b* is any number such that $b>0$,$b \neq 1$,*and* $x>0$, denotes logarithmic function. When $b>1$, the values of the logarithm increase; and for $0<b<1$, the values decrease.

The logarithmic function slowly grows towards positive infinity as the variable increases and slowly goes towards negative infinity as it approaches 0. Three special cases of logarithm are quite well-known and useful: (i) b=10, called decimal or common logarithm and is more used in science and engineering

applications. (ii) b = e, approximately equal to 2.718281828 is called natural logarithm and used frequently in Mathematics and physics, and (iii) base =2, is called binary logarithm, and is frequently used in computer science. For all bases b, including the three mentioned here, log b l= 0

In the logarithmic function, the domain is all positive real numbers (never zero) while the range is all real numbers. The graph is asymptotic to the y-axis -gets very, very close to the y-axis but does not touch it or cross it (Fig 4.33).

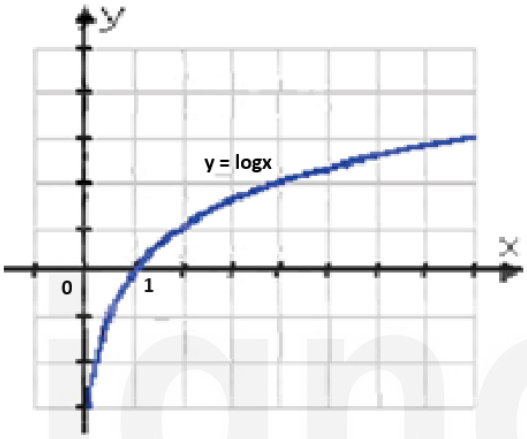


Fig. 4.33: Graph of Logarithmic Function

Check Your Progress -2

1) Draw the graph for the function, $y=x^2-2x$.

……

……

……

2) Let f be a rational function defined by $f(x) = \frac{x+1}{x-1}$. Find the x and y intercepts of the graphs of the function.

……

……

……

3) Draw the graph of an average fixed cost curve. Why do you think it is asymptomatic?

……

……

……

4.7 GRAPHING FUNCTIONS WITH TWO VARIABLES

In the above discussion we have seen that for a function of one variable, $y=f(x)$, the graph of f is the set of all points (x, y) in 2-dimensional space such that $y=f(x)$. In general, these points lie on a curve in the plane. Now consider a function of two variables.

The graph of a function of two variables is a *surface in three dimensions.* Thus we deal with a 3D (**three-dimensional graph).** In geometry and physics, **3D** stands for **three-dimensional**. It describes a space or object that has three measurements: **length**, **width**, and **depth** (or height).

While a 2D shape (like a square on paper) is flat, a 3D object (like a cube) has "volume"—it takes up physical space. Though discussion on 3D is not in the scope of this course, here you are introduced to 3D graphs for the sake of completeness. Further, if you need to understand more about this topic, you may explore it on your own.

To define a point in 3D space, we use a co-ordinate system with three axes, usually labeled as x, y, and z.

- X-axis (Length): Left to right.
- Y-axis (Height): Up and down.
- Z-axis (Depth): Forwards and backwards.

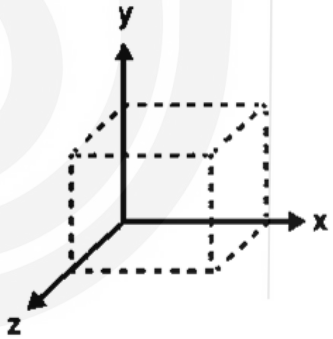


Fig. 4.34 : 3D Coordinate System

The graph of a function $z = f(x, y)$ of two variables, or the graph of a relationship $g(x, y, z)$ among three variables. For example, here (Fig. 4.35) is the graph of $z=2x^2+2y^2-4$.

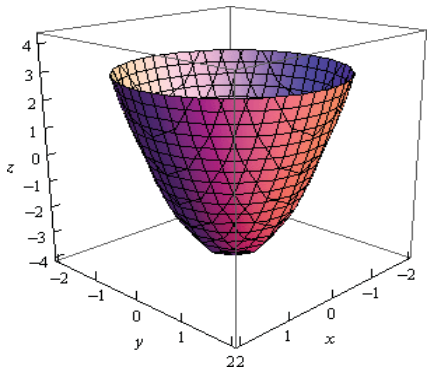


Fig. 4.35: Three-Dimensional Graph

The information in a three-dimensional graph is often collapsed into a two-dimensional graph with the use of contour lines. The procedure is: The *x* and *y* axes are retained, but instead of depicting *z* or *f(x,y)* axis, all *x*, *y* combinations giving rise to the same *z* or *f(x, y)* values are connected with a contour line; an arbitrary number of these may be shown for various values of *z* of *f(x,y)*.

4.6.4 Level Curves

Let *f* be a function of two variables, and *c is* a constant. The set of pairs *(x, y)* such that $f(x, y)=c$ is called the **level curve** (or, contour lines) of *f* for the value *c*.

Example: Let $f(x, y)=x^2+y^2$ for all *(x, y).* The level curve of *f* for
Value l is the set of all pairs (*x*, *y*) such that $x^2+y^2=1$, a circle of radius 1. This set is shown in Fig. 4.36.

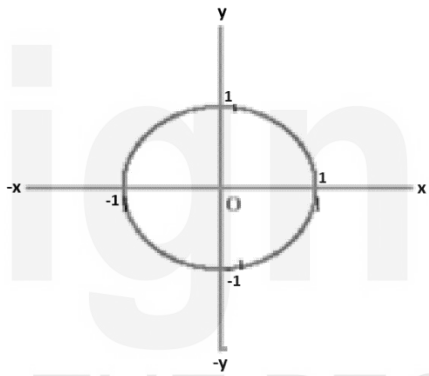


Fig.4.36: Level curve

Economists call the level curves of a utility function *indifference curves* and those of a production function *is o quants.* Thus, level curves have algebraic equations of the form $f(x, y) = k$ for all possible values of *k*. A contour diagram or contour map of a function $f(x, y)$ is a 2-dimensional graph (Fig. 4.37) showing several level curves in the xy-plane corresponding to several values of k.

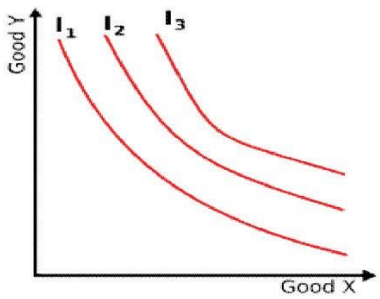


Fig.4.37: Indifference Curve Map

Example: Cobb-Douglas Production Model

Given P=the monetary value of all goods produced, K = total capital investment over the same period, and L = the total labor force, the following relationship between these quantities holds: $P = 1.01L^{0.75}K^{0.25}$

The contour diagram shown in Fig. 4.38 below shows that reducing labour increases capital investment in order to maintain the same level of production i.e., supposedly more investment is needed in machinery to make up for the lack of available labour. Conversely, increasing labour reduces capital investment since supposedly the work is being done by the labor force.

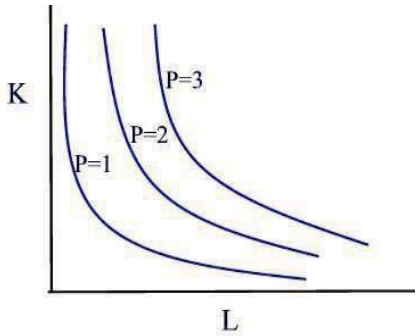


Fig. 4.38: Indifference Curve Map- Cobb Douglas Example

Example:

Let $f(x,)=x^2-y^2$. We will study the level curves $c=x^2-y^2$.

First, look at the case c=0. The level curve equation $x^2 - y^2 = 0$ factors to $(x - y) . (x + y) = 0$.This equation is satisfied if either $y = x$ or $y=-x$. Both these are equations for lines, so the level curve for c=0 is two lines.

See that this is the equation for a hyperbola. If c is positive, the hyperbolas open to the left and right. If c is negative, the hyperbolas open up and down.

For example, if c=l, the equation is $x^2-y^2=1$. If c=-1, the equation is $x^2-y^2=1$. - Several level curves are plotted in Fig. 4.39.

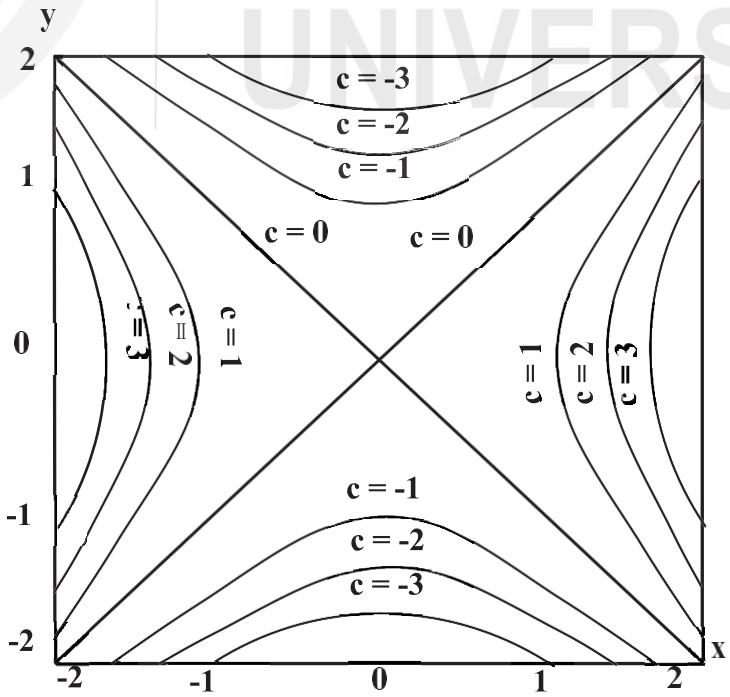


Fig.4.39: Level Curves

Check Your Progress -3

1) Explain the meaning of level curve.

…….

…….

…….

2) Let $f(x,y) = xy$ for all (x,y).Show a collection of level curves of f.

…….

…….

…….

3) Suppose a function is defined by $f(x,y)=1$ for all (x,y).Comment on the shape of level curves.

…….

…….

…….

4.8 SUMMARY

In this unit, we mainly discussed the method of sketching graphs of functions. However, first we discussed how to obtain algebraic equations for well-known geometric figures. In the reverse direction, for the purpose of graphing functions, starting with the relationship between elements of domain and co-domain, we recalled the basis of functional relationships. An operational method for dealing with a function was presented to identify dependent and independent variables. Before proceeding with the process of sketching the graph of a function, the use of the vertical line test to determine whether a relationship is a functional relationship was discussed. According to this test, the equations of circles, ellipses, etc., fail to meet its requirement of a functional relationship.

While discussing the method of graphing, this unit considered functions in five groups, viz., linear, curve, asymptotic, piecewise, and curve with branches. Linear graphs were drawn when the functions were linear, absolute value, or step-like. In cases of graphs showing curves, functions included were odd and even, quadratic and cubic. Among the graph of asymptotic types, functions included were square root, exponential and logarithmic. Graphs appearing in

pieces in the xy-plane comprised rational functions, piecewise functions and discontinuous functions. Finally, in this unit, in the last group of graphs shaving *curve* with branch functions concerned with hyperbola and rectangular hyperbola were discussed.

4.9 KEYWORDS

Abscissa : *x*-coordinate (horizontal) of a point m a 2-dimensional co-ordinate system.

Absolute Value Function : Obtaining the *absolute value* (or *magnitude)* of a number.

Asymptote : A line that a curve approaches, as it heads towards infinity.

Asymptotic : Graph approaching a point but never touching the **Discontinuity** point.

Axis of Symmetry : A line of symmetry for a graph. The two sides of a graph on either side of the axis of symmetry look like mirror images of each other.

Cobb-Douglas Production Function : Production function which is applied to describe how much output two inputs into a production process make.

Contour Diagram : Graph that provides one way of visualizing (contour map) functions of two variables.

Co-ordinates : Any of a set of two or more numbers used to determine the position of a point, line, curve, or plane in a space of a given dimension with respect to a system of lines or other fixed references.

Cubic Function : A polynomial function of degree 3.

Directrix : A line perpendicular to the axis of symmetry of a parabola.

Discontinuous: A function which for certain values or between certain values of the variable does not vary continuously as the variable increases.

Even Function: University function is *f(x)* is said to be even provided that *f(x)=f(-x)*.

Exponential Function : Function defined by ex $p(z)=e^z$.

Hyperbola : A hyperbola is a curve where the distances of any point from: a fixed point (the focus), and a fixed straight line (the directrix) are always in the same ratio.

Indifference Curve : A group of indifference curves for two commodities

Level Curves : A contour diagram or contour map of a function f(x;y) is a 2 dimensional graph showing several level curves in the *xy-plane.*

Logarithmic function : A function (as $y = \log_a x$ or $y = \ln x$) that is the inverse of an exponential function (as $y=a^x$ or $y=e^x$) so that the independent variable appears in a logarithm.

Odd Function : Univariate function $f(x)$ is said to be odd provided that $f(-x)=-f(x)$.

Piecewise Function : A function can be in Pieces. (split function)

Point Discontinuity : Graph of the function has a hole at $x=a$.

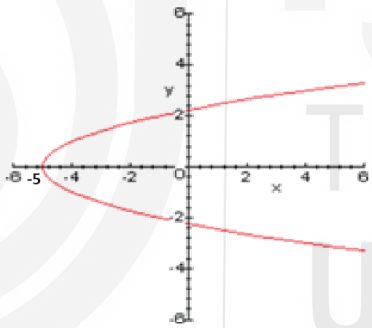
Quadratic Function : A function that can be described by an equation of the form(x)=ax^2+bx+c, where $a\neq 0$.

Rational Function : A function that is the ratio of two polynomials.

4.10 ANSWERTO CHECK YOUR PROGRESS

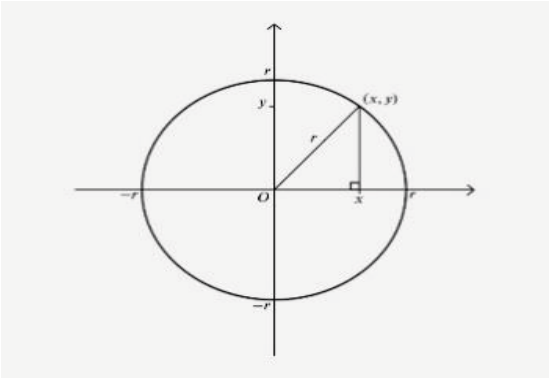
Check Your Progress -1

1) The graph is



By the vertical line test, this graph is not the graph of a function, because There are many vertical lines that intertsects the graph more than once.

2) As in the case of straight line, for translating the geometric figure a circle, create a co-ordinate system around the object, with centre of circle as origin of co-ordinate system, as shown below.



Mark an arbitrary point, say P, on the circle, as (x, y). Draw a perpendicular from Ponto X-axis, meeting it in say pointQ. Then length of perpendicularis of length y. and x is the length of the intercept on X-axis. Then from conventional/ synthetic geometry, we know

$x^2 + y^2 = r^2$

is an algebraic equation of the circle with radius r.

3) First prepare the t-chart which is Table :t-chart of *y*=(-5/3) x -2

x	*Y*=(-5/3) x-2
-6	8
-3	-2
0	2
3	-7

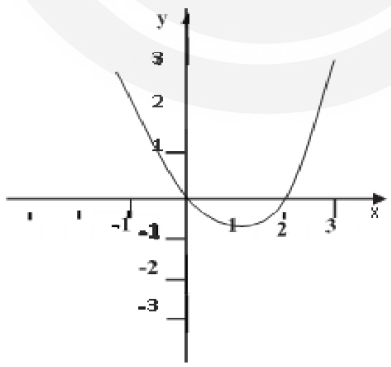
Then you draw the graph.

4) *x* = 4/3 or x = -8/3

5) $f(x) = \begin{cases} 25 + 0.30x, \text{for } 0 < x \leq 500 \\ 100 + 0.15x, \text{for } 500 < x < l000 \end{cases}$

CheckYourProgress-2

1)



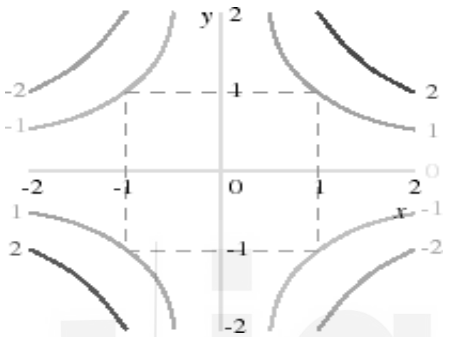
2) The x intercept is found by solving *f (x)*=0or, *x* + l=0. The *x* intercept is at the point (-1, 0). The y intercept is at the point (0, *f* (0)) = (0,-1).

3) Since total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases without t becoming zero.

Check Your Progress -3

1) A curve in two dimensions on which the value of a function f(x, y) is a constant.

2) The following figure may be seen. The number beside each curve is the value of the function to which it corresponds. The level curve for the value 0 consists of the axes.



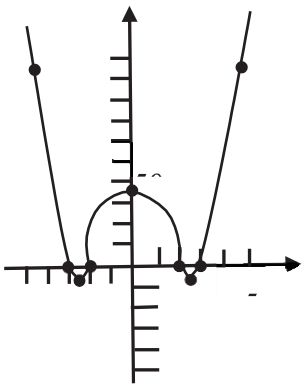
3) A level curve is defined as a *set*, and indeed some level curves of some functions are not″ curves″ at all. In the given function the value of *f* is 1 for all values of *x* and *y*. As a result, you get straight lines, not curves. Moreover, the level curve of this function for the value 2 is empty as there are *no* values of (*x*,*y*) such that *f*(*x*,*y*)=2.

4.11 SOME MORE EXERCISES

Q1. Let y = x4 -13x2+36. Draw its graph.

Ans. This is a positive even power (″to the fourth″), so the graph will be upon both ends. Factoring the polynomial, we get *y*=(*x*+3). (*x*-3) (*x*+2) (*x*-2), so the zeroes (*x*-intercepts) are-3, -2, 2, and 3. If you plot a few other points on your T-chart, it will be no trouble to graph this:

x	$y=x^4 -13x^2+36$
-4	84
-2.5	-6.19
-1	24
0	36
1	24
2.5	-6.19
4	84

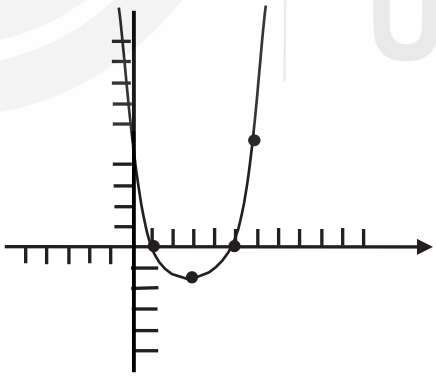


Graph of $y=x^4-13x^2+36$

Q2. A quadratic function is given as y=x2-6x+5. Draw its graph.
Ans. We need to plot more than three points to get the correct shape. Find the intercepts, which in this case are at (1, 0), (5, 0), and (0, 5); find the vertex, which in this case is at (3, -4). Remember that parabolas look like ″smilies″, not like straight lines. So, we want to plot a few more points:

Table: Chart of $y=x^2+6x+5$

x	$y=x^2+6x+5$
0	5
1	0
3	-4
5	0
6	5



Graph of $y=x^2+6x+5$

4.12 REFFERENCES AND FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011

3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MST-001: Foundation in Mathematics and Statistics
   Unit-2 Functions
   (https://egyankosh.ac.in/handle/123456789/92122)
5. MEC-203 : Quantitative Methods
   Unit-4 Co-ordinate Geometry and Representation of Functions
   (https://egyankosh.ac.in/handle/123456789/98141)

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